Paediatric Urology 2 Foundations, Bladder & Reconstruction

Campbell-Walsh-Wein · Chapters 32–40, 42–48

For MCh Urology theory & viva · NEET-SS / competitive MCQs

How to use (total ≈ 8h 53min even at a slow, careful pace)

- Each chapter reads top-to-bottom in the order you would write an essay answer: definition → pathology → diagnosis → management → complications → follow-up. The same material serves theory and viva.
- Boxed **ALGORITHM** and shaded **KEY POINT** panels carry the reasoning examiners ask about ("why do we…?").
- **MCQ one-liners** end each chapter: re-read these the day before the paper.
- The last section puts the big treatment choices side by side for essay questions.

Conventions: **underlined bold** = number to memorise · MCQ = exam trivia only · → *Ch 48* = cross-reference · † = not in the source chapter notes (added because it is essential in practice). Abbreviations are spelled out at first use in each chapter.

Chapter	Topic	Slow-reader time
32	Embryology of the Genitourinary Tract	27 min
33	Urologic Aspects of Paediatric Nephrology	20 min
34	Perinatal Urology	44 min
35	Urologic Evaluation of the Child	24 min
36	Paediatric Urogenital Imaging	25 min
37	Infection & Inflammation of the Paediatric GU Tract	39 min
38	Perioperative Management in Children	25 min
39	Laparoscopic & Robotic Surgery in Children	19 min
40	Clinical & Urodynamic Evaluation of LUTD	27 min
42	Bladder Anomalies in Children	16 min
43	Exstrophy–Epispadias Complex	63 min
44	Prune-Belly Syndrome	27 min
45	Posterior Urethral Valves	45 min
46	Neuromuscular Dysfunction of the LUT (Neurogenic Bladder)	29 min
47	Functional Bladder & Bowel Disorders, Enuresis	45 min
48	Lower Urinary Tract Reconstruction in Children	49 min
★	Putting it together: common paediatric pathways	9 min
	Total (37184 words, at 70 words/min)	**8h 53min**

32 Embryology of the Human Genitourinary Tract

Reading time ≈ 27 min (slow pace)

1 · OVERVIEW

- Urinary and reproductive systems are most closely linked early; development begins in the first trimester and is mostly complete by end of the second.
- **Indifferent stage** ≈ **8–9 wk** (Carnegie collection); thereafter androgens drive the sexes apart.
- Normal development is the map for every congenital anomaly and the first step in planning reconstruction.

2 · KIDNEY AND URETER

Three successive kidneys

Kidney	Features
†Pronephros	Cervical, week 4; non-functional, regresses completely
Mesonephros (second kidney)	†Thoracolumbar, functions transiently; its **duct (Wolffian)** persists. Tubules → efferent ductules, paradidymis (male); epoöphoron, paroöphoron (female)
Metanephros (permanent)	†Week 5; ureteric bud + metanephric blastema (intermediate mesoderm) → nephrons (glomerulus, proximal tubule, loop of Henle, distal tubule)

Reciprocal induction †

- The **ureteric bud** arises from the Wolffian duct and invades the metanephric mesenchyme. Bud stalk → ureter; repeated branching → pelvis, calyces, collecting tubules.
- Mesenchyme secretes **glial cell line-derived neurotrophic factor (GDNF)**, which binds **RET** receptor on the bud and drives budding and branching. **WT1** (Wilms tumour 1) makes mesenchyme competent; **PAX2** maintains bud and duct; loss → renal agenesis/hypoplasia (renal-coloboma syndrome).
- Bud tips induce adjacent mesenchyme to condense and undergo mesenchymal-to-epithelial transition → renal vesicle → comma → S-shaped body → nephron, which joins the collecting duct. Nephrogenesis ends ≈ 34–36 wk; no new nephrons after birth.

> **KEY POINT** Why one defect gives both dysplasia and an abnormal orifice: bud position on the Wolffian duct sets where it meets the blastema (kidney quality) and where its orifice lands in the bladder (see §6).

Ascent and rotation †

- The kidney ascends from pelvis to lumbar position (weeks 6–9), taking successively higher arterial supply; hilum rotates 90° from anterior to medial.
- Failure → pelvic/ectopic kidney, malrotation, multiple renal arteries; lower poles fusing before ascent → **horseshoe kidney**, trapped beneath the inferior mesenteric artery.

3 · CLOACA, BLADDER, TRIGONE AND URETHRA

- **Weeks 5–6**: the **urorectal septum** divides the cloaca into dorsal anorectal canal and ventral **urogenital sinus (UGS)**. Anorectal malformations are thought to arise from the **cloacal membrane**, not the septum.
- UGS + **allantois** → bladder and urethra. Bladder is initially continuous with the allantois → **urachal** remnant (†failed obliteration → patent urachus, cyst, sinus, diverticulum).
- Wolffian duct entry into the cloaca divides: cephalad **vesicourethral canal** → bladder and pelvic urethra; caudal UGS → phallic urethra (male), urethra and vestibule (female).
- Bladder and ureter share a wall: urothelium, lamina propria, outer smooth muscle, serosa. Sexes alike except the larger prostatic urethra in males.

Trigone

- Weeks 5–6: **common excretory ducts** (Wolffian duct distal to bud origin) dilate and join the UGS; **apoptosis** disconnects ureters from the mesonephric ducts so they fuse with the bladder.
- Orifices are incorporated and migrate **cranially and laterally**.
- Classic view: trigone = ureteral tissue extending into bladder. **Current (mouse lineage)**: mostly **bladder smooth muscle**, minor ureteral contribution.

Weigert–Meyer rule (duplex kidneys)

	Lower-pole ureter	Upper-pole ureter
Abnormal bud origin	Too **low** on Wolffian duct	Too **high**
Incorporation	Premature, excess migration	Delayed/incomplete
Orifice	Lateral	Caudal and medial (ectopic)
Result	**Vesicoureteral reflux (VUR)** — short intramural tunnel	**Ectopic insertion**: male — bladder neck, verumontanum, vas; female — Gartner duct cyst, vestibule

- Upper-pole orifice rotates dorsally relative to the lower-pole orifice. Male ectopia stays in Wolffian territory (above external sphincter) → no continuous wetting; †female ectopia below the sphincter → continuous dribbling with normal voids.
- **Ectopic vas**: vas joins the ureter instead of the verumontanum; both drain into the prostatic urethra.

4 · GONADS

- **Germinal ridge** at 4 wk on the medial urogenital ridge. Indifferent gonad = coelomic epithelium + mesenchyme + adjacent mesonephros + **primordial germ cells (PGCs)**.
- PGCs: yolk-sac wall (week 3) → dorsal hindgut mesentery → gonadal ridges. Primary sex cords at week 6; human seminiferous cords appear to derive from mesonephric tubules.
- Gonads identical at 7–8 wk; then testis becomes spherical, ovary stays oblong.

Testis

- **SRY** (sex-determining region Y, short arm of Y) → **Sertoli cells** enclose PGCs in cords (≈ **6 wk**). Without SRY protein → ovary.
- Cords canalise → seminiferous tubules → rete testis → 5–12 mesonephric tubules (efferent ductules) → epididymis; continuity by month 3. Thick tunica albuginea forms.
- **Leydig cells** from 8 wk make testosterone (placental chorionic gonadotrophin early, fetal pituitary later).

	Müllerian-inhibiting substance (MIS / AMH)	Testosterone
Cell	Sertoli	Leydig
Target	Müllerian (paramesonephric) duct	Wolffian duct, UGS, genital tubercle
Effect	Regression **8–10 wk** (sensitivity window 9–13 wk)	Wk 8–12 → epididymis, vas, seminal vesicle
Male remnants	Appendix testis, prostatic utricle	Appendix epididymis, paradidymis
If absent	Uterus, tubes persist	Wolffian ducts regress

Ovary — the default

- No SRY → no Sertoli cells, MIS or androgen → female. Cortex/medulla segregate ≈ 8 wk; germ-cell **nests** 10–11 wk.
- FOXL2+ cells surround oocytes → primordial follicle → primary (cuboidal, ≈ 20 wk) → secondary (term) → antral (postnatal). Oocytes arrest in meiosis I until puberty.

> **KEY POINT** Two independent testicular signals act on two independent duct systems: MIS decides Müllerian, testosterone decides Wolffian and external genitalia. Every disorder of sex development (DSD) below is a combination of the two.

5 · GONADAL DESCENT

- Gonad held by **cranial suspensory ligament (CSL)** above and **gubernaculum** below.
- **Transabdominal phase (10–15 wk)**: androgen → CSL regression; **INSL3** (insulin-like peptide 3, Leydig) → gubernacular swelling (androgen-independent); cord shortening positions testis at internal ring. *Insl3*-null mice: isolated intra-abdominal cryptorchidism.
- **Inguinoscrotal phase (from 7th month)**: gubernaculum migrates 3–5 cm to scrotum, hollowed by the **processus vaginalis**; driven by intra-abdominal pressure, steered by androgen via the **genitofemoral nerve → calcitonin gene-related peptide (CGRP)**. Androgen-dependent.
- Female: CSL persists; gubernaculum → **ovarian ligament + round ligament**. Patent processus → indirect hernia.

6 · GENITAL DUCTS

Male accessory glands

- **Prostate**: buds at **10 wk** from FOXA1+ endodermal UGS epithelium below the ureteric insertion; six stages (bud → branch → canalise → differentiate). Most buds in gutters lateral to the **verumontanum** (Wolffian ducts + utricle open on it; mesoderm–endoderm interface).
- Androgen acts through **androgen receptor (AR)** in **mesenchyme**; type 2 5α-reductase (5αR2) makes **dihydrotestosterone (DHT, ~10× AR affinity)** in stroma. 5αR2 deficiency → rudimentary prostate + severe hypospadias. Mesenchyme can induce prostate even from bladder epithelium.
- Oestrogen → squamous metaplasia via ESR1 in **both** stroma and epithelium.
- **Seminal vesicle**: Wolffian diverticulum at 10–12 wk; main anomaly = cyst.

Female tract

- Week 5: coelomic invagination → Müllerian ducts, which use the **Wolffian duct as guidewire** to reach the UGS (hence combined anomalies).
- Week 8 fuse → uterovaginal canal with midline septum; septum gone by week 9. Unfused cranial ducts → tubes; canal → corpus, cervix.
- **Vagina**: PAX2+ Müllerian plate is progressively replaced (14–21 wk) by FOXA1+ UGS epithelium → **entire vaginal epithelium and hymen are UGS-derived** (supports Bulmer over Koff). Canalisation 15–19 wk.
- Female Wolffian remnants: epoöphoron, paroöphoron, **Gartner duct**.

7 · EXTERNAL GENITALIA

- Order: ↑ anogenital distance (first sign) → tubercle elongation → urethral plate. Genital tubercle: ectoderm → skin/prepuce; mesoderm → corpora; **endoderm → urethral plate**.
- **"Double zipper"**: **opening** = proximal-to-distal canalisation of plate → urethral groove (proliferation, no apoptosis; androgen-**independent**, both sexes); **closing** = fusion of urethral folds → tubular urethra + raphe (androgen-**dependent**).
- **Glanular urethra** forms by direct canalisation within the glans (endodermal throughout; Glenister's ectodermal ingrowth theory outdated). Meatus = ecto–endodermal junction.
- Timeline: coronal sulcus 11–12 wk; shaft fused **16 wk**; complete 18 wk; fetal ventral curvature resolves 20 wk.
- **Prepuce** forms with glanular urethra and depends on it; preputial lamina delaminates postnatally.
- **Nerves**: dorsal nerves at 11 and 1 o'clock, fanning to the spongiosal junction; **nerve-free zone at 12 o'clock**, where tunica is thickest → dorsal plication site for mild–moderate curvature.

- **Female**: vestibular plate canalises but folds do not fuse → open vestibule; clitoris = penis without urethra, same 12 o'clock zone.

8 · DEVELOPMENTAL STEP → ANOMALY

Step	Anomaly
†Bud fails / no induction (GDNF–RET, PAX2)	Renal agenesis (absent hemitrigone)
†Abnormal bud–blastema contact	Dysplasia, hypoplasia, multicystic dysplastic kidney
†Early bud bifurcation / two buds	Bifid ureter / complete duplication
Bud too low / too high	VUR / ectopic ureter (Weigert–Meyer)
†Ascent or rotation fails	Pelvic kidney, malrotation, horseshoe kidney
Cloacal membrane / septation	Anorectal malformation; †cloacal anomalies, exstrophy
Allantois persists	Urachal remnants
Fold fusion arrests (shaft–glans junction commonest)	Hypospadias
Urethral plate splits longitudinally (Sonic hedgehog)	Urethral duplication (complete from bladder; Y along shaft)
Androgen excess in 46,XX	Vestibular fusion → virilisation (congenital adrenal hyperplasia, CAH)
CSL retained	Intra-abdominal testis
Gubernacular attachment abnormal / ectopic	Canalicular testis / ectopic (perineal, femoral, prepenile)
Gubernacular migration fails	Congenital cryptorchidism
Processus fails to close	Hydrocele, inguinal hernia
Processus fails to involute (fibrous remnant)	Acquired (ascending) testis
One Müllerian duct absent	Unicornuate uterus
Ducts fail to fuse	Uterus didelphys
Septum persists	Uterine obstruction/duplication
Vaginal plate fails to canalise	Transverse vaginal septum (middle–upper third)
Inferior end fails to open	Imperforate hymen — commonest vaginal obstruction
Müllerian + Wolffian both abnormal	Obstructed hemivagina and ipsilateral renal anomaly (OHVIRA)

9 · CLINICAL CORRELATIONS

- **Hypospadias** (≈ 1 in 250): ectopic meatus, dorsal hooded prepuce, ventral skin deficiency ± curvature. Abortive spongiosal elements lie alongside → **arrest, not deformation**; rest of penis normal. Cause mostly unknown (genetic susceptibility + endocrine disruptors).
- **CAH**: autosomal recessive cortisol defect → 46,XX virilisation; may be male-identical. Clues: **impalpable gonads + hyperpigmentation**; mimics hypospadias with bilateral impalpable testes.
- **Streak gonad** (45,X; 46,XX or 46,XY dysgenesis): no MIS, no testosterone → female phenotype. Hormone replacement; **remove streak if 46,XY** (gonadoblastoma).
- **Ovotesticular DSD**: ducts follow the ipsilateral gonad; female fertility possible, spermatogenesis not reported.
- **Persistent Müllerian duct syndrome (PMDS)**: 46,XY with uterus/tubes (hernia uteri inguinale), usually MIS-receptor defect; found at exploration for bilateral abdominal testes.
- **OHVIRA**: post-menarche cyclic pain, discharge, pyocolpos. Kidney is often **dysplastic with an ectopic ureter into the obstructed hemivagina**, not agenesis → persistent leakage after septum resection if missed.

KEY POINT Homologue map predicts remnants: Littre ↔ Skene, Cowper ↔ Bartholin, spongiosum ↔ vestibular bulb, gubernaculum ↔ ovarian + round ligaments, labioscrotal swellings ↔ labia majora. Traps: vagina is UGS-derived; appendix testis is Müllerian, appendix epididymis Wolffian.

10 · MCQ ONE-LINERS

1. Indifferent stage of external genitalia: 8–9 weeks.
2. First sign of masculinisation: increased anogenital distance.
3. Entire male urethral epithelium: urogenital sinus endoderm.
4. Canalisation of urethral plate is androgen-independent; fold fusion is androgen-dependent.
5. Commonest hypospadiac meatal site: shaft–glans junction.
6. Penile nerve-free zone: 12 o'clock — dorsal plication site.
7. Prostatic buds appear at 10 weeks from FOXA1+ UGS epithelium.
8. 5αR2 deficiency: rudimentary prostate and severe hypospadias.
9. DHT has about tenfold higher AR affinity than testosterone.
0. SRY lies on the short arm of Y; acts in pre-Sertoli cells.
1. MIS from Sertoli cells; Müllerian regression 8–10 weeks.
2. Appendix testis: Müllerian; appendix epididymis: Wolffian.
3. INSL3 knockout: isolated intra-abdominal cryptorchidism.
4. Gubernacular swelling is androgen-independent.
15. Inguinoscrotal descent: androgen via genitofemoral nerve CGRP.
16. Acquired ascending testis: failed involution of processus vaginalis.
17. Entire vaginal epithelium and hymen derive from UGS (FOXA1).
18. Commonest cause of vaginal obstruction: imperforate hymen.
19. Transverse vaginal septum: junction of middle and upper thirds.
20. Uterus didelphys: failed fusion of caudal Müllerian ducts.
21. Urorectal septum divides cloaca at weeks 5–6.
22. Trigone muscle: mainly bladder smooth muscle (lineage tracing).
23. Upper-pole duplex ureter: caudal, medial, ectopic; lower pole: lateral, refluxing.
24. CAH clues in "male": impalpable gonads, hyperpigmentation.
25. 46,XY streak gonads: remove (gonadoblastoma).
26. PMDS: usually defective MIS receptor; hernia uteri inguinale.
27. †GDNF (mesenchyme) binds RET (ureteric bud) to drive branching.
28. †Horseshoe kidney ascent arrested by inferior mesenteric artery.

33 Urologic Aspects of Pediatric Nephrology

Reading time ≈ 20 min (slow pace)

1 · RENAL FUNCTION IN THE CHILD

- Congenital anomalies of the kidney and urinary tract (CAKUT) = **commonest cause of end-stage kidney disease (ESKD) in children**.
- Renal blood flow rises from 3–4% to 12–16% of cardiac output in year 1, because renal vascular resistance falls (flow shifts to outer cortical glomeruli; angiotensin II constricts the efferent arteriole → ↑ glomerular pressure).
- Glomerular filtration rate (GFR): doubles in the first 2 weeks, adult level by **2 years**; term infant ≈40 mL/min/1.73 m².
- Creatinine: first 48 h reflects **maternal** function; by day 7 the neonate's own (**<0.5 mg/dL** at term).
- † Bedside Schwartz estimate: eGFR = 0.413 × height (cm) / serum creatinine (mg/dL).
- Neonate dilutes to 50 mOsm/L but concentrates poorly: ADH (antidiuretic hormone) unresponsiveness, short loops, low Na^+-K^+-ATPase, low medullary urea.
- Sodium: negative balance in week 1 (atrial natriuretic peptide). Potassium: higher serum K^+ (aldosterone high but tubule unresponsive). Bicarbonate: lower than adults; matures over 3–4 weeks.

> **KEY POINT** The neonatal kidney is short and "hormonally deaf": maternal creatinine, salt wasting, hyperkalaemia, low bicarbonate and poor concentration all reflect immaturity, not disease.

2 · HAEMATURIA

- Microscopic: **>5 red blood cells (RBCs)/high-power field** on fresh spun urine; ≈1% of school children, mostly resolves within 1 year.
- **Fresh microscopy is the critical step**: red/bloody → non-glomerular; brown "tea-coloured" with RBC casts → glomerular. Heme-positive dipstick without RBCs → haemoglobinuria/myoglobinuria.
- Urinary tract infection (UTI) = commonest cause of gross haematuria → culture. Trauma, pain or clots → urological work-up, often computed tomography (CT). Stone history → stone imaging → *Ch 54*.
- Suspected glomerulonephritis (GN): creatinine, electrolytes, blood count, albumin, complement C3/C4, antistreptolysin O (ASO)/anti-DNase B → nephrology.
- Isolated microscopic haematuria: ultrasound (US) low-yield; **voiding cystourethrography (VCUG) and cystoscopy not indicated**.
- Gross-only causes: urethrorrhagia, Wilms/bladder tumour, renal vein thrombosis, papillary necrosis, nutcracker. Microscopic-only: thin basement membrane, interstitial nephritis, sickle cell.

3 · PROTEINURIA

- Normal urine protein-to-creatinine ratio (UPCR): **<0.2 mg/mg** (<0.7 newborn); timed <4 mg/m²/h; adolescents <150 mg/24 h.
- **Transient** (fever, exercise) >75% of isolated cases. **Orthostatic**: normal first-morning UPCR, raised after hours upright — benign. **Fixed** (glomerular, tubular, reflux nephropathy, obstructive uropathy, dysplasia) → full evaluation ± biopsy.

4 · GLOMERULAR DISEASE

Age filters: <1 year → congenital/genetic (Denys-Drash, Frasier); 1–10 years → minimal change, post-streptococcal GN, Henoch-Schönlein purpura (HSP), haemolytic uraemic syndrome (HUS); ≥7 years → lupus, IgA, membranoproliferative GN (MPGN), Alport.

Nephrotic syndrome

- Proteinuria **>40 mg/m²/h or UPCR >2.0** + hypoalbuminaemia + hyperlipidaemia; oedema of eyelids, scrotum, labia.
- Congenital (onset <3 months): genetic/infective; immunosuppression unhelpful.
- Primary: **steroids first, no biopsy** — minimal change nephrotic syndrome (MCNS) is 90% under 10 years and ~90% steroid-sensitive.

Category	Definition	Action
Steroid-sensitive	Remission within 8 weeks	~50% relapse
Frequently relapsing	≥2 relapses in 6 months or ≥4 in 12	Steroid-sparing: cyclophosphamide, mycophenolate, rituximab, calcineurin inhibitors
Steroid-dependent	2 consecutive relapses on steroids	
Steroid-resistant	No remission after 6–8 weeks	Biopsy → usually focal segmental glomerulosclerosis (FSGS), which recurs after transplant

Nephritic syndrome

Haematuria + proteinuria + hypertension + acute kidney injury (AKI). Complement narrows the differential:

C3	C4	Diagnoses
Low	Normal	Post-infectious GN, C3 glomerulopathy
Low	Low	Lupus nephritis, MPGN
Normal	Normal	IgA nephropathy, HSP, Alport, ANCA (antineutrophil cytoplasmic antibody) GN, anti-GBM (glomerular basement membrane) disease

- **Post-streptococcal GN**: weeks after pharyngitis/pyoderma; C3 <50% normalises by **8–12 weeks**; persistently low C3 → biopsy.
- **IgA nephropathy**: commonest primary GN; painless gross haematuria **concurrent with** respiratory infection (vs weeks' latency in post-infectious GN).
- **Alport**: type IV collagen (mostly X-linked COL4A5); haematuria + high-frequency sensorineural deafness; ~50% of males ESKD by 25 years.
- **Anti-GBM**: rare after shockwave lithotripsy and in Alport patients after transplant.
- **HUS**: commonest cause of severe AKI in a healthy young child. Shiga-toxin E. coli (bloody diarrhoea) → supportive care, ~half dialysed. Atypical HUS (complement dysregulation) → eculizumab.

5 · TUBULOPATHIES

Renal tubular acidosis (RTA): hyperchloraemic (normal anion gap) acidosis with inappropriately high urine pH.

Type	Defect	Key features
Proximal / Fanconi	Bicarbonate leak	Urine acidifies once HCO_3^- falls; Fanconi adds glucosuria, aminoaciduria, phosphaturia (cystinosis, drugs)
Distal	Distal acidification failure	Hypokalaemia, hypercalciuria, **nephrocalcinosis/stones**, rickets
Type 4	Aldosterone deficiency/resistance	**Hyperkalaemia**; obstructive uropathy is a cause → urological correction

6 · CYSTIC DISEASE

- Multicystic dysplastic kidney (MCDK): non-communicating cysts, non-functional; bilateral is lethal. Associated vesicoureteral reflux (VUR) or pelviureteric junction obstruction in up to 25% → VCUG,

especially with hydroureteronephrosis. Follow function, blood pressure (BP) and US 3-monthly in year 1 → 6-monthly to involution or ≥5 years; trend to annual review into adulthood.
- Autosomal recessive polycystic kidney disease (ARPKD, *PKHD1*): enlarged kidneys, hepatic fibrosis; >75% ESKD by 5 years. Autosomal dominant (ADPKD, *PKD1/PKD2*): usually adult.
- Nephronophthisis: polyuria, anaemia → ESKD by adolescence; syndromic forms (Senior-Løken, Joubert, Bardet-Biedl).

7 · HYPERTENSION

- Repeated BP **>95th percentile** for age, sex, height; adolescents normal <120/80. Up to a quarter white-coat.
- Age predicts cause: infants → coarctation/renovascular; early childhood → Wilms, neuroblastoma; school age → **renal parenchymal incl. reflux nephropathy**; obese >6 years and adolescents → essential.
- Initial tests: urinalysis, electrolytes, urea, creatinine, lipids. No screening renal US >6 years unless urinalysis/function abnormal. Echocardiogram when drugs start.
- Lifestyle first (stage 1, no end-organ damage); drugs for stage 2, symptoms, end-organ damage or secondary cause. Target <90th percentile; chronic kidney disease (CKD) **<50th**.
- Emergency = severe BP with end-organ injury; urgency = without.

8 · ACUTE KIDNEY INJURY

KDIGO stage	Creatinine	Urine output
I	↑ ≥0.3 mg/dL in 48 h or 1.5–1.9×	<0.5 mL/kg/h for 6–12 h
II	2.0–2.9×	<0.5 for ≥12 h
III	≥3× or ≥4.0 mg/dL or clearance <35 (<18 years)	<0.5 for ≥24 h or <0.3 for 12 h

- KDIGO = Kidney Disease: Improving Global Outcomes. Creatinine lags GFR by 24–72 h and rises only after ~50% loss.
- Fractional excretion of sodium (FeNa) **<1%** = prerenal (tubule intact); >3% = intrinsic; on diuretics use urea <35%.
- Prerenal → vigorous isotonic fluid, no potassium until urine flows. Oliguria → insensible losses + output. Dose drugs as if GFR <15.
- Renal replacement therapy (RRT) for diuretic-resistant overload, refractory hyperkalaemia/acidosis, uraemia, nutrition; continuous RRT is standard when unstable. Two-thirds develop CKD later.

9 · CHRONIC KIDNEY DISEASE

- Abnormality >3 months; stages (children >2 years) by GFR: I ≥90, II 60–89, III 30–59, IV 15–29, V **<15** or dialysis.
- Young children: CAKUT (hypo/dysplasia, posterior urethral valves, prune belly) and cystic disease; adolescents: glomerular disease.
- Complications: growth failure (growth hormone **resistance**), CKD-mineral bone disease (rickets, slipped epiphysis), anaemia (erythropoietin deficiency → stimulating agents + iron), acidosis, cardiovascular death.
- **Do not restrict protein**; strict BP control (50th percentile) is the only proven way to slow progression.

10 · DIALYSIS AND TRANSPLANTATION

- Peritoneal dialysis first if **<9 years / <20 kg** (venous access fails); prior vesicostomy acceptable; contraindicated by omphalocele, gastroschisis, diaphragmatic hernia, bladder exstrophy.

- Transplant is the goal, often pre-emptive living-related; many wait for **≥10 kg and 65 cm** (graft thrombosis).
- **Urologist's role**: reconstruct the lower tract before transplant. Native nephrectomy for recurrent pyelonephritis, refractory hypertension or uncontrolled nephrotic syndrome.
- Living donor: delayed graft function 5% vs 15%; half-life 20 vs 12.5 years. Risks: cytomegalovirus, Epstein-Barr, BK virus; malignancy ≈7-fold.

11 · MCQ ONE-LINERS

1. Commonest cause of paediatric ESKD: CAKUT.
2. GFR reaches adult levels by 2 years.
3. Neonatal creatinine in first 48 h: maternal.
4. Microscopic haematuria: >5 RBCs/high-power field.
5. Commonest cause of gross haematuria in children: UTI.
6. Heme-positive dipstick, no RBCs: haemoglobinuria or myoglobinuria.
7. Normal UPCR in older children: <0.2 mg/mg.
8. Nephrotic-range proteinuria: >40 mg/m²/h or UPCR >2.0.
9. Commonest nephrotic lesion in children: minimal change.
0. Steroid-resistant nephrotic syndrome: usually FSGS.
1. Low C3 and C4: lupus nephritis or MPGN.
2. C3 normalises after post-streptococcal GN by 8–12 weeks.
13. Commonest primary GN worldwide: IgA nephropathy.
14. Alport: COL4A5, X-linked, sensorineural deafness.
15. Atypical HUS treatment: eculizumab.
16. Distal RTA: hypokalaemia, nephrocalcinosis.
17. Hyperkalaemic RTA (type 4) may be caused by obstructive uropathy.
18. ARPKD gene: PKHD1; with congenital hepatic fibrosis.
19. Hypertension in school-age child: renal parenchymal, including reflux nephropathy.
20. Prerenal AKI: FeNa <1%.
21. CKD stage V: GFR <15 mL/min/1.73 m².
22. Growth failure in CKD: growth hormone resistance, not deficiency.
23. Only proven measure slowing paediatric CKD: strict BP control.
24. Peritoneal dialysis preferred <9 years and <20 kg.
25. Transplant size threshold: ≥10 kg and 65 cm.

34 Perinatal Urology

Reading time ≈ 44 min (slow pace)

1 · FETAL RENAL DEVELOPMENT AND PRENATAL IMAGING

- Metanephric kidney appears at **day 28**; urine production begins in earnest at **14 weeks**. After **~16 weeks** amniotic fluid is mainly fetal urine — the basis of oligohydramnios in urinary obstruction and of pulmonary hypoplasia, since lungs need fluid to develop.
- Urinary tract dilation (UTD; antenatal hydronephrosis) = **second commonest prenatal anomaly** after cardiac, 1–3% of pregnancies; most never need surgery.
- Screening ultrasound (US) at 18–20 weeks. Amniotic fluid index (AFI) = sum of four largest pockets: normal 8–18 cm; oligohydramnios **<5–6 cm**.
- US is sensitive for dilation but a poor predictor of diagnosis: severity tracks pelviureteric junction (PUJ) obstruction and posterior urethral valves (PUV), not vesicoureteral reflux (VUR) or duplex anomalies. Exstrophy and neural tube defects are usually diagnosed correctly; genital ambiguity is not.
- Fetal magnetic resonance imaging (MRI): only for anatomical detail — cloacal anomaly, urogenital sinus, exstrophy, duplex anomalies with outlet obstruction.

> **KEY POINT** The value of prenatal detection is triage and counselling (differential, postnatal plan, natural history); there is no proof it improves outcome.

2 · GRADING SYSTEMS

Anterior–posterior renal pelvic diameter (APD)

- Objective; transverse image at mid-kidney using the hilum. Classic error: measuring in the sagittal plane.
- APD rises with gestational age (GA), as third-trimester urine output rises, so **a value means nothing without its GA**.

Degree	2nd trimester	3rd trimester
Mild	4 to <7 mm	4 to <9 mm
Moderate	7–10 mm	9–15 mm
Severe	>10 mm	>15 mm

- Probability of any postnatal pathology rises from ≈12% (mild) to ≈88% (severe); PUJ obstruction and PUV rise with grade, but **VUR shows no trend**.

Society for Fetal Urology (SFU) grading — postnatal only

Grade	Finding
0	No dilation
1	Urine barely splits the renal sinus
2	Full pelvis; major calyces dilated
3	Minor calyces uniformly dilated; parenchyma spared
4	Parenchymal thinning

- All SFU 3 kidneys keep differential renal function (DRF) >40%; ~70% of SFU 4 do. Early surgery mostly SFU 4. Weakness: subjective; ignores ureter and bladder.
- Medullary pyramidal thickness (PT) **≤3 mm** predicts pyeloplasty and reduced function; reliable between raters.

UTD classification (multidisciplinary consensus)

Six parameters: APD, calyceal dilation (central vs peripheral), parenchymal thickness, parenchymal appearance, ureter, bladder (+ unexplained oligohydramnios prenatally). Normal APD ceiling: **<4 mm (16–27 weeks) → <7 mm (≥28 weeks) → <10 mm (postnatal >48 h)**, with all other parameters normal.

Feature	APD	SFU	UTD
Nature	Objective	Subjective	Both
Period	Prenatal	Postnatal	Both (A1, A2–3; P1–P3)
Assesses	Pelvis	Pelvis, calyces, parenchyma	Whole tract + amniotic fluid

3 · PRENATAL RISK STRATIFICATION AND FOLLOW-UP

Parameter	A1 — low risk	A2–3 — increased (any one)
APD 16–27 weeks	4 to <7 mm	≥7 mm
APD ≥28 weeks	7 to <10 mm	≥10 mm
Calyces	Central or none	Peripheral
Parenchyma, ureter, bladder	Normal	Abnormal
Oligohydramnios	None unexplained	Unexplained

ALGORITHM

A1 before 28 weeks → repeat US after 32 weeks → postnatal US after 48 h

A2–3 → US every 4 weeks to delivery (lesion may progress and change delivery plan; second anomaly likelier) → postnatal US after 48 h

- Multidisciplinary prenatal consultation for: exstrophy, cloaca/urogenital sinus, autosomal recessive polycystic kidney disease (ARPKD), outlet obstruction (PUV, prune-belly syndrome [PBS], urethral atresia), myelomeningocele.

4 · FETAL LOWER URINARY TRACT OBSTRUCTION (LUTO) AND OLIGOHYDRAMNIOS

- Suspect in a male fetus with bilateral hydroureteronephrosis, distended non-cycling bladder, **"keyhole" sign** (dilated bladder + posterior urethra) and oligohydramnios. Imaging cannot separate PUV, urethral atresia and PBS; only fetal cystoscopy can.
- "Pop-offs" (urinary ascites, urinoma, VUR, unilateral dysplasia) mark a system under tension.
- Poor prognosis: oligohydramnios **before 27 weeks**, renal cortical cysts, increased echogenicity. Oligohydramnios after 27 weeks → favourable lungs.
- Fetal urine sodium, chloride, osmolality and β2-microglobulin may predict need for renal replacement but are **not validated**; European consensus (2022): fluid volume, echogenicity and urine biomarkers are not accurate predictors. † Classic "favourable" values on serial bladder taps: Na <100 mEq/L, Cl <90 mEq/L, osmolality <200 mOsm/L, β2-microglobulin <4 mg/L.

Ruano stage	Fluid / kidneys	Urine biochemistry	Intervention
I — mild	Normal fluid, normal echogenicity	Favourable	Not indicated
II — severe, function preserved	Oligo/anhydramnios; hyperechoic; no cysts/dysplasia	Favourable within 3 samples	**Indicated** — lungs and kidneys
III — severe, function abnormal	Anhydramnios; hyperechoic ± cysts, dysplasia	Unfavourable after 3 samples	Maybe — lungs only

Options and technique

- Early delivery (lungs mature), vesicoamniotic shunt (VAS), fetal cystoscopy, serial amnioinfusion. Window **18–24 weeks**; need ≈1 in 60,000.

- Operating room, maternal epidural; fetal analgesia/paralysis via umbilical vein; percutaneous US-guided access.
- Complications ≈**44%**: incomplete drainage/shunt failure (commonest), preterm labour, membrane rupture, urinary ascites, chorioamnionitis, bowel herniation. Rodeck shunt lies flush to limit migration.

Feature	VAS	Fetal cystoscopy
Diagnosis of cause	No	Yes — views posterior urethra
Mechanism	Bladder → amniotic bypass	Laser valve ablation → bladder cycles
Specific complications	Blockage, migration, bowel herniation	Urethrocutaneous/rectourethral fistula (≤10%)
Survival	Improved	Improved
Renal function (PUV)	Not improved	Short-term improvement only

- Timing: one series found VAS before 16 weeks gave better lung and renal outcome than later placement; newer shunts appear safe even before 14 weeks.
- Serial amnioinfusion: for lung development in bilateral renal agenesis (no urine, so shunting cannot work); RAFT trial ongoing.
- Long-term after VAS: over half renal insufficiency, a third renal replacement; a third need clean intermittent catheterisation (CIC).

> **KEY POINT** Counsel three facts: intervention improves perinatal survival (lungs); complications exceed 40%; it does not reliably prevent long-term renal replacement, because dysplasia is often established before the needle goes in.

5 · POSTNATAL IMAGING

- **First US after 48 h**: newborns are relatively dehydrated until third-spaced fluid returns, so earlier scans under-read dilation. Delay up to 2 weeks is harmless.

Parameter	P1 — low	P2 — intermediate	P3 — high
APD	10 to <15 mm	≥15 mm	≥15 mm
Calyces	Central	Peripheral	Peripheral
Parenchyma	Normal	Normal	**Abnormal**
Ureter	Normal	Abnormal	Abnormal
Bladder	Normal	Normal	Abnormal
SFU equivalent	1–2	3	4

Classify by the highest-risk feature: parenchymal thinning makes P3 at any APD.

- **Voiding cystourethrogram (VCUG)**: VUR in up to 31%, and up to 25% with VUR have a normal first US. Order VCUG by **UTI risk** — females, uncircumcised males, high-grade dilation, dilated ureter, abnormal bladder. Technique: ALARA (as low as reasonably achievable) radiation, non-balloon catheter, cycling, scout + filling + voiding films. Outpatient unless outlet obstruction suspected (bladder wall ≥0.5 cm, distended bladder, bilateral hydroureteronephrosis). Active surveillance of known VUR is safe in stable infants; benefit of finding asymptomatic VUR is unproven.
- Contrast-enhanced voiding urosonography (CeVUS): no radiation, possibly more sensitive for VUR, but poor urethral view → VCUG when posterior urethra matters.
- **Technetium-99m mercaptoacetyltriglycine (MAG3)** diuretic renogram: preferred over DTPA (diethylenetriaminepentaacetic acid) for better extraction; "well-tempered" protocol standardises hydration, bladder catheter, diuretic dose/timing. Surgery considered if **DRF <40%**, T½ >20 min, or retention on delayed upright images.

- MAG3 pitfalls: false obstruction from dehydration or furosemide given before peak uptake into a capacious pelvis; region-of-interest placement; needs intravenous line + catheter.
- **Dimercaptosuccinic acid (DMSA)**: cortical agent — dysplasia/defects in VUR, confirming non-function in multicystic dysplastic kidney (MCDK), upper-pole function in duplex.
- Magnetic resonance urography (MRU): anatomy + function without radiation; for ectopic ureter, megaureter, crossing vessels. Drawbacks: sedation, cost.

6 · POSTNATAL MANAGEMENT, UTI AND PROPHYLAXIS

Element	P1 (SFU 1–2)	P2 (SFU 3)	P3 (SFU 4)
Repeat US	3–6 months	Individualised	~1 month
Continuous antibiotic prophylaxis (CAP)	No	Discretion; **yes if ureter >7 mm**	Yes
VCUG	No	Discretion (outpatient)	Yes
MAG3	No	Discretion	If persistent
Course	Resolves by 2–4 years	Variable	Highest UTI, function loss, surgery

- Dissent: reflux is raised at every grade, so some base VCUG on ureteral dilation or LUTO features rather than P grade alone.
- CAP without VUR in P3 continues until pyeloplasty or **1 year**, as ~90% of UTIs occur in year 1.
- Urinary tract infection (UTI) in 8–22%. Agreed risk factors: **female sex, high-grade dilation, intact foreskin**; debated: ureteral dilation, VUR, no CAP.
- Circumcision protects: number needed to treat (NNT) = **10**; odds ratio ≈0.3.
- Evidence for CAP is weak because UTI is uncommon; the shift is from blanket CAP to targeting high-grade UTD, females, uncircumcised males and dilated ureter/VUR. SFU registry (mostly low-grade, UTI 8%) found no overall benefit; McMaster registry (mostly high-grade, 19%) found benefit — case mix, not contradiction.

Prophylactic agent	Minimum age
Amoxicillin; first-generation cephalosporin	Newborn
Nitrofurantoin; trimethoprim	**≥8 weeks** (immature hepatic metabolism)

ALGORITHM

Prenatal UTD → US after 48 h → normal (<10 mm, all parameters normal) → transient, insignificant

P1 → US 3–6 months, no CAP/VCUG → P2 → CAP if ureter >7 mm; VCUG/MAG3 at discretion → P3 → CAP + VCUG + US ~1 month → persistent → MAG3

Bilateral severe dilation ± oligohydramnios → treat as outlet obstruction: decompress → CAP → urgent evaluation

7 · SPECIFIC ANTENATAL DIAGNOSES

Aetiology of prenatal UTD	Incidence
Transient	50–70%
PUJ pattern	10–30%
VUR	10–40%
Ureterovesical junction (UVJ) pattern	5–15%
Duplex, ureterocele, ectopic ureter	5–7%
MCDK	2–5%
LUTO (PUV, PBS)	0.1–0.5%

Transient dilation

Mild dilation resolves postnatally in 90–98%; SFU 1 by ~2.5 years, SFU 2 ~95% by 4 years. Warn: undiagnosed VUR possible.

PUJ obstruction pattern

- Commonest **surgical** cause. Third-trimester APD **15 mm**: sensitivity and specificity ≈80% for pyeloplasty; postnatal APD predicts better than prenatal. Even severe UTD: only ~a fifth need surgery.
- Plan: early US; consider CAP for P3; VCUG to exclude VUR and confirm a normal posterior urethra in boys; MAG3 at 1 month for persistent P3, or serial US (P2, APD <20 mm, PT >3 mm → DRF >40% likely).
- Surgery if: rising dilation, DRF <40% with poor drainage, falling DRF, delayed upright retention, symptoms (stone, pyelonephritis, haematuria) → *Ch 53*.

Vesicoureteral reflux

Feature	Prenatally detected	Presenting with febrile UTI
Sex	Male	Female
Grade	High, often bilateral	Lower
Renal damage	Congenital dysplasia (up to 40%)	Acquired scarring
Resolution	Often early (infant bladder dynamics settle)	Standard rates

Screen by UTI risk: girls with high-grade UTD and uncircumcised boys yes; circumcised boys with low-grade UTD no.

Megaureter / UVJ obstruction

- Types: obstructed, non-obstructed, refluxing, refluxing-obstructed. Exclude single-system ectopic ureter (dysplasia; inserts beyond bladder neck or into vagina).
- Most resolve. Surgery predicted by ureter **>13 mm**, DRF <30%.
- British Association of Paediatric Urologists: prenatal ureter **>7 mm after 30 weeks** → postnatal US, CAP, VCUG for all; consider MAG3.
- Surgery for DRF <40%, febrile UTI, pain. **Under 1 year → temporise** (cutaneous ureterostomy, pyelostomy, refluxing reimplant); definitive reimplant after 1 year †(tapering a large ureter into a small infant bladder risks obstruction). Follow into adulthood.

> **KEY POINT** Across PUJ, VUR and megaureter, dilation alone is not an indication: operate for loss of function, infection or progression; grade sets surveillance intensity.

LUTO conditions

- **PUV**: commonest LUTO (1.4–2.2/10,000); VCUG shows dilated posterior urethra → *Ch 45*.
- **PBS**: megacystis/upper tract dilation + deficient abdominal musculature + bilateral impalpable testes; may coexist with urethral atresia or megalourethra → *Ch 44*.
- **Urethral stenosis/atresia**: catheter will not pass; membrane at membranous/prostatic urethra → vesicostomy → progressive augmented dilation of anterior urethra (PADUA).
- **Megacystis (massive reflux)**: laterally placed orifices, urine cycles bladder ↔ ureters; correcting reflux restores dynamics. No oligohydramnios.
- **Megacystis–microcolon–intestinal hypoperistalsis syndrome (MMIHS)**: distended non-obstructed bladder; vesicostomy or CIC; CAP only for recurrent UTI.

> **KEY POINT** Bilateral dilation with normal fluid = urine still leaves the fetus (reflux, megacystis); with low fluid = true outlet obstruction.

Neonatal management of LUTO

ALGORITHM
Team at delivery → (pulmonary resuscitation if needed) → 5 Fr feeding tube, finger in rectum to guide over high bladder neck → confirm on kidney–ureter–bladder (KUB) film → fails → US-guided suprapubic catheter (Seldinger)

- **Avoid balloon catheters**: the balloon falls into the posterior urethra, obstructs it and compresses the ureteric orifices.
- No immediate VCUG in a decompressing neonate with renal insufficiency (catheter may be hard to replace). Decompression length guided by **creatinine nadir**.
- Warn of concentrating defect + post-obstructive diuresis. Oliguria after decompression may reflect a thick bladder obstructing tortuous ureters → antimuscarinics.
- US at maximal decompression → VCUG → definitive surgery.

Multicystic dysplastic kidney

- Non-communicating cysts, no large central cyst; can mimic P3 UTD. VUR ≈20%; contralateral PUJ obstruction.
- **No association with Wilms tumour** → routine nephrectomy abandoned. No CAP. VCUG and DMSA optional (not evidence based).
- Serial US for involution and contralateral hypertrophy; monitor blood pressure, proteinuria through puberty (hyperfiltration). Persistent cyst with solid component → remove.

Duplex anomalies, ureterocele, ectopic ureter

- **Meyer–Weigert rule**: upper-pole ureter inserts inferomedially → obstruction/ectopia; lower pole superolaterally → short tunnel → reflux ("drooping lily").
- Suspect duplex when ureter is more dilated than pelvis. Ureterocele = thin-walled intravesical cyst; ectopic ureter shows two walls with a muscle plane. In girls a ureterocele can cause LUTO.
- Work-up: examine introitus (prolapsed ureterocele), US with bladder, CAP from birth, VCUG, scintigraphy for upper-pole function.
- Management: endoscopic puncture (increasingly first-line despite creating reflux; mandatory for outlet obstruction/sepsis); upper-tract approach (nephron-sparing or ablative, increasingly laparoscopic); common-sheath reimplant for high-grade lower-pole VUR. No VUR → upper-tract approach suffices; high-grade VUR → usually ureterocele excision.

Exstrophy

Feature	Bladder exstrophy	Cloacal exstrophy (OEIS)
Non-visualised bladder, low umbilicus, pubic diastasis	Yes	Yes
Omphalocele, imperforate anus	No	Yes
Spine, kidneys, limbs	Usually normal	Lumbosacral, renal, limb anomalies
Postnatal imaging	Renal US	KUB, renal + spinal US ± MRI
Urgency	Not an emergency	Dictated by organ systems

OEIS = omphalocele–exstrophy–imperforate anus–spinal defects. Bladder exstrophy detected prenatally in ~50%; non-visualised bladder alone does not distinguish the two.

Cloaca, urogenital sinus and congenital adrenal hyperplasia (CAH)

- Bladder + vaginal distension = urine trapped in the vagina → outlet obstruction; reflux through uterus and tubes → urinary ascites.
- **Cloaca** (females, ~1/50,000): single channel; pelvic cyst, calcified meconium, bifid uterus/vagina. Polyhydramnios → suspect tracheo-oesophageal fistula. Planned delivery with neonatal, surgical

and urological teams.

- **Urogenital sinus**: no bowel obstruction, normal external genitalia.
- **CAH**: 21-hydroxylase (chromosome 6p), autosomal recessive; classic salt-wasting 75%. Only **1 in 8** at-risk pregnancies yields an affected girl, so prenatal treatment is controversial. Postnatal: electrolytes first; 17-hydroxyprogesterone + genetics; decompress if LUTO.

Myelomeningocele

Diagnosed by alpha-fetoprotein + US. Fetal closure before 26 weeks reduces ventriculoperitoneal shunting and improves motor/cognitive outcome; risks preterm labour, uterine dehiscence; **no bladder benefit**.

8 · NEONATAL UROLOGICAL EMERGENCIES

Urinary ascites and retention

- Urine causes up to 30% of neonatal ascites: respiratory distress, oliguria, hyponatraemia, hyperkalaemia, raised creatinine. **PUV commonest**; also bladder rupture (spontaneous usually heals with catheter; umbilical catheter injury → repair), cloaca, urogenital sinus.
- Retention >24 h: boys → PUV, urethral atresia; girls → urogenital sinus, ureterocele. Catheterise; if fails, angiocatheter (16 G → 3.5 Fr; 14 G → 5 Fr tube). Persistent vaginal/bladder distension → vesicostomy ± vaginostomy.
- Delayed voiding: most void within 24 h; later is usually benign.
- Massive upper-tract obstruction compromising breathing/feeding → percutaneous drainage.

Abdominal masses

- Renal cystic: MCDK, ARPKD, ADPKD (autosomal dominant), cystic nephroma. Solid: **congenital mesoblastic nephroma** commonest <6 months (hypertension, maternal polyhydramnios); Wilms usually 1–5 years.
- Pelvic cystic: distended bladder, hydrometrocolpos, urogenital sinus/cloaca, ovarian cyst (most resolve by 6 months).

Feature	Adrenal haemorrhage	Neuroblastoma
Doppler	No flow	Vascular
Serial US 1–2-weekly	Shrinks	Persists/grows
Calcification	Peripheral eggshell	Stippled

Adrenal haemorrhage: right-sided, bilateral 10–15%; birth trauma, hypoxia, sepsis; may track via patent processus to mimic acute scrotum.

Anorectal malformation and ambiguous genitalia

- Anorectal malformation: genitourinary/spinal anomalies up to **75%** → renal + spinal US (tethered cord); CAP until videourodynamics at 6 weeks.
- Ambiguous genitalia: multidisciplinary disorders of sex development (DSD) team; defer gender assignment. Commonest: CAH (46,XX DSD) — impalpable gonads, uterus on US. **First exclude salt-wasting crisis**; karyotype, electrolytes, 17-hydroxyprogesterone, testosterone, dihydrotestosterone. Rarer 46,XY, ovotesticular or sex-chromosome DSD may need gonadal biopsy *→ Ch 59*.

Interlabial masses

Mass	Position	Management
Imperforate hymen (hydrometrocolpos)	Midline bulging membrane	Hymenal incision
Urethral prolapse	Midline oedematous collar at meatus	Conservative
Prolapsed ureterocele	Off-midline, from meatus; hymen intact behind	US + VCUG
Paraurethral cyst	Off-midline, displaces meatus	Most resolve

Gross haematuria

Confirm with urinalysis (exclude oestrogen withdrawal bleeding, urate crystals, circumcision bleeding) → Doppler US. Vascular causes lead; haematuria + hypertension = severe disease.

Feature	Renal vein thrombosis	Renal artery thrombosis	Cortical necrosis
Risk	Maternal diabetes, prematurity, umbilical venous catheter, polycythaemia; prothrombotic disorder ≤50%	Umbilical artery catheter	Hypoperfusion (sepsis, anoxia, haemorrhage)
Presentation	**Triad**: haematuria + flank mass + thrombocytopenia	Hypertension, heart failure	Renal failure
Treatment	Hydration ± anticoagulation	Remove catheter ± thrombolysis	Supportive

Scrotal masses and torsion

- Differential: extravaginal torsion, hernia, hydrocele (transilluminates), meconium peritonitis (stippled calcification), adrenal haematoma, tumour seeding via processus, testis tumour (yolk sac, teratoma, juvenile granulosa).

Feature	Prenatal torsion	Postnatal (first 30 days)
At birth	Hard, non-tender, discoloured, fixed	Normal, then new swelling/tenderness
Salvage	Nearly nil	Up to 44%
Surgery	Urgent contralateral orchidopexy ± exploration	Immediate exploration + contralateral fixation

The main rationale for exploring is preventing asynchronous contralateral torsion; spinal anaesthesia avoids inhalational agents.

9 · LANDMARK TRIALS

Trial	Design	Result	Take-home
PLUTO	randomised controlled trial (RCT), VAS vs conservative in fetal LUTO; 31 randomised, stopped early	Fewer perinatal deaths with VAS; no renal benefit; 44% complications	Shunting saves lungs, not kidneys
Great Ormond Street pyeloplasty RCT	APD ≥15 mm, DRF >40%: pyeloplasty vs observation	Most observed stable/improved; minority (APD 20–40 mm) lost function	Observation safe with monitoring
MOMS†	RCT, prenatal vs postnatal myelomeningocele closure	Less shunting, better motor; no urodynamic benefit	Fetal closure not for bladder

10 · MCQ ONE-LINERS

1. Second commonest prenatal anomaly: urinary tract dilation.
2. Amniotic fluid mainly fetal urine after ~16 weeks.
3. Oligohydramnios: AFI <5–6 cm.
4. Normal APD: <4 mm before 28 weeks, <7 mm after, <10 mm postnatal.
5. Severe APD: >10 mm second trimester, >15 mm third trimester.
6. Prenatal APD grade does not predict VUR.
7. SFU grade 4: parenchymal thinning.
8. Pyramidal thickness ≤3 mm predicts pyeloplasty.
9. First postnatal US after 48 hours.
10. P3 at any APD if parenchyma abnormal.
11. P1: no prophylaxis, no VCUG.
12. P2 with ureter >7 mm: prophylaxis.
13. Circumcision NNT to prevent one UTI: 10.
14. Nitrofurantoin/trimethoprim avoided before 8 weeks.
15. Commonest surgical cause of prenatal UTD: PUJ obstruction.
16. Third-trimester APD 15 mm predicts pyeloplasty.
17. MAG3 surgical threshold: DRF <40%, T½ >20 min.
18. Megaureter in infancy needing surgery: temporising diversion; reimplant after 1 year.
19. Poor fetal LUTO prognosis: oligohydramnios before 27 weeks.
20. Ruano stage II: intervention indicated.

21. Fetal cystoscopy specific complication: urethrocutaneous/rectourethral fistula.
22. PLUTO: survival benefit, no renal benefit.
23. Avoid balloon catheter in PUV neonate.
24. Commonest cause of neonatal urinary ascites: PUV.
25. MCDK: no increased Wilms tumour risk.
26. Duplex: upper pole obstructs, lower pole refluxes (Meyer–Weigert).
27. Omphalocele distinguishes cloacal from bladder exstrophy.
28. Commonest solid renal tumour <6 months: mesoblastic nephroma.
29. Renal vein thrombosis triad: haematuria, flank mass, thrombocytopenia.
30. Renal artery thrombosis: umbilical artery catheter.
31. Anorectal malformation: genitourinary anomalies up to 75%.
32. Prenatal torsion: urgent contralateral orchidopexy; salvage nearly nil.

35 Urologic Evaluation of the Child

Reading time ≈ 24 min (slow pace)

1 · TRIAGE BY ACUITY

Triage asks one question: **what is lost by waiting** — an organ, a life, or nothing time-critical? Guardians share decisions; age and size constrain instruments and options.

Tier	Conditions
Emergent	Acute scrotum; acute abdomen; acute kidney injury (AKI); obstruction with urinary tract infection (UTI) and/or AKI; retention; incarcerated hernia; paraphimosis; priapism; trauma; newborn ambiguous genitalia, bladder/cloacal exstrophy, renal vein thrombosis
Urgent (24 h)	Gross haematuria **without infection**; newborn bilateral or solitary-kidney hydronephrosis; renal or testicular tumour; spina bifida
Semiurgent (1–3 wk)	After treated febrile UTI; postnatal dilation (no outlet obstruction); reducible hernia; interlabial mass; amenorrhoea
Routine	Hydrocele; circumcision; daytime incontinence/enuresis; hypospadias; labial adhesions; meatal stenosis; microscopic haematuria; recurrent UTI; undescended testis; varicocele

> **KEY POINT** Context shifts the tier: hernia emergent if incarcerated, semiurgent if reducible; hydronephrosis urgent if bilateral/solitary.

2 · HISTORY

- Engage the child; dispel fears so the child participates. Adolescents: offer private history.
- **Prenatal** (key for hydronephrosis): conception (in vitro fertilisation), growth restriction, maternal diabetes, anomalies (cardiac, dysraphism, imperforate anus, cloacal), discordant karyotype vs ultrasound genitalia. Serial scans: gestation at detection, progression, **amniotic fluid** — a proxy for fetal renal function because it is largely fetal urine.
- **Birth**: gestational age, neonatal intensive care stay, circumcision. Medications, family and sibling history (familial conditions).
- **Social**: who holds legal consent (foster care, caseworker); **consent from guardian, assent from child**; all carers need training (catheterisation); bullying/abuse may surface during voiding/sexual history.
- Neurogenic bladder: mobility, hand dexterity, sensory level — determine self-catheterisation. Note behavioural comorbidity (anxiety, autism) in bladder–bowel complaints.

3 · EXAMINATION

Vital signs and growth

- Heart and respiratory rate fall, blood pressure (BP) rises with age. Minimum systolic BP: **≥60** neonate; ≥70 at 1 mo–1 yr; **70 + 2×age** at 1–10 yr; ≥90 above 10 yr.
- **Shock presents as tachycardia and tachypnoea; hypotension means decompensation.** Children compensate longer than adults.
- Weight every visit (drug dosing) and growth charts.

Abdomen

- Distended bladder, stool, lax wall (prune belly); costovertebral angle tenderness → pyelonephritis or colic; infant renal mass/hydronephrosis often palpable.

Male genitalia

- Explain first; parent at head (prepubertal) or documented chaperone; privacy for adolescents.
- Prepubertal: supine frog-leg; adolescent: standing with Valsalva (hernia, varicocele).

- Foreskin retractability, adhesions/skin bridges; testicular position, size, consistency, symmetry; hydrocele, mass.
- **Undescended testis**: relaxed frog-leg, sweep from anterior superior iliac spine along canal, other hand delivers testis to scrotum; soap/lubricant helps; obese child — cross-legged or "catcher's" squat. Palpate canal, mons, perineum, upper thigh (ectopic).
- **Tanner staging**: first male pubertal sign = testicular/scrotal enlargement (II), not hair; hair to medial thighs = V in both sexes.
- **Micropenis**: stretched length (pubis to glans tip) < mean − 2.5 SD — term newborn **1.9 cm**, adult 9.3 cm. Length barely changes 5–11 yr; preterms need gestation-specific values.

Female genitalia

- Frog-leg/butterfly; **caudal traction on labia majora** (outward, minimal lateral) opens the vestibule. Inspect clitoris, meatus, introitus, adhesions, pooling, discharge.

Neurological

- Tone, head circumference (infant); gait, symmetry (≥12 mo).
- **Lumbosacral stigmata**: **simple dimple** (within gluteal cleft, <5 mm, base visible, <25 mm above anus) → observe, no imaging. Image for palpable lipoma, coccygeal tag >5 mm, deep/atypical dimple, gluteal fold deviated by a mass. Sacral dimples are over-referred.

4 · URINE AND BLOOD TESTS

Urine collection and UTI

- Contamination rises as invasiveness falls. In febrile children <24 months, obtain **catheter or suprapubic aspirate** for both urinalysis and culture; **bag specimens have very high false-positive rates**.
- American Academy of Pediatrics (AAP) definition: pyuria and/or bacteriuria + **≥50,000 colony-forming units (CFU)/mL** of a uropathogen (catheter/suprapubic), in a symptomatic child.
- †Toilet-trained: clean-catch midstream is adequate. A bag or clean-catch specimen is useful only if negative.

Thresholds

Test	Threshold
Microscopic haematuria	**>5 red cells (RBC)/high-power field** (adult ≥3), >2 occasions
Calcium/creatinine (hypercalciuria)	>0.4 age 2–5 yr; >0.2 age >5 yr
Protein/creatinine normal	<0.7 first year; <0.4 to 4 yr; <0.2 after 5 yr; >3.0 nephrotic
Specific gravity	<1.010 dilute (water excess, diabetes insipidus); >1.020 dehydration

- Dipstick blood → **always confirm by microscopy** (haemoglobinuria/myoglobinuria give false positives).
- Proteinuria: use **first-morning** sample (orthostatic proteinuria is common). Persistent non-orthostatic on two samples = pathological → nephrology. Many RBCs falsely raise dipstick protein.
- Haemoglobin may be normal early after acute bleeding, then fall abruptly.

5 · IMAGING

Only ultrasound and magnetic resonance avoid radiation; keep doses as low as reasonably achievable (ALARA).

Modality	Answers
Ultrasound (US)	Anatomy: kidney, bladder, scrotum, torsion, spine
Voiding cystourethrogram (VCUG)	Reflux, urethra, genitogram
Computed tomography (CT)	Severe trauma, stones

Modality	Answers
Magnetic resonance imaging (MRI)	Tumours, ectopia, duplication, Müllerian/sinus anatomy; urogram gives function
Mercaptoacetyltriglycine (MAG3) renogram	**Obstruction** + differential function
Dimercaptosuccinic acid (DMSA)	**Scarring**/pyelonephritis + differential function

6 · EMERGENT PRESENTATIONS

Acute scrotum

- Commonest cause: **torsion of appendix testis** (prepubertal; point tenderness; US shows epididymal hyperaemia mimicking epididymitis) → supportive care.
- **Cord torsion** (most after age 10): severe pain, vomiting, high-riding/abnormally oriented testis, absent cremasteric reflex (presence does not exclude). **Clinical diagnosis**: if suspected, skip imaging and explore; manual detorsion does not obviate exploration; fix the contralateral testis.

TWIST item	Points	Score → action
Testicular swelling	2	0 → no US
Hard testis	2	1–5 → scrotal US
Absent cremasteric reflex; nausea/vomiting; high-riding testis	1 each	**≥6** → explore without US

- **Neonatal (extravaginal) torsion**: firm, painless enlarged testis; often prenatal (nubbin); authors prefer urgent exploration.

Other emergencies

- **Ambiguous genitalia**: stabilise, diagnose, assign sex of rearing (multidisciplinary). Exclude **salt-wasting congenital adrenal hyperplasia (CAH)** (21-hydroxylase ≈ 95%): crisis day **4–15** → do not discharge early; check **17-hydroxyprogesterone + electrolytes**.
- **Exstrophy**: tie cord with silk (no plastic clamp), cover plate with cling film, saline irrigation each change, transfer to a specialist centre.
- **Infected obstruction** (commonly stone): drain promptly — stent or nephrostomy; neonatal options pyelostomy/ureterostomy, ureterocele decompression, vesicostomy for valves.
- **Renal vein thrombosis**: newborn haematuria + flank mass + thrombocytopenia; Doppler.
- **Paraphimosis**: always ask circumcision status; exclude **hair tourniquet**. Compress glans with distal traction under analgesia (ice, Coban wrap, sugar, needle punctures) → dorsal slit if fails.
- **Priapism** (≥4 h): ischaemic, usually **sickle cell disease** → oxygen, fluids, analgesia; cold packs only if sickle cell not suspected. Corporal gas confirms.
- **Trauma**: haematuria and vital signs unreliable → **mechanism** drives imaging (CT). Straddle injury: always consider **sexual abuse**; meatal blood in a boy → retrograde urethrogram, no blind catheter.
- **Retention**: expected capacity (mL) = **(age + 2) × 30**. Try analgesia, sitz bath, disimpaction before catheter; no cause → neurological evaluation.

7 · COMMON REFERRAL PRESENTATIONS

Antenatal hydronephrosis follow-up

- Scan **before discharge** if bilateral, solitary kidney or suspected outlet obstruction (third-trimester anteroposterior diameter >7 mm or Urinary Tract Dilation A2–3).
- Scan after **48 h** (earlier underestimates dilation, neonatal oliguria) unless outlet obstruction suspected. Creatinine reflects maternal value for 48 h → follow trend.
- Boy with bilateral dilation, thick bladder, dilated posterior urethra → **VCUG before discharge**; catheter pending imaging. Voiding does not exclude obstruction.

Haematuria

	Glomerular	Urological
Look	Tea-coloured	Bright red, clots, terminal
Urine	Dysmorphic RBC, casts, proteinuria	No protein
Next	C3, C4, antinuclear antibody, antistreptolysin O	US; persistent → cystoscopy + retrogrades

- All gross haematuria: urinalysis, culture, renal–bladder US. Haemorrhagic cystitis after marrow transplant (BK virus) or cyclophosphamide.
- Isolated microhaematuria: usually benign (thin basement membrane, IgA nephropathy, hypercalciuria) → repeat over 2–3 wk; persistent → US + calcium/creatinine.

Bladder bowel dysfunction (BBD) and enuresis

- Spectrum: frequency, urgency, incontinence, enuresis, straining, non-febrile UTI, constipation, encopresis. Validated questionnaires track response.
- **Always examine the back** for occult dysraphism, gait, toe-walking.
- Treatment: behavioural → medical → biofeedback → pelvic floor physiotherapy → psychology; surgery rare.

Scrotal and inguinal

- **Hernia/hydrocele**: patent processus; communicating hydrocele smaller in the morning. Reliable parental history or photograph suffices; both treated by high ligation at internal ring.
- **Undescended testis**: US **not recommended**. Hypospadias or micropenis with it → suspect disorder of sex development (DSD).
- **Varicocele** (≈15% adolescents): measure volume (Lambert: L×W×H×**0.71**). Non-decompressing supine or right-sided → retroperitoneal US for tumour.
- **Testicular tumour**: prepubertal mostly benign (teratoma, epidermoid; malignant = yolk sac) → α-fetoprotein, β-hCG (human chorionic gonadotrophin), lactate dehydrogenase guide testis-sparing surgery.

Genital

- **Phimosis**: physiological — never forcibly retract; pathological/lichen sclerosus (balanitis xerotica obliterans) → treat. Steroid if UTI risk (vesicoureteral reflux, prior UTI).
- **Meatal stenosis**: circumcised boys; dorsally deflected thin stream; meatus <5 Fr → meatotomy (topical EMLA).
- **Labial adhesions** (peak 13–23 mo): observe; symptomatic (post-void dribble, UTI) → oestrogen/steroid cream.
- **Interlabial mass**: first exclude upstream obstruction by US (hydrometrocolpos, ureterocele); consider rhabdomyosarcoma.
- **Primary amenorrhoea** with cyclic pain → outflow obstruction (septum, imperforate hymen).

> **KEY POINT** Red flags: absent cremasteric reflex with acute pain; newborn ambiguous genitalia with impalpable gonads; bilateral hydronephrosis in a boy; fever with obstruction; hypotension (late shock); abdominal mass; meatal blood; lumbosacral lipoma or deviated cleft over a mass.

8 · CHILD COMFORT

- Distraction (tablet ≈ midazolam), midazolam before VCUG, intranasal ketamine and topical EMLA (lidocaine–prilocaine; adhesions, meatotomy) avoid general anaesthesia.

9 · MCQ ONE-LINERS

1. Commonest cause of paediatric acute scrotum: torsion of testicular appendage.
2. TWIST ≥6: explore without ultrasound.
3. Commonest cause of newborn ambiguous genitalia: CAH (21-hydroxylase).
4. Salt-wasting crisis: days 4–15 of life.

5. Expected bladder capacity (mL) = (age + 2) × 30.
6. Paediatric microhaematuria: >5 RBC/hpf on more than two occasions.
7. AAP UTI culture threshold: ≥50,000 CFU/mL by catheter or aspirate.
8. Bag urine specimens: high false-positive rate.
9. Postnatal scan ideally after 48 hours.
0. Commonest renal tumour under 15: Wilms, peak 1–5 years.
1. Commonest prepubertal malignant testis tumour: yolk sac tumour.
2. Commonest cause of ischaemic priapism in children: sickle cell disease.
13. Neonatal priapism: benign, resolves in about 4–5 days.
14. Renal vein thrombosis triad: haematuria, flank mass, thrombocytopenia.
15. Micropenis: stretched length below mean − 2.5 SD (term ≈1.9 cm).
16. First male pubertal sign: testicular enlargement (Tanner II).
17. Minimum systolic BP 1–10 years: 70 + 2 × age.
18. Ultrasound not recommended to locate undescended testis.
19. Simple sacral dimple: no MRI needed.
20. MAG3 answers obstruction; DMSA answers scarring.

36 Pediatric Urogenital Imaging

Reading time ≈ 25 min (slow pace)

1 · PRINCIPLES — ALARA AND THE THREE HARMS

- **ALARA** ("as low as reasonably achievable", International Commission on Radiological Protection) is the consensus standard. Individual radiosensitivity cannot be predicted, so **every child is treated as radiation sensitive** (pregnancy, fetus, childhood especially).
- Ionising radiation harms children more: radiosensitive developing tissue and higher cumulative lifetime dose → stochastic risk of malignancy after long latency.
- **Sedation/anaesthesia risk usually exceeds contrast risk** — higher than nephrogenic systemic fibrosis from gadolinium (poor or immature neonatal renal function) or iodinated-contrast nephropathy.
- Magnetic resonance imaging (MRI): sedation in young children; fetal MRI avoided in first trimester. Ultrasonography (US): **safe at all ages and gestations**.

> **KEY POINT** Every choice trades **radiation, sedation and contrast** against yield; the child's long lifespan tips it towards US, and the question is which test fills US's functional gap (obstruction, reflux, pyelonephritis) most cheaply.

2 · ULTRASONOGRAPHY

Fetal and neonatal kidney

- Fetal urine production begins at **10–12 weeks**; by **20 weeks** US detects most urological anomalies.
- Check **cord insertion and abdominal wall** (exstrophy).
- **Newborn trap**: prominent hypoechoic pyramids with slightly echogenic cortex mimic hydronephrosis. Hydronephrosis is **underestimated in the first 2 days** (physiological oliguria) and in dehydration.

Kidney/ureter — hydronephrosis

Society for Fetal Urology (SFU) grade	Postnatal US criterion
0	No central renal dilatation
1	Renal pelvis only visible
2	Major calyces identified
3	Major and minor calyces
4	Grade 3 + **parenchymal thinning**

- Grade 3 is heterogeneous; grade 3→4 is defined by parenchyma, not further dilatation. Reporting varies → **review the images yourself**.
- US is anatomical; obstruction vs vesicoureteral reflux (VUR) vs normal variant needs a functional test. Obstructive vs non-obstructive megaureter/ureterocele → diuretic renography or MR urography (MRU).
- **Bilateral hydroureteronephrosis + thick bladder + poor emptying → voiding cystourethrogram (VCUG)** to exclude posterior urethral valves (PUV). Otherwise diuretic renography is the gold standard.

Cysts, masses, stones

Lesion	Side	US hallmark
Multicystic dysplastic kidney (MCDK)	Unilateral	Non-communicating random cysts, scant echogenic parenchyma

Lesion	Side	US hallmark
Autosomal recessive polycystic kidney disease (ARPKD)	Bilateral	Large, homogeneously **bright** kidneys (dilated collecting tubules), not discrete cysts
Autosomal dominant polycystic kidney disease	Bilateral	Parenchyma almost replaced by cysts
Multilocular cystic nephroma	Unilateral	Like MCDK but more parenchyma; infancy
Cystic Wilms tumour	Unilateral	Overlaps above; 2–4 years

- Cysts do not communicate; dilated calyces communicate with the pelvis.
- Solid masses by age: neonatal solid mass → **congenital mesoblastic nephroma**; toddler → Wilms; neonate with diffuse enlargement + acute kidney injury (AKI) + hypertension → **renal vein thrombosis**. Contrast computed tomography (CT) usually follows US.
- **Stones**: US first and for follow-up (lifelong imaging → minimise dose). Stone = echogenic **with** shadowing (fat: without); colour Doppler **twinkling artefact** confirms calculus.

Infection and trauma

- Urinary tract infection (UTI): not routine at start of treatment; **failure to improve on antibiotics → US** (abscess, obstruction). Insensitive for low-grade VUR, duplication, acute pyelonephritis (APN) and scars; inferior to technetium-99m dimercaptosuccinic acid (DMSA). **A normal US neither risk-stratifies APN nor predicts VUR.**
- Trauma: US compares favourably with contrast CT for renal injury and follow-up.

Contrast-enhanced voiding urosonography (CeVUS)

- Sulfur hexafluoride microbubbles; five-grade VUR system analogous to VCUG; **no radiation**.
- Meta-analysis: area under curve (AUC) **0.97**, sensitivity ~92%, specificity ~91–94%, first- and second-generation agents equal → recommended for VUR detection.
- Detects **intrarenal reflux** (mostly during voiding, lower pole, rises with grade) — its polar distribution mirrors DMSA scars: the anatomical substrate of pyelonephritic scarring.

Bladder/urethra

- Ureterocele, diverticula, calculi, debris; pre- and post-void volumes. **Keyhole sign** = thick bladder + dilated posterior urethra → PUV.
- Clot vs tumour in haematuria: scan supine then prone — **clot moves**. US cannot give histology.
- Viral cystitis: thick wall mimicking mass; upper respiratory prodrome, gross haematuria, frequency.

Scrotum and genitalia

- US is the first and usually only modality — but most scrotal problems are clinical; **never delay exploration for suspected torsion**.
- **Whirlpool sign** (twisted cord) raises torsion sensitivity to ~96% vs ~76% for colour Doppler; specificity ~99%. No arterial flow + heterogeneous parenchyma → non-viable.
- Epididymo-orchitis = hyperaemia (also after spontaneous detorsion); appendage torsion = avascular nodule with surrounding hyperaemia; idiopathic scrotal oedema = **fountain sign**.
- Prepubertal tumours: benign **teratoma** commonest (solid/cystic/calcified); yolk sac = homogeneous vascular mass; epidermoid = **onion rings**, no flow.
- **Do not image routine undescended testis** (American Urological Association Choosing Wisely) — exception: obese child, to find an inguinal testis. US is primary for internal structures in differences of sex development (DSD).

3 · VOIDING CYSTOURETHROGRAPHY

- No sedation. Scout film may show spinal anomalies. Check catheter tip: non-midline → utricle/ectopic ureter; excess catheter → knotting.

- **Timing after UTI**: once on antibiotics several days and improving; the old 4–6-week wait is unnecessary (significant VUR rarely disappears). Very early VCUG may **overgrade** VUR (endotoxin-induced ureteric dilatation).
- Indications: hydronephrosis with **dilated ureter(s)** (not all hydronephrosis); febrile/recurrent UTI; suspected PUV; ectopic ureter/utricle in boys with epididymitis; any UTI in a circumcised boy may merit evaluation.
- Shows trabeculation, ureterocele (may evert), diverticula, PUV, stricture, utricle, polyps, emptying.
- **Occult VUR**: VCUG finds VUR in only ~40% of DMSA-proven APN (cycle variation, infection-induced junction dysfunction). **Positional instillation of contrast (PIC)** cystography — cystoscopic contrast stream from 1 m aimed at the orifice — for negative VCUG + recurrent febrile UTI.
- **Trauma**: blood at meatus → **retrograde urethrogram before any catheter** → then fill bladder by gravity to capacity. Expected capacity (mL) = **(age + 2) × 30**, max 400. "Pie-in-the-sky" bladder = pelvic fracture with extraperitoneal injury.
- **Genitography** in DSD/cloaca/urogenital sinus: catheter in the single opening defines urethrovaginal confluence (Shopfner classification).

4 · NUCLEAR MEDICINE

DMSA cortical scintigraphy

- Taken up by proximal tubules (flow-dependent); APN → photon-deficient areas.

Feature	Acute pyelonephritis	Scar
Contour	Preserved (reniform)	Volume loss, concavity
Timing	Scan within days; resolves over ~5 months	Lesion at **≥6 months** = scar

- Acute lesions persist as scars in roughly a third to a half of kidneys.

Mercaptoacetyltriglycine (MAG3) diuretic renography

- Gold standard for obstructive vs non-obstructive dilatation; tracer MAG3 (commoner) or diethylenetriamine pentaacetic acid (DTPA). † DTPA is purely glomerular-filtered with low extraction, so images are noisier in the immature neonatal kidney; MAG3 (tubular secretion, high extraction) is preferred.

"Well-tempered" element	How	Pitfall if wrong
Hydration	Intravenous (IV) line + oral fluids; avoid fasting	Poor diuresis → false obstruction
Region of interest (ROI)	Tight ROI on cortex (differential function) or dilated system (drainage); careful background	Spurious values; ROI choice localises level (ureteropelvic vs ureterovesical junction)
Bladder catheter	Urethral catheter	Full bladder/VUR impair drainage; gonadal dose
Furosemide	**1 mg/kg IV** at maximal filling	Too early → drainage measured before system full

- Outputs: differential renal function, washout curve, **T½ timed from maximal filling**, not from injection (otherwise includes cortical transit).
- **No validated paediatric T½ cut-off** → decide on age, function, T½, curve shape, upright drainage, with serial studies. Predictors of progression: **biphasic curve** (drains then stops) and poor gravity-assisted (upright) drainage.
- Intermittent obstruction (Dietl's crisis): cortical retention up to 48 h after an attack; a scan between attacks may be normal.

Radionuclide cystography and testicular scan

- **Direct** radionuclide cystography: catheter needed; more sensitive and lower dose than VCUG but poor anatomy and coarse (mild/moderate/severe) grading — good for follow-up. **Indirect** (after renography): no catheter, insensitive → not routine.
- Testicular scan (pertechnetate): cold = torsion, hot = inflammation; only if equivocal and no surgical delay.

5 · MR UROGRAPHY

- Anatomy + function without radiation: differential function and drainage, ideal for complex anomalies (ectopic ureter, duplication, calyceal diverticulum — often precontrast T2 suffices).
- Protocol: IV hydration + catheter → **furosemide 1 mg/kg (max 20 mg)** → T2 fast spin-echo (3D fat-saturated) → slow gadolinium → dynamic T1 until ureters fill; **Patlak** analysis for function. Hydration + diuretic give better distension and fewer artefacts.
- Limits: up to 1 h; sedation/anaesthesia often needed <6 years; complex post-processing; equivalence to renography unproven.
- Poor for stones, infection, trauma, acute scrotum. Best for **bladder rhabdomyosarcoma** (low T1, heterogeneous high T2/enhancement; no subtype prediction) — acquire non-enhanced sequences before gadolinium. Preferred for **local staging of renal tumours** and synchronous contralateral lesions.

6 · CT AND CONVENTIONAL RADIOGRAPHY

- CT is chosen for speed and availability, not superiority — trauma, polytrauma, tumour staging; little role in hydronephrosis.
- **Trauma: arterial-phase images miss collecting-system/ureteric injury → delayed excretory images mandatory.** CT cystography classifies intraperitoneal vs extraperitoneal rupture (also augment rupture), which drives management.
- Stones: best quantification, rarely changes care (ALARA). Infection: abscess or fistula when US/MR inconclusive.
- Kidney–ureter–bladder (KUB) radiograph: radiopaque stones, constipation in voiding dysfunction. Intravenous pyelography (IVP) obsolete. **Retrograde pyelography**: anaesthesia; free retrograde flow does not prove antegrade patency (e.g. postoperative flaps).

7 · CHOICE OF TEST BY CLINICAL QUESTION

Question	First line	Next / problem-solver	Avoid
Hydronephrosis — obstruction?	US	MAG3 renogram; MRU	CT, IVP
Hydroureteronephrosis ± thick bladder	US	VCUG (PUV, VUR)	—
VUR detection/follow-up	VCUG	CeVUS; direct nuclear cystogram; PIC if occult	Indirect cystogram
APN / scar	DMSA (early = APN; ≥6 mo = scar)	—	Grey-scale US; MRI
Stones	US	Low-dose CT; KUB	Reflex CT
Renal mass	US	Contrast CT; MRI	—
Bladder mass	US (full bladder)	MRI	—
Abdominal/pelvic trauma	CT with delayed phase; US	Retrograde urethrogram; cystography	Catheter before urethrogram if meatal blood

Question	First line	Next / problem-solver	Avoid
Acute scrotum	Clinical ± Doppler/whirlpool	Scintigraphy if equivocal	Any imaging delaying exploration
Undescended testis	Examination	US only if obese	Routine US, MRI
DSD	US + genitography	MRI (very complex)	CT

8 · MCQ ONE-LINERS

1. SFU grade 4 = grade 3 plus parenchymal thinning.
2. Large bright kidneys bilaterally without discrete cysts → ARPKD.
3. Neonatal solid renal mass → congenital mesoblastic nephroma.
4. Neonate with renal enlargement, AKI and hypertension → renal vein thrombosis.
5. Twinkling artefact on colour Doppler → calculus.
6. Keyhole sign → posterior urethral valves.
7. Whirlpool sign → testicular torsion; raises sensitivity to ~96%.
8. Fountain sign → acute idiopathic scrotal oedema.
9. Onion-ring avascular testicular lesion → epidermoid cyst.
0. CeVUS agent → sulfur hexafluoride microbubbles; AUC 0.97.
11. Expected bladder capacity = (age + 2) × 30 mL.
12. Blood at meatus → retrograde urethrogram before catheterisation.
13. "Pie-in-the-sky" bladder → pelvic fracture, extraperitoneal injury.
14. VCUG during acute febrile UTI may overgrade VUR.
15. DMSA lesion persisting at 6 months = scar.
16. Diuretic renogram furosemide dose → 1 mg/kg IV at maximal filling.
17. T½ is timed from maximal filling, not injection.
18. MRU furosemide → 1 mg/kg, maximum 20 mg.
19. Arterial-phase CT alone misses collecting-system injury.
20. Genitogram classification in DSD → Shopfner.

37 Infection and Inflammation of the Paediatric Genitourinary Tract

Reading time ≈ 39 min (slow pace)

1 · OVERVIEW AND THE PARADIGM SHIFT

- Urinary tract infection (UTI) affects ~**1% of boys, 3–5% of girls** in childhood; 30–50% recur. Commonest bacterial infection in febrile children (~7% of febrile infants).
- Old view: recurrent febrile UTI with vesicoureteral reflux (VUR) causes scarring, hypertension and chronic kidney disease (CKD) → aggressive imaging and treatment. **Current view**: congenital renal dysplasia plays the larger role → **selective imaging, prompt treatment, prophylaxis only for the high-risk child**.

2 · PATHOGENESIS AND VIRULENCE

- **Ascending route**: gut uropathogen → perineum/periurethra → bladder → adherence to umbrella cells → invasion → ascent → renal damage → bacteraemia. Complicated UTI (e.g. catheter): fibrinogen deposits on catheter bind uropathogens → biofilm.
- Uropathogenic *Escherichia coli* (UPEC) causes most UTIs. Virulence factors solve four problems: **adhere, feed, hide, damage** (plus persist and ascend).

Factor	Mechanism	Association
Type 1 fimbriae (mannose-sensitive)	FimH tip binds uroplakins → bacterial entry	Cystitis, asymptomatic bacteriuria
P fimbriae (mannose-resistant)	Bind Galα1–4Galβ digalactoside (P blood group)	**Pyelonephritis**
Aerobactin	Siderophore strips iron from host proteins	Growth in iron-poor urine
α-Haemolysin, cytotoxic necrotising factor 1	Lyse cells; cytoskeletal disruption	Inflammation, invasion
Capsule (K antigen)	Antiphagocytic, anticomplementary	Pyelonephritis
Biofilm, flagella	Barrier to immunity/antibiotics; motility	Recurrence; ascent

- Host response: attachment → Toll-like receptor 4 (TLR4) signalling → cytokines → neutrophil killing. No step 1 → asymptomatic carriage; failed step 2 → destructive inflammation.
- **Intracellular bacterial communities (IBC)**: UPEC invade umbrella cells, form biofilm-like pods, flux out, then persist as a **quiescent intracellular reservoir** for months → reseed the lumen during urothelial turnover → recurrence despite apparently adequate antibiotics.

> **KEY POINT** Why does the same organism recur? IBCs and biofilms shield bacteria from neutrophils and antibiotics; a biofilm survives 1000–1500× normal killing concentrations because of poor penetration, slow growth and altered targets.

3 · HOST RISK FACTORS

Age	Boys	Girls
First year	**3.7%** (higher)	2%
Cumulative by 6 years	2%	7%

- **Sex/age**: boys exceed girls only in infancy; febrile girl 2 months–2 years ≈ 2.3× risk of a febrile boy. UTI commoner in White than Black children.
- **Genetics/blood group**: P1 phenotype (P-fimbrial receptor) in girls with recurrent pyelonephritis; Lewis Le(a−b−); non-secretor status → scarring.

- **Circumcision**: reduces UTI in the first 6 months **~10-fold** (preputial colonisation peaks in infancy). Classic US Army cohort: circumcised 0.11%, uncircumcised 1.12%, girls 0.57% MCQ. Number needed to circumcise: **111** normal boys, 11 recurrent UTI, **4** high-grade VUR. Protective in antenatal hydronephrosis.
- **Faecal–perineal colonisation**: same strain cycles in faecal reservoir; antibiotics (especially trimethoprim–sulfamethoxazole, TMP-SMX) destroy protective flora.
- **VUR**: 1–2% of newborns, **25–40%** after first UTI; most pyelonephritis occurs without VUR, but grade III–IV doubles pyelonephritic change → *Ch 41*.
- **Bladder and bowel dysfunction (BBD)** = dysfunctional elimination syndrome: infrequent voiding, incomplete emptying, constipation, overactivity. Treating it reduces recurrent UTI and improves VUR resolution.
- **Neurogenic bladder**: ~2.5 UTIs/year. On clean intermittent catheterisation (CIC) 40–80% have bacteriuria — **asymptomatic bacteriuria (ASB) without VUR needs no antibiotics**; daily prophylaxis only breeds resistance; catheter type irrelevant; timely, frequent catheterisation prevents UTI.
- **Catheter-associated UTI**: commonest nosocomial infection; bacteriuria 3–10% per catheter-day → catheters for necessity, remove early.
- **Immunity**: serum immunoglobulin G nadir at 1–3 months, urothelial immunoglobulin A nearly absent → highest risk early infancy. Obesity also raises risk.

> **KEY POINT** Every host factor either raises inoculum and dwell time (prepuce, faecal flora, stasis, constipation, catheter) or lowers defence (immature immunity, receptor-bearing blood groups); prevention = reverse whichever applies.

4 · CLASSIFICATION

- Axes: **complicated vs uncomplicated** (anatomical/functional abnormality or foreign body; all neonatal/infant UTI presumed complicated); upper vs lower; first vs recurrent.
- **Febrile UTI (presumed pyelonephritis)**: high fever, vomiting, flank pain, lethargy. **Cystitis**: afebrile, dysuria, frequency, urgency, suprapubic pain. Clinical localisation is poor: only **50–66%** of febrile children show acute changes on dimercaptosuccinic acid (DMSA) scan (gold standard); fever does not exclude cystitis.

Recurrence type	Urine between episodes	Cause	Remedy
Unresolved bacteriuria	Never sterile	Non-compliance, malabsorption, resistant organism	Culture-directed retreatment
Bacterial persistence	Sterile; same organism returns	Nidus: stone, anomaly, necrotic papilla, catheter, stent	**Remove nidus**
Reinfection	Sterile; usually new organism	New ascent	Prevention

Asymptomatic bacteriuria

- Kass: two consecutive cultures **≥10^5 colony-forming units (CFU)/mL**, same organism, no symptoms. Low-virulence strains.
- Infant (<1% term, 3% preterm): treat and obtain renal–bladder ultrasound (RBUS) — may signal anomaly. Preschool/school-age girls: **do not treat or screen** if no VUR/anomaly — treatment gives no lasting cure, selects resistance and may replace a harmless strain.

5 · DIAGNOSIS

Clinical

- Fever ≥38 °C (rectal). Toxic appearance, age <30 days and temperature ≥39.4 °C predict bacteraemia. Neonates/infants: nonspecific — irritability, poor feeding, jaundice, failure to thrive,

vomiting.

- Useful under 2 years: fever >40 °C, previous UTI, suprapubic tenderness, **uncircumcised penis**. Older verbal child: dysuria, frequency, new incontinence, abdominal/back pain.
- **Another source of fever (e.g. otitis) only halves UTI risk** — classic history: recurrent "otitis", then DMSA scars.
- Examine: bladder/flank mass, costovertebral tenderness, genitalia (phimosis, meatal stenosis, labial adhesions, ectopic ureter), spine (sacral dimple, gluteal cleft → neurogenic bladder), constipation. Adolescents: consider *Neisseria gonorrhoeae*, *Chlamydia*.

Urine collection — contamination rises as invasiveness falls

Method	Contamination	Role
Bag	50–60% (false-positive culture up to 75%)	**Reliable only when negative**; urinalysis screen only, never culture
Clean catch (midstream)	~26%	Toilet-trained children; preferred by NICE when possible
Catheter	10–15%	Sensitivity 95%, specificity 99%; **standard in non-toilet-trained <2 years**
Suprapubic aspiration (SPA)	<1%	Reference standard; ultrasound guidance raises success to 80–90%

- Collect **before antibiotics** unless delay endangers a sick child (National Institute for Health and Care Excellence, NICE). American Academy of Pediatrics (AAP), 2–24 months: bag sample for urinalysis → if positive, culture from catheter/SPA.
- Catheter technique in girls: two-person, labia drawn gently outward (excess lateral pull hurts); lidocaine gel; discard first drops. SPA: 22-gauge needle 1–2 cm above pubis.
- Examine within 1 h (room temperature) or 4 h refrigerated.

Dipstick vs microscopy vs culture

Test	Sensitivity	Specificity	Use
Leukocyte esterase (LE)	~79%	~87%	Negative helps **rule out**
Nitrite	~50%	**~98%**	Positive **rules in**
LE or nitrite or microscopy positive	99.8%	70%	Best rule-out
Pyuria ≥5 white blood cells/high-power field (WBC/HPF)	~74%	~86%	No advantage over LE
Bacteria on fresh uncentrifuged urine (Gram)	~91%	92–96%	Most reliable rapid test

- **Nitrite false negatives**: needs hours of bladder dwell to reduce dietary nitrate (infants void frequently), dilute urine, gram-positive organisms (Enterococcus) → low sensitivity. First-morning urine best.
- LE: inflammation of any cause gives false positives; pyuria less frequent with Klebsiella/Enterococcus → culture a suspected UTI even without pyuria. True UTI without pyuria is rare — helps separate ASB.
- Pyuria threshold **≥10 WBC/HPF** more reliable under 2 years.
- **Culture is essential** (≥18 h; final 2–3 days); unnecessary if urinalysis entirely normal.

Guideline	Specimen	Significant count (CFU/mL)
Kass (historical)	Voided	$\geq 10^5$
AAP 2011 (2–24 months)	Catheter/SPA	**≥50,000 + pyuria**
European Association of Urology	Midstream/catheter	10^3–10^4 monoculture
Royal Children's Hospital (RCH) Melbourne	SPA / catheter	Any gram-negative / $>10^3$
Italian Society of Pediatric Nephrology (ISPN)	Catheter / midstream	$>10^4$ / $>10^5$

- Rule: **the cleaner the specimen, the lower the accepted count**. Contamination suggested by low counts, mixed growth, lactobacilli, coagulase-negative staphylococci, diphtheroids.
- Serum markers: fever, white count, erythrocyte sedimentation rate do not localise; C-reactive protein sensitive, not specific; **procalcitonin ≥0.5 ng/mL** best predicts acute pyelonephritis (APN) and later scarring, moderate accuracy.

6 · IMAGING AFTER UTI

- No evidence routine imaging prevents scarring, hypertension or renal failure. Rule: **order a test only if its result will change management**.
- Ultrasound: radiation-free first study; abnormal in ~15% after first febrile UTI but only 1–2% need action; **urgent if unwell or not responding at 48 h** (obstruction, pyonephrosis, abscess). Pitfalls: oedema enlarges kidney, lobar nephronia mimics tumour, endotoxin dilates pelvis. Poor for VUR and scars.
- Voiding cystourethrogram (VCUG): gold standard for VUR grading, bladder and urethra; do when urine sterile and child well. Cyclic filling raises sensitivity; record volume at reflux onset (predicts resolution). Contrast-enhanced voiding urosonography: radiation-free alternative, especially girls and follow-up.
- DMSA (± single-photon emission computed tomography): proximal tubular uptake, ~1 millisievert (mSv). APN sensitivity maximal within **1 week**; scar assessment at **≥6 months**; ~15% of acute lesions scar. Scar is indistinguishable from congenital dysplasia.
- Computed tomography (CT): 10–15 mSv — only for stones not seen on ultrasound or mass vs inflammation. Magnetic resonance imaging: sedation, cost.

Guideline	Ultrasound	VCUG	Late DMSA
AAP (2–24 months)	Yes, after first febrile UTI	If ultrasound abnormal (or recurrence)	No
NICE <6 months	Yes	Atypical/recurrent	Atypical/recurrent
NICE >6 months	Atypical/recurrent	No	Atypical/recurrent
Top-down approach	No	If acute DMSA abnormal	If acute DMSA abnormal
RCH Melbourne	If <3 months or seriously ill	No	No

- **NICE "atypical" UTI**: seriously ill, poor urine flow, abdominal/bladder mass, raised creatinine, septicaemia, no response within **48 h**, non-*E. coli* organism. NICE late DMSA at 4–6 months.
- Trade-off: NICE/AAP miss about half of grade III–V VUR, but spare most children VCUG; high-grade VUR usually declares itself at recurrence, and prompt treatment then limits damage. AAP Section on Urology still accepts VCUG as an option (Randomized Intervention for Vesicoureteral Reflux, RIVUR: prophylaxis halves recurrences).

7 · TREATMENT

- Goals: eradicate infection, prevent scarring, relieve symptoms. Urine sterile within 24 h of effective therapy. **Delay matters**: odds of new scarring double if antibiotics start after 72 h vs within 48 h.
- Start empirically after urine is collected; adjust to culture.

Oral vs intravenous; admit or not

- **Oral = intravenous (IV)** for defervescence and scar prevention in randomised trials (urinary levels far exceed serum) for non-toxic children >2 months who tolerate oral drugs with reliable families.
- **Admit**: age <1 month (some <2–6 months), toxic, dehydrated, vomiting, doubtful compliance. Neonates: full septic work-up (bacteraemia ~20%, hyponatraemia, hyperkalaemia) → IV → oral after ≥24 h afebrile; ≤2 days IV then oral is safe in young infants.

- 90% afebrile by 48 h; **not improving at 48 h → renal and bladder ultrasound (RBUS) and broaden cover**.

Infection	Duration
Febrile UTI	**7–14 days** (shorter courses inferior)
Afebrile cystitis	**2–4 days** (better than single-dose; equals 7–14 days)
Acute focal bacterial nephritis (lobar nephronia)	≥3 weeks
Renal abscess	Antibiotics; drain if no response
Infected obstruction / pyonephrosis	Antibiotics + tube or stent drainage

Choice of antibiotic

- Base empirical choice on **local antibiogram** (revised yearly). TMP-SMX and amoxicillin are poor empirical agents (high *E. coli* resistance). Narrow choices: **first-generation cephalosporin (cephalexin)**, nitrofurantoin for cystitis. Susceptibility uses serum breakpoints; urinary levels may be >100× higher, so do not escalate needlessly to third-generation agents.
- Organisms: *E. coli* 83% in outpatient girls but only ~37% in inpatient boys; Enterococcus, Klebsiella, Pseudomonas rise in inpatients; **Proteus is a boys' organism**.
- **Neonates**: cover Enterococcus (cephalosporin-resistant) → **ampicillin + third-generation cephalosporin or aminoglycoside**.
- Broad cover if previous UTI, recent antibiotics/hospitalisation, anomaly; aminoglycoside (once daily) if Pseudomonas risk. Fluoroquinolones: reserve for resistant organisms; Food and Drug Administration restricts to complicated UTI/pyelonephritis.

Agent	Dose (mg/kg/day)	Restriction
Cephalexin	50–100 in 4	—
Cefixime	8 once daily	—
Amoxicillin–clavulanate	20–40 in 3	—
TMP-SMX	6–12 (TMP) in 2	**<6 weeks/premature**: kernicterus (bilirubin displacement)
Nitrofurantoin	3–5 in 2	**<3 months**, glomerular filtration rate (GFR) <50%, glucose-6-phosphate dehydrogenase (G6PD) deficiency (haemolysis); **never for febrile UTI** (low tissue levels)
Ceftriaxone (IV)	75 once daily	—
Gentamicin (IV)	7.5 (once daily acceptable)	Nephro-/ototoxicity †

Repeat culture after treatment is unnecessary if the organism was sensitive.

8 · RENAL SCARRING AND SEQUELAE

- Scars favour the **poles** (compound papillae permit intrarenal reflux); risk rises with VUR grade and with APN → focal nephritis → abscess.

Pyelonephritic episode	1st	2nd	3rd	4th	5th
Scar risk	5%	10%	20%	40%	60%

- Small kidney with diffusely reduced uptake and low function = usually **dysplasia**, not acquired scar.
- Steroids: no consistent benefit — not routine.
- Long term: childhood UTI accounts for **<1% of CKD**; end-stage risk after childhood UTI ~1 in 10,000. Bilateral scars lower GFR over decades; scarring raises risk of hypertension and pre-eclampsia.

- **Follow-up**: height, weight, blood pressure after febrile UTI; bilateral scars or reduced function → lifelong blood pressure, renal function, proteinuria; renin–angiotensin blockade for proteinuria.

9 · PREVENTION AND PROPHYLAXIS

- Recurrence **10–30%**, highest in first 3–6 months; higher after index UTI >1 year and in girls (up to 40%).
- **Continuous antibiotic prophylaxis (CAP)**: not after first UTI (AAP, NICE). Benefit in **high risk**: girls with dilating VUR (grade ≥III), BBD, recurrent febrile UTI. Low-risk: circumcised boy, no BBD, normal imaging, no dilating VUR.
- Dose ~¼ **therapeutic**, at bedtime; agent should differ from the treatment agent (prior exposure triples resistance).
- Agents: TMP-SMX/trimethoprim (not <6 weeks), nitrofurantoin (little effect on faecal flora; not for APN), first-generation cephalosporin (young infants).
- **BBD**: assess in every child; VUR + BBD → recurrence ~45% vs 15% without. Ladder: timed voiding + constipation treatment → anticholinergics (overactivity), biofeedback (dysfunctional voiding), CIC → *Ch 47*.

Non-antibiotic measure	Mechanism	Verdict
Circumcision	Removes preputial colonisation	Medical indication in high-risk boys: recurrent febrile UTI, high-grade VUR, hydroureteronephrosis, obstructive uropathy
Topical steroid for phimosis	Allows retraction, hygiene	Fewer recurrences — reasonable
Cranberry	Blocks adherence	Mixed evidence; not routine
Probiotics	Restore flora	Adjunct only
D-mannose	Competitively blocks type 1 fimbriae	Unproven in children

10 · SPECIAL INFECTIONS

Entity	Defining feature / imaging	Treatment
Acute focal bacterial nephritis (lobar nephronia)	Localised suppuration; CT wedge-shaped hypoenhancement, mimics mass	Antibiotics ≥3 weeks
Pyonephrosis	Pus in obstructed dilated system; high-attenuation fluid	**Drain** + antibiotics
Renal abscess	Urine culture negative in up to 20% → radiological diagnosis; ultrasound detects ≥1 cm; CT most accurate	Antibiotics; drain if no response
Xanthogranulomatous pyelonephritis (XGP)	Chronic, obstructive, Proteus/*E. coli*; mass with perinephric extension mimics tumour; calculi 38–70%	Diffuse → **nephrectomy**; focal → partial nephrectomy/medical

- XGP presents vaguely (fever, flank mass, weight loss, failure to thrive, anaemia); nephrectomy is hard — inflammation encases psoas and great vessels.
- **Viral haemorrhagic cystitis**: adenovirus 11 (and 21); in bone marrow transplant recipients severe disease carries 50–80% mortality. **BK polyomavirus** reactivates in immunosuppression → intravenous/intravesical **cidofovir**.
- **Fungal (Candida)**: risk — antibiotics, prematurity, central/umbilical lines, parenteral nutrition, immunocompromise; kidney is the organ most involved in disseminated candidiasis. *C. glabrata* often fluconazole-resistant. Persistent funguria → ultrasound for **fungus balls**: alkalinise + antifungal → percutaneous drainage with local amphotericin B → endoscopic/open removal.

Catheter funguria: change catheter, stop antibiotics; treat if repeated counts >10,000–15,000 CFU/mL.

- † **Epididymitis** in a prepubertal boy: usually sterile/post-infectious; if urine culture positive or recurrent, image for ectopic ureter (into vas/seminal vesicle), dilated utricle or urethral obstruction with ultrasound and VCUG. Exclude torsion first.
- † **Balanitis/balanoposthitis**: preputial inflammation in uncircumcised boys; warm soaks and hygiene, topical antibiotic/antifungal; recurrent episodes or pathological phimosis → circumcision.

11 · LANDMARK TRIALS

Trial	Design	Result	Take-home
RIVUR	607 children, VUR I–IV, TMP-SMX vs placebo, 2 years	Recurrent febrile/symptomatic UTI ~halved (number needed to treat 8); **no difference in scarring**	Prophylaxis prevents recurrence, not scars; greatest benefit with BBD and febrile index UTI
Swedish Reflux Trial	Randomised, dilating VUR, 1–2 years	Prophylaxis reduced recurrence in girls	CAP for girls with dilating VUR

12 · MCQ ONE-LINERS

1. Boys outnumber girls with UTI only in the first year.
2. Circumcision reduces infant UTI about tenfold.
3. Number needed to circumcise in high-grade VUR → 4.
4. P fimbriae bind Galα1–4Galβ → pyelonephritis.
5. Type 1 fimbriae are mannose-sensitive; FimH binds uroplakins.
6. Aerobactin is a siderophore.
7. Quiescent intracellular reservoir explains same-strain recurrence.
8. Bag urine is reliable only when negative.
9. Suprapubic aspiration contamination → <1%.
0. Most specific dipstick test → nitrite (~98%).
1. Gram-positive organisms give false-negative nitrite.
2. AAP catheter threshold → ≥50,000 CFU/mL plus pyuria.
3. Kass ASB → two cultures ≥10^5 CFU/mL, same organism.
4. ASB in schoolgirls without VUR → do not treat.
5. Gold standard for acute pyelonephritis → DMSA scan.
6. DMSA scar assessment → ≥6 months after infection.
17. Procalcitonin ≥0.5 ng/mL predicts APN and scarring.
18. NICE atypical UTI includes non-*E. coli* organism.
19. Top-down approach → DMSA first, VCUG if abnormal.
20. Febrile UTI course → 7–14 days; cystitis → 2–4 days.
21. Oral antibiotics equal IV in non-toxic children >2 months.
22. Neonatal empirical regimen → ampicillin plus gentamicin or cefotaxime.
23. Nitrofurantoin contraindicated in febrile UTI, <3 months, G6PD deficiency.
24. TMP-SMX avoided <6 weeks → kernicterus.
25. Prophylaxis dose → one-quarter therapeutic, at bedtime.
26. RIVUR → fewer recurrences, no scar reduction.
27. Scar risk doubles if antibiotics start after 72 h.
28. Childhood UTI causes <1% of CKD.
29. Renal abscess urine culture negative in up to 20%.
30. XGP organisms → Proteus or *E. coli*; diffuse → nephrectomy.
31. Adenovirus 11 → haemorrhagic cystitis.
32. BK virus cystitis → cidofovir.

38 Core Principles of Perioperative Management in Children

Reading time ≈ 25 min (slow pace)

1 · PERIOPERATIVE SURGICAL HOME AND ENHANCED RECOVERY

- **Perioperative surgical home** (American Society of Anesthesiologists, ASA): continuity-of-care model with standardised pathways for pre-, intra-, post-operative and post-discharge phases, led jointly by surgeon and anaesthesiologist.
- **Enhanced recovery after surgery (ERAS)**: counselling, shorter starvation, judicious intravenous (IV) fluid, no routine tubes or drains, opioid-sparing analgesia, early feeding and mobilisation. Paediatric urology studies: shorter stay, earlier feeding, no rise in complications. Each element minimises disturbance of the child's baseline; recovery follows from what is not disturbed.

2 · PREOPERATIVE ASSESSMENT

- Risk raised by upper respiratory tract infection (URTI), asthma exacerbation, dehydration, uncontrolled diabetes, prematurity, fever. Common events: nausea and vomiting, laryngospasm, bronchospasm, emergence delirium; rare: anaphylaxis, malignant hyperthermia.
- **Preoperative anxiety** (up to half of children) predicts more pain, emergence delirium and new behaviour problems (night crying, enuresis) → reducing it is a clinical intervention: child life specialists, age-appropriate explanation, games, virtual-reality theatre tours.
- **Jehovah's Witness**: duty is to the child — a child is not allowed to die for lack of blood (court order). Correct anaemia and coagulopathy beforehand.
- **Investigations are indication-driven**: no routine bloods, chest radiograph or electrocardiogram (ECG) in healthy children; a murmur alone is not an ECG indication. Urine human chorionic gonadotrophin (hCG) in females of menstrual age — MCQ false positive after augmentation cystoplasty.
- **Urine culture** before any procedure opening the urinary tract; treat positives by sensitivity and reschedule until sterile. Always culture in neurogenic bladder; treat asymptomatic bacteriuria before endourology (mucosal injury → sepsis).
- **URTI**: airway hyperreactive for **6–8 weeks** → laryngospasm, bronchospasm, desaturation; laryngeal mask safer than endotracheal tube. Postpone elective surgery ≥ 4, ideally 6 weeks.
- **Sepsis**: antibiotics within **1 h** of septic shock, fluids ± vasopressors; avoid neuraxial blocks (hypotension, coagulopathy). **Trauma**: always a full stomach → rapid-sequence induction.

3 · FASTING (NIL PER OS, NPO)

Minimum fast	Material
2 h	Clear liquids, carbohydrate drinks (European and Canadian 2022: 1 h)
4 h	Breast milk
6 h	Infant formula, non-human milk, light meal, tube feeds
8 h	Regular meal

- Order follows gastric emptying. Trap: non-human milk counts as a light meal. Children dehydrate faster than adults → schedule youngest first; inpatients receive IV maintenance while fasting; medicines allowed with clear fluid to 2 h.
- **Glucose while fasting**: ketogenic diet — withhold carbohydrate (drugs, suspensions); glycogen storage and urea cycle disorders — glucose-containing fluid throughout; mitochondrial disease — avoid lactated Ringer and propofol.
- **Type 1 diabetes**: first case, glucose before induction; metformin stopped 24 h before (lactic acidosis); insulin adjusted, not stopped (half long-acting dose the night before); long case or

emergency → IV insulin infusion.
- Omit angiotensin-converting enzyme (ACE) inhibitors on the morning; stop warfarin/aspirin 7–10 days before; never stop clonidine abruptly (rebound hypertension).

4 · ANAESTHESIA CONSIDERATIONS

Neonate and preterm infant

- Every organ has less reserve: surfactant deficiency, bronchopulmonary dysplasia, small blood volume (small loss → shock), retinopathy of prematurity (accept saturation in low 90s), hypothermia, poor urinary concentration.
- **Postoperative apnoea** peaks in the first 12–23 h and may be fatal → admit for 23-h cardiorespiratory monitoring if postconceptional age is **< 50–60 weeks** (ex-preterm) or **< 44 weeks** (term). Hence elective surgery is deferred beyond these windows.

Anaesthetic neurotoxicity debate

- US Food and Drug Administration (FDA) warning (2016): repeated or lengthy (**> 3 h**) anaesthesia or sedation at **< 3 years** or in third-trimester pregnancy may affect brain development — based on animal neuroapoptosis.
- Randomised and sibling-matched human studies of single short exposures are negative (§11); only registries lacking exposure duration show small deficits.
- Counselling: inform of the warning, reassure that a single exposure < 1 h has not been shown harmful; do not delay necessary surgery — inadequate anaesthesia and analgesia themselves cause harm.

Comorbidities

- **Asthma**: continue inhalers, pre-induction β_2-agonist; stress steroids if on systemic or daily inhaled steroids. **Congenital heart disease**: highest risk with single ventricle, pulmonary hypertension, left ventricular outflow obstruction, cardiomyopathy.
- **Chronic kidney disease (CKD)**: anaemia, uraemic platelet dysfunction, hyperkalaemia; avoid surgery within 4–6 h of haemodialysis; avoid non-steroidal anti-inflammatory drugs (NSAIDs) in CKD 3–4.
- **Sickle cell disease**: remove every sickling trigger — dehydration (most important: first case, IV fluid during fast), hypoxia, acidosis, cold, stasis (no tourniquets). Acute chest syndrome is the main cause of death.
- **Phaeochromocytoma**: α-blockade *before* β-blockade (unopposed α → crisis) plus volume loading; post-resection hypoglycaemia (rebound hyperinsulinaemia).
- **Congenital adrenal hyperplasia**: no stress cortisol response → IV stress-dose hydrocortisone with glucose-containing fluid.

Spina bifida

- Confirm ventriculoperitoneal (VP) shunt function — suspected malfunction → postpone. Scoliosis → restrictive lungs; contractures → careful padding.
- **Latex allergy**: anaphylaxis (bronchospasm, vasodilatation, capillary leak) — fatal without immediate treatment. † Sensitisation follows repeated mucosal latex exposure (catheters, operations), so all spina bifida patients are managed latex-free from birth.
- **Autonomic dysreflexia** (lesion ≥ **T6**): bladder distension or noxious stimulus → severe hypertension, bradycardia, stroke. † Stop stimulus, drain bladder, sit up, rapid antihypertensive.

5 · FLUID AND ELECTROLYTE MANAGEMENT

- Fluid = deficit (fasting, bowel preparation) + maintenance + losses; both restrictive and liberal extremes harm.
- **Holliday–Segar 4-2-1 rule**: **4 mL/kg/h** for first 10 kg + **2 mL/kg/h** for next 10 kg + **1 mL/kg/h** per kg above 20. E.g. 22 kg → 40 + 20 + 2 = 62 mL/h.

- Large-volume normal saline → hyperchloraemic acidosis, so some prefer lactated Ringer. Add dextrose under 2–4 months (immature liver); no potassium in dialysed patients.

Maintenance fluid	Historical	Current
Tonicity	Hypotonic (0.45% saline)	Isotonic (0.9% saline or lactated Ringer) with dextrose
Potassium	10 mEq/L if < 10 kg; 20 mEq/L if > 10 kg	

> **KEY POINT** **Why isotonic?** Surgical stress, pain, nausea and hypoxia release antidiuretic hormone (ADH) independent of osmolality → aquaporin-2 water retention. Hypotonic fluid then supplies free water → hyponatraemia (< 135 mmol/L). † The child's large brain-to-skull ratio leaves little room for cerebral oedema, so hyponatraemic encephalopathy can be fatal.

6 · BLOOD LOSS AND TRANSFUSION †

- Estimated blood volume (EBV): preterm 90–100, term neonate 80–90, infant 75–80, child 70–75 mL/kg — a modest absolute loss is a large fraction in an infant.
- Maximum allowable blood loss = EBV × (starting − target haematocrit) / starting haematocrit; crossmatch when expected loss approaches it.
- Transfuse red cells at haemoglobin **< 7 g/dL** in stable children (higher in neonates, cyanotic heart disease, active bleeding); 10–15 mL/kg raises haemoglobin ≈ 2–3 g/dL.

7 · TEMPERATURE

- Infants lose heat fast: high surface-area ratio, thin skin, little brown fat. Hypothermia → vasoconstriction, acidosis, coagulopathy, apnoea.
- Warm theatre, †forced-air warming and warmed fluids; monitor core temperature. Laparoscopy adds gas-related cooling → *Ch 39*.

8 · ANTIBIOTIC PROPHYLAXIS AND INTRAOPERATIVE CRISES

- American Urological Association: give prophylaxis when host defence, likelihood of bacterial invasion and morbidity of surgical site infection justify antibiotic risk.
- **Timing**: within **60 min** of incision; vancomycin and fluoroquinolones within **120 min** (long infusion).
- Doses: cefazolin **30 mg/kg** (redose 4 h); ampicillin 50 mg/kg (2 h); gentamicin 2.5 mg/kg, vancomycin 15 mg/kg, metronidazole 15 mg/kg, ciprofloxacin 10 mg/kg (not redosed — long half-life).

	Anaphylaxis	Malignant hyperthermia (MH)
Triggers	Neuromuscular blockers, β-lactams, latex	Succinylcholine, volatile agents (not nitrous oxide)
First sign	Collapse, bronchospasm, hives minutes after induction	Unexplained hypercarbia, tachycardia, masseter spasm; late — acidosis, hyperkalaemia, rigidity
Treatment	**Adrenaline**, 100% oxygen, fluids; tryptase	Stop triggers, **dantrolene**, clean circuit; admit 24–48 h (recrudescence)

9 · ANALGESIA

- Self-report is the gold standard but impossible < 3 years → observational scales: FLACC (Face, Legs, Activity, Cry, Consolability) for non-verbal children, confounded by emergence delirium.
- **Regional blocks under general anaesthesia** are safe, standard and opioid-sparing: dorsal penile or penile ring block, **caudal**, spinal, epidural (from 2 kg in experienced hands, for large open incisions). Ultrasound-guided pudendal block for hypospadias: less pain and opioid than caudal. † Caudal 0.25% bupivacaine 0.5 mL/kg (sacral) to 1 mL/kg (lower thoracic); maximum bupivacaine 2.5 mg/kg.

- **Multimodal, scheduled (not as-needed)**: acetaminophen 15 mg/kg 6-hourly + ibuprofen 10 mg/kg 8-hourly; ketorolac 0.5 mg/kg IV 8-hourly for ≤ 72–96 h (renal toxicity) — NSAIDs not < 6 months. Adjuncts: IV lidocaine (count in local anaesthetic total), ketamine, dexmedetomidine.
- Opioids for breakthrough; patient-controlled analgesia from ≈ 4 years. **Codeine contraindicated**: poor metabolisers get no analgesia, ultrarapid metabolisers make lethal morphine. Methadone needs ECG (QT).
- Discharge opioids: 3–5 days' supply; teach storage and disposal — leftovers cause childhood poisoning and adolescent misuse.

10 · POSTOPERATIVE CARE, VTE AND DAY-CASE DISCHARGE

- Laryngospasm → positive-pressure ventilation → propofol → reintubation. Bronchospasm → nebulised salbutamol → adrenaline, steroids, magnesium.
- **Emergence delirium** (2–8 years, within 30 min): child does not recognise parents and is inconsolable, whereas a child in pain recognises parents and localises. Treat with low-dose propofol, dexmedetomidine or fentanyl.
- **Venous thromboembolism (VTE)** †: rare before puberty; adolescents with immobility, obesity, oestrogen pills, central venous catheter, malignancy, prolonged pelvic surgery or prior VTE → early mobilisation and mechanical prophylaxis; add low-molecular-weight heparin if several risk factors.
- **Day-case**: †ASA I–II, outside apnoea-risk age bands, short low-blood-loss procedure. Discharge at baseline function, stable haemodynamics, pain controlled — Aldrete score or Post-Anesthetic Discharge Scoring System; †tolerating fluids, responsible adult at home.

11 · LANDMARK TRIALS

Trial	Design	Result	Take-home
GAS	Randomised; infant hernia repair, awake regional vs sevoflurane < 1 h	No cognitive difference at 2 and 5 years	Single short anaesthetic is safe
PANDA	Sibling-matched; single exposure < 3 years	No difference in intelligence quotient at 8–15 years	Repeated exposure untested
MASK	Cohort; exposure < 3 years	Negligible deficit after single exposure	Reassuring

12 · MCQ ONE-LINERS

1. Fasting: clear 2 h, breast milk 4 h, formula 6 h, regular meal 8 h.
2. Non-human milk is fasted as a light meal.
3. FDA warning: anaesthesia > 3 h or repeated, age < 3 years.
4. GAS trial: no neurodevelopmental difference at 5 years.
5. Apnoea monitoring: ex-preterm < 50–60 weeks, term < 44 weeks postconceptional age.
6. 4-2-1 rule: 22 kg → 62 mL/h.
7. Maintenance fluid: isotonic with dextrose, not 0.45% saline.
8. Postoperative hyponatraemia: non-osmotic ADH release.
9. Avoid lactated Ringer and propofol in mitochondrial disorders.
0. Post-URTI airway hyperreactivity lasts 6–8 weeks.
1. Prophylaxis within 60 min; vancomycin within 120 min.
12. Cefazolin 30 mg/kg, redosed 4-hourly.
13. False-positive urine hCG after augmentation cystoplasty.
14. Earliest MH sign: unexplained hypercarbia; treat with dantrolene.
15. Nitrous oxide does not trigger MH.
16. Autonomic dysreflexia: lesions at or above T6.
17. Spina bifida: latex-free care.
18. Phaeochromocytoma: α-blockade before β-blockade.
19. Sickle cell: avoiding dehydration matters most; acute chest syndrome kills.
20. Codeine contraindicated — ultrarapid metabolisers.
21. Ketorolac limited to 72–96 h; NSAIDs not under 6 months.
22. Methadone requires ECG for QT.
23. Emergence delirium: child does not recognise parents.
24. Metformin stopped 24 h preoperatively.
25. Avoid surgery within 4–6 h after haemodialysis.

39 Principles of Laparoscopic and Robotic Surgery in Children

Reading time ≈ 19 min (slow pace)

1 · EVOLUTION AND INDICATIONS

- Sequence: diagnostic laparoscopy for non-palpable testis (**1976**, first paediatric urological laparoscopy) → nephrectomy (1994) → pyeloplasty, first reconstructive minimally invasive surgery (MIS) (1995) → first paediatric robotic case (2002).
- Each stage was limited by intracorporeal suturing and ergonomics; the wristed robot removed that bottleneck, so paediatric robotics has grown in **reconstruction**.
- **Indications**: pyeloplasty (commonest), ureteric reimplantation, (hemi)nephrectomy, ureteroureterostomy, intra-abdominal testis, urachal and diverticulum excision; deep pelvis — utricle, urogenital sinus, bladder neck reconstruction; Mitrofanoff and augmentation cystoplasty in few centres.
- **Selection is physiological**: the only strict contraindication is **inability to tolerate pneumoperitoneum**. Age and size are not contraindications; significant cardiopulmonary disease needs anaesthetic and specialist optimisation.

2 · PHYSIOLOGY OF PNEUMOPERITONEUM

Raised intra-abdominal pressure (IAP) causes changes absent in open surgery. Almost all are pressure-dependent and reversed by desufflation, so the strategy is the **lowest age-appropriate pressure**: keep IAP < 12 mm Hg; ≈ 7 mm Hg is feasible.

Age	0–1 month	2–12 months	1–2 years	2–8 years
Maximum IAP	**≤ 6**	**≤ 8**	**≤ 10**	**≤ 12 mm Hg**

- **Cardiovascular**: IAP > 10 mm Hg ↓ venous return and cardiac index; catecholamines and renin–angiotensin–aldosterone activation ↑ systemic vascular resistance. Peritoneal stretch at insufflation may trigger a **vagal reflex** (bradycardia, instability) → warn anaesthetist, desufflate, reinsufflate slowly.
- **Respiratory**: high chest wall compliance and low functional residual capacity in infants; diaphragm pushed up → ↓ tidal volume and compliance, ↑ peak airway pressure; Trendelenburg worsens it.
- **Carbon dioxide (CO_2)**: gas of choice — non-combustible, soluble, exhaled. Absorption needs higher minute ventilation; infants are most affected; with lung disease → hypercarbia, acidosis, arrhythmia.
- **Intracranial**: hypercarbia ↑ cerebral blood flow. Ventriculoperitoneal (VP) shunts: feared obstruction and raised intracranial pressure, but MIS is generally safe.
- **Renal**: parenchymal and renal vein compression → transient oliguria, most marked < 1 year. It is reversible and not a guide to resuscitation → **do not treat initial oliguria with fluid boluses**.
- **Hypothermia** †: cold, dry insufflated gas and high flows cool infants (large surface-area ratio) → warmed humidified gas, low flows, minimise leaks, forced-air warming.
- **CO_2 embolism** †: gas entering a vein → right ventricular outflow block → sudden falling end-tidal CO_2, desaturation, collapse, "mill-wheel" murmur. Stop insufflation, desufflate, left lateral decubitus (right side up) with head down, 100% oxygen, aspirate via central line.

3 · ACCESS

	Open (Hasson)	Closed (Veress needle)
Principle	Fascia and peritoneum opened under vision; port placed, then insufflate	Blind entry with spring-loaded blunt needle; insufflate, then port
Checks	Visual	Aspirate (vessel, bowel) → saline drop test → low opening pressure
Evidence	No difference in entry failure or vascular/visceral injury (Cochrane); both safe in children	

- Veress site: umbilicus; after scarring or failure, **Palmer's point** — left midclavicular line ≈ 3 fingerbreadths below costal margin.
- Saline that does not drop means malposition or inadequate relaxation. Preperitoneal tip → high opening pressure → subcutaneous emphysema.
- What protects the child is confirmation of intraperitoneal position, not the method.

4 · PORTS, INSTRUMENTS AND PLACEMENT

- Ports 3, 5, 8, 10, 12 mm; **3- and 5-mm** laparoscopic ports suit infants; most robotic platforms need **8 mm**. Valveless trocars keep constant pressure and use less CO_2 but can cause subcutaneous emphysema.
- Laparoscopic instruments have 4 degrees of freedom (DOF); robotic wristed instruments 7 DOF, making suturing easier.
- Energy: monopolar (any exposed metal conducts → insulation failures cause remote burns); bipolar and ultrasonic sealers seal vessels up to 7 mm with less collateral spread.
- **Placement**: triangulation is essential in pure laparoscopy; robotic cases tolerate linear placement (more space, fewer collisions). **Hidden incision endoscopic surgery (HIdES)** moves ports below the Pfannenstiel line for scars hidden by clothing.

Positioning

- **Renal**: modified flank at bed edge, elevated 30–45° so bowel falls away; lower leg bent, upper straight, padding between; test table rotation before draping. Transmesenteric approach needs less tilt.
- **Pelvic**: supine (frog-leg for genital access), steep Trendelenburg, secure strapping, face protected from instruments.

5 · APPROACHES

	Transperitoneal	Retroperitoneal	Vesicoscopic
Space	Largest; familiar anatomy	Limited	Limited; small bladder predicts complications
Strength	Complex reconstruction, adjacent organs seen	Avoids peritoneum; preserves peritoneal dialysis	Intravesical access
Limitation	Colon mobilised or mesenteric window	Unfamiliar view; peritoneal tear collapses space	Learning curve; port-site closure
Uses	Testis, kidney, ureter, bladder	Pyeloplasty, (hemi)nephrectomy, lymph node dissection	Cross-trigonal reimplantation

- **Left-sided pyeloplasty**: transmesenteric window avoids colonic mobilisation — advantageous when feasible.
- **Retroperitoneoscopy**: † open entry below the 12th rib tip, space created by finger or balloon dissection, child in flank or prone position. Transperitoneal and retroperitoneal nephrectomy outcomes are equivalent; choose retroperitoneal for peritoneal dialysis patients (no dialysis interruption).

- **Vesicoscopy**: camera port through bladder dome under cystoscopic view → two working ports → urethra occluded → bladder insufflated with CO_2. Robotic version limited by 8-mm ports.

6 · ROBOTICS IN CHILDREN

- **Working space** is the infant problem: ≈ **1 L at 1 year vs 6 L in adults** → arm collisions, reduced range. Aids: **port spacing** as wide as the small abdomen allows, linear placement, "burping" (lifting ports against the abdominal wall); minimise inhalational anaesthetic and use preoperative enema and rectal tube to reduce bowel distension.
- Infant robotic pyeloplasty is safe; versus open: shorter stay, less narcotic.
- **Limitations**: no 5-mm robotic ports (8-mm minimum); da Vinci single-port needs a 25-mm cannula and ≈ **10 cm** from cannula to target — prohibitive in small children; no haptic feedback; cost; harder to teach; often sited in adult theatres away from paediatric equipment.
- **Benefits**: shorter stay (hours for pyeloplasty, days for Mitrofanoff/augmentation); better cosmesis — incisions < 15 mm contract, larger ones enlarge; pain benefit is least proven.

7 · COMPLICATIONS AND PREVENTION

Complication	Key fact	Prevention / management
Major vascular injury	**Right common iliac** — directly behind umbilicus (Veress)	Aspirate, low opening pressure; open access in thin children; early recognition, conversion
Minor vascular injury	**Inferior epigastric** at port placement	Port insertion under vision
Bowel, bladder injury	Rare; may present late	Decompress bladder and stomach †; inspect on exit
Subcutaneous emphysema	Preperitoneal insufflation	Check opening pressure
Port-site hernia	0.5–3%; highest around 1 year	**Close fascia** at every port, especially young children
Physiological	Bradycardia, hypercarbia, oliguria, hypothermia	Age-appropriate IAP, communicate at insufflation

- Complications cluster at entry and exit. Surgeon experience is the strongest predictor; surgeons doing > 12 cases a year have fewer. Pyeloplasty complication rates do not differ between open, laparoscopic and robotic.
- Same-day discharge is feasible after uncomplicated robotic renal or ureteric surgery without drains; stent and catheter drainage remain standard for pyeloplasty.

> **KEY POINT** **Ergonomics**: laparoscopy strains hands, thumb and shoulders (fulcrum effect, wrist deviation) → table at 70–80% elbow height, monitor at eye level. The robot offloads the upper limb but fixes the neck; master the clutch to keep elbows on the armrest.

8 · MCQ ONE-LINERS

1. First paediatric urological laparoscopy: non-palpable testis, 1976.
2. First paediatric reconstructive MIS: laparoscopic pyeloplasty, 1995.
3. Maximum IAP: neonate 6, infant 8, 1–2 years 10, 2–8 years 12 mm Hg.
4. Only strict contraindication: inability to tolerate pneumoperitoneum.
5. Oliguria during pneumoperitoneum: do not give fluid boluses.
6. Insufflation may provoke vagal bradycardia — desufflate.
7. Commonest major vessel injured at Veress entry: right common iliac.
8. Commonest minor vascular injury: inferior epigastric.
9. Palmer's point: left midclavicular line, 3 fingerbreadths below costal margin.
10. No outcome difference between Hasson and Veress access.

1. Smallest multiport robotic port: 8 mm.
2. Single-port robot needs ≈ 10 cm working distance.
3. Infant working space ≈ 1 L vs 6 L adult.
4. Laparoscopic instruments 4 DOF; robotic 7 DOF.
5. Retroperitoneal approach preferred in peritoneal dialysis.
6. Vesicoscopic approach: cross-trigonal reimplantation.
17. Port-site hernia risk highest near 1 year — close fascia.
18. HIdES places ports below Pfannenstiel line.
19. da Vinci lacks haptic feedback.
20. CO_2 embolism: falling end-tidal CO_2, mill-wheel murmur.

40 Clinical and Urodynamic Evaluation of Lower Urinary Tract Dysfunction in Children

Reading time ≈ 27 min (slow pace)

1 · PRINCIPLES AND TERMINOLOGY

- Most lower urinary tract (LUT) disease is diagnosed by **history, examination and non-invasive tests**; invasive urodynamics is selective.
- **Diagnostic sequence**: exclude an anatomical or neurological cause → then define the type of bladder (± bowel) dysfunction, primarily **storage vs emptying**.
- Terminology: International Children's Continence Society (ICCS) standardisation (2014 update).
- **Expected bladder capacity (EBC)**: infant ≤1 y = **weight (kg) × 7 mL**; older child = **(age in years + 2) × 30 mL**. Small capacity <65% EBC; large >150% (some use >130%). Functional capacity (largest diary void, or voided volume + post-void residual [PVR]) is normal at ≈70–115% EBC.

2 · HISTORY

- **Storage symptoms**: urgency, frequency, incontinence (continuous vs intermittent, day vs night, how often). **Secondary enuresis** = new wetting after **≥6 months** dry → look for stressors. Caffeine is a bladder irritant.
- **Voiding symptoms**: boys — stream deviation/narrowing, straining, dribble. Girls — **toilet posture**: legs together, perching, hovering or dangling feet activate the pelvic floor → raised outlet resistance → high voiding pressure, incomplete emptying, urinary tract infection (UTI). Fix: footstool, or sit "backwards" facing the cistern.
- **Holding manoeuvres** (potty dance, leg-crossing, squatting, perineal grabbing) suppress detrusor overactivity (DO): perineal pressure inhibits the detrusor through the sacral reflex arc (gate control) — the same mechanism as neuromodulation.
- **Bowel**: shared S2–S4 innervation; bowel dysfunction may be the *sole* cause of urinary symptoms. **Bristol Stool Scale** 1 (hard pellets) – 7 (watery); target **Bristol 3–4 daily**, painless.
- **Rome IV functional constipation** (developmental age ≥4 y, ≥1 month): **≥2 of** — ≤2 stools/week; ≥1 soiling/week; retentive posturing; painful/hard stools; large rectal mass; stools that block the toilet.
- **UTIs**: bagged specimens are reliable only if negative; febrile culture-proven UTI suggests stasis.
- **Neurology red flags**: perineal numbness, gait or limb sensory change with new LUT symptoms (LUTS).

ALGORITHM
Bowel and bladder dysfunction (BBD): Rome IV → 7-day diary + urinalysis + pelvic ultrasound (US) → **treat bowel first** → LUTS persist → uroflow + electromyography (EMG) → urotherapy → anticholinergic → β3-agonist → botulinum toxin / biofeedback / neuromodulation.

Bladder diary and questionnaires

Tool	Records	Use
7-day bowel–bladder diary	Void and stool timing, Bristol type, wetting day/night	Elimination habits; objective constipation; removes recall bias
48-h frequency–volume chart	Intake and voided volumes (measuring "hat"); 2 × 24-h, need not be consecutive	Functional/maximum capacity vs EBC, polyuria, drink effects. **Mandatory** when first-line therapy fails

- Symptom scores: **Dysfunctional Voiding Symptom Score (DVSS)** best tracks treatment; **Dysfunctional Voiding and Incontinence Scoring System (DVISS)** best for diagnosis.

Paediatric Incontinence Quality of Life score (PinQ) measures burden, not response.

- Psychological comorbidity: enuresis < daytime < **faecal incontinence (highest)**; screen and refer.

3 · PHYSICAL EXAMINATION

- **Vitals every visit**: height, weight, blood pressure (BP). Confirmed raised BP → renal US, creatinine/cystatin C.
- **Neurological**: reflexes, power, sensation, gait, perineal sensation, anal tone. Hand function decides feasibility of self-catheterisation. Functional neurological level and ambulation **do not correlate**.
- **Sacral reflexes**: anal wink (perianal stroke) and bulbocavernosus (glans/clitoral squeeze) → external anal sphincter contracts. Absent = interrupted S2–S4 arc.
- **Spine**: a simple sacral dimple (<2.5 cm from anus) is low risk → spinal US up to 6 months. High risk (hair tuft, lipoma/mass, vascular malformation, multiple stigmata, vertebral–anal–cardiac–tracheo-oesophageal–renal–limb [VACTERL] features) → **magnetic resonance imaging (MRI)**. US works only until the arches ossify (≈10–12 months).
- **Genitalia**: damp underwear. Boys — meatal stenosis matters only with stream deviation. Girls — labial fusion (resolves at puberty; topical oestrogen); in **continuous incontinence look for an ectopic ureteric orifice**.

4 · LABORATORY TESTS

- **Urinalysis for every child with LUTS**: glucose, protein, leucocytes, nitrite, blood, specific gravity; negative reassures.
- **Culture is not routine** — false positives and over-treatment in colonised bladders. On clean intermittent catheterisation (CIC) culture only with fever, abdominal/suprapubic pain, malaise, new/worse incontinence or spasm.
- Blood tests only if upper-tract or renal disease is suspected.

5 · NON-INVASIVE URODYNAMICS

Uroflowmetry

- Key variables: **voided volume, maximum flow rate (Qmax), pattern** (also average flow, Qavg). Qmax counts only if the peak lasts **≥2 s** (spikes are artefact).
- **Valid only if**: toilet-trained child; voided volume **≥50% EBC** (pattern needs ≈50–100 mL); >1 curve; read as emptying data only.

Pattern	Shape	Suggests
Bell	Smooth, symmetrical	Normal
Tower	High, short, peaked	Overactive bladder (explosive detrusor)
Staccato	Irregular, never reaches zero	Intermittent sphincter/pelvic-floor overactivity (dysfunctional voiding)
Interrupted	Discrete spurts, flow falls to zero	Underactive detrusor with abdominal straining, or detrusor–sphincter dyssynergia (DSD); EMG separates
Plateau	Flat, low, prolonged	Anatomical obstruction (valves, stricture) or underactive detrusor

> **KEY POINT** The curve reflects detrusor vs outlet. Too much detrusor → tower; too much outlet → plateau; outlet activity that comes and goes → staccato; detrusor effort that comes and goes → interrupted. Patch EMG tells which side is at fault.

Patch EMG

- Perineal/perianal surface electrodes; pelvic floor should be **silent during voiding** (synergic).

- **Lag time** (pelvic-floor relaxation → onset of flow): **<2 s** = DO; **2–6 s** normal; **>6 s** = primary bladder-neck dysfunction [MCQ].

Pelvic ultrasound and PVR

- Bladder wall thickness abnormal **>3 mm full / >5 mm empty** (increased detrusor workload; DO, pelvic-floor overactivity).
- Rectal diameter **>30 mm** without urge = faecal loading.
- **PVR** (repeat — very variable): abnormal at 4–6 y if **>30 mL or >21%** of capacity; 7–12 y **>20 mL or >15%**; repeated >20 mL is abnormal. Valid only if measured **<5 min** after void and bladder 50–115% EBC — a late or over-distended PVR is invalid, not abnormal.

6 · INVASIVE (FORMAL) URODYNAMICS

Indications

- **Neurogenic**: spinal dysraphism, tethered cord, sacral agenesis, sacrococcygeal teratoma, transverse myelitis, spinal cord injury (SCI), possible neurogenic bladder with new LUTS.
- **Severe obstruction**: posterior urethral valves (PUV), urethral atresia, large ectopic ureterocele.
- Non-neurogenic LUTS: only after failed 2nd/3rd-line therapy. Rule: study only if the result will change management.

Technique — paediatric modifications

- Calm setting, parent present; anxiety peaks at catheter insertion; if very difficult, place catheters under anaesthesia and study once awake.
- **Avoid sedation/general anaesthesia** (alters detrusor); if unavoidable, ketamine preferred to midazolam.
- Routine pre-study culture/prophylaxis not needed — post-study UTI ≈ **1.4%**; target high-risk children (CIC, recent UTI, immunosuppression, symptoms).
- Empty bladder → lidocaine gel → smallest **6 Fr dual-lumen** urethral catheter + rectal catheter; EMG patches at 3 and 9 o'clock perianally (needles add pain, not accuracy). Vesicostomy: Foley on traction.
- **Fill rate 5–10% of EBC per minute**; infusate room-to-body temperature, never cold.
- **Stop filling** at strong urge, discomfort, voiding, leak > fill rate, detrusor pressure (Pdet) persistently **>40 cmH_2O**, or volume **>150% EBC**.
- First fill gives the highest Pdet; repeat cycles optional — reporting the first fill is the kidney-protective choice.
- **Natural-fill cystometry**: smaller capacity, higher voiding pressure than conventional.
- **Videourodynamics**: contrast filling adds bladder shape/trabeculation, diverticula, vesicoureteric reflux (VUR) and the volume/pressure at which it starts, bladder-neck configuration.

Interpretation — storage phase

- Vesical (Pves) and abdominal (Pabd) pressures zeroed to atmosphere at the symphysis; **Pdet = Pves − Pabd**. Normal filling: flat Pdet.
- **DO**: phasic involuntary Pdet rises (neurogenic or idiopathic); exclude cough/strain artefact. **Underactivity**: large, low-pressure, acontractile bladder; voids by straining.
- **Compliance = ΔV/ΔPdet** (mL/cmH_2O), calculate in every study: **>20 normal; 10–20 moderately; <10 severely impaired**. In small contracted bladders absolute Pdet (<30–40) is the better safety guide. An **early** pressure rise means the child spends most of the day at high pressure.

	Detrusor leak point pressure (DLPP)	Abdominal leak point pressure (ALPP)
Leak with	Rising Pdet, no strain or contraction	Cough/Valsalva, no detrusor contraction
Reflects	Outlet resistance the bladder must overcome — **upper-tract risk**	Sphincter competence

	Detrusor leak point pressure (DLPP)	Abdominal leak point pressure (ALPP)
Threshold	**>40 cmH₂O** (or no leak at high pressure) = danger	No paediatric norms (adult <60 intrinsic sphincter deficiency)

KEY POINT **McGuire (1981)**, myelodysplasia: leak pressure >40 cmH₂O → hydronephrosis or secondary VUR in **85%**. Above 40 cmH₂O, bladder pressure is transmitted up the ureters to the kidneys. Aim for the lowest achievable pressure.

Voiding phase (pressure–flow)

- Valid if ≥50–100 mL and ≥50% functional capacity voided. Record Qmax, pattern, Pdet at Qmax, PVR. **High Pdet + low Qmax = obstruction**.
- Anatomical obstruction: plateau, high Pdet, **quiet EMG**. Functional (DSD/dysfunctional voiding): staccato/interrupted, **active EMG**.
- Normal Pdet at Qmax ≈ boys 66, girls 57 cmH₂O; infant boys up to 118 cmH₂O (physiological) MCQ .

KEY POINT **Findings threatening the upper tracts**: compliance <20 (esp. <10); DLPP >40 cmH₂O; sustained high filling Pdet; DSD with poor emptying; high voiding pressure with poor flow. Interobserver agreement is poor.

Surveillance frequency

- Urologic Management to Preserve Initial Renal Function (UMPIRE) protocol (spina bifida): urodynamics at ≈**3 months** → 1, 2, 3 years → yearly to 5 y if VUR or hostile bladder.
- Hydronephrosis is a late, insensitive marker; any clinical change → re-study; repeat before reconstruction.

7 · EVALUATION IN SPECIFIC CONDITIONS

Condition	Typical findings	When to study
Non-neurogenic LUTD / enuresis	DO, dysfunctional voiding	Only if refractory
Spinal dysraphism	UMPIRE risk by DLPP/end-fill Pdet: **<25 low; 25–39 or DO intermediate; ≥40 high**	Baseline, serial, pre-surgery
Sacral agenesis	Areflexia + weak sphincter, or DO + DSD; level does not predict; ⅓ diagnosed late (flat buttocks, short cleft)	If voiding dysfunction
Anorectal malformation	Tethered cord common	Baseline ≈1 y after repair; on change
PUV	DO → poor compliance → myogenic failure; polyuria worsens it ("valve bladder" ≈15%)	Serial
Cerebral palsy	Upper motor neuron; DO, large or small capacity, DSD rare	Only if conservative therapy fails
SCI	Spinal shock → retention; higher lesions → more DSD	Video-urodynamics 3–6 months after injury
Transverse myelitis	DO > poor compliance > DSD > DLPP >40; motor recovery does not predict bladder	Early, with CIC

KEY POINT UMPIRE dropped DSD from its high-risk group because reviewers could not agree who had it — poor agreement, not proof that DSD is benign. Goals in every neurogenic bladder: normal capacity, low storage pressure, complete emptying, upper-tract protection.

8 · LANDMARK TRIALS

Trial	Design	Result	Take-home
McGuire 1981	Myelodysplasia urodynamic cohort	DLPP >40 cmH_2O → upper-tract damage in 85%	40 cmH_2O = safety threshold

9 · MCQ ONE-LINERS

1. EBC in children >1 year: (age + 2) × 30 mL.
2. EBC in infants: weight (kg) × 7 mL.
3. ICCS small capacity <65% EBC; large >150%.
4. Secondary enuresis: wetting after ≥6 months of dryness.
5. Rome IV functional constipation needs ≥2 criteria for ≥1 month, age ≥4.
6. Bell-shaped flow = normal.
7. Tower flow = overactive bladder.
8. Staccato flow never reaches zero; interrupted flow does.
9. Qmax is valid only if sustained ≥2 seconds.
0. Flow interpretable only if voided volume ≥50% EBC.
1. EMG lag time >6 s suggests primary bladder-neck dysfunction.
2. Abnormal PVR at 7–12 y: >20 mL or >15% capacity.
13. PVR must be measured within 5 minutes of voiding.
14. Filling rate in paediatric cystometry: 5–10% EBC/minute.
15. Post-urodynamic UTI rate ≈1.4%; routine prophylaxis unnecessary.
16. Compliance <10 mL/cmH_2O = severely impaired.
17. DLPP >40 cmH_2O → 85% upper-tract deterioration (McGuire).
18. Anatomical obstruction: high Pdet, low flow, quiet EMG.
19. Normal Pdet at Qmax: boys ≈66, girls ≈57 cmH_2O.
20. UMPIRE high risk: DLPP/end-fill pressure ≥40 cmH_2O.
21. Continuous incontinence in a girl: search for ectopic ureter.
22. SCI: baseline video-urodynamics 3–6 months after injury.

42 Bladder Anomalies in Children

Reading time ≈ 16 min (slow pace)

1 · CLASSIFICATION AND EMBRYOLOGY

- **Structural** (exstrophy, duplication, agenesis, urachal, diverticula) or **functional** (usually secondary); prune-belly syndrome (PBS) is both.
- Weeks 4–6: urorectal septum divides cloaca → ventral urogenital sinus (UGS) + dorsal rectum. Cranial UGS → bladder + pelvic urethra; caudal UGS → phallic urethra (male) / distal vagina (female).
- Urine production from week 10.
- Months 4–5: bladder descends; allantoic duct cannot keep pace → elongates, narrows, obliterates → **urachus → median umbilical ligament**. Every urachal anomaly = failure of this one obliteration at a different level.

2 · FETAL BLADDER ON ULTRASOUND (US)

- Wall ≤3 mm; seen in nearly all by 13 weeks.
- Empties every **15–20 min** → if not seen, **repeat scan in the same sitting**. Amniotic fluid is mainly fetal urine only after **16 weeks**.
- Non-visualised bladder: absent, malformed or not filling — kidneys and amniotic fluid discriminate.

	Agenesis	Hypoplasia	Exstrophy
US	Absent after repeat scan	Present, small, never reaches capacity	No filling; template on abdominal wall
Core defect	Cranial UGS atrophy or trigone not formed	Inadequate filling/storage	Open bladder plate
Outcome	Survives only with ectopic ureteric drainage	May grow; augmentation often	Reconstruction → *Ch 43*

- **Agenesis** (~1:600,000, female majority): hindgut normal. Ureters drain to müllerian structures (female) or rectum (male); associated renal dysplasia/agenesis, absent prostate/seminal vesicles/penis/vagina. Confirm with contrast; magnetic resonance imaging (MRI) or computed tomography (CT) for anatomy → ureterosigmoidostomy or stoma.
- **Hypoplasia causes**: low outlet resistance (severe epispadias), UGS anomalies, bilateral renal dysplasia/agenesis, ureteric ectopia (urine bypasses bladder).
- Exstrophy: absent filling with **normal amniotic fluid**.

3 · MEGACYSTIS

- **Massive vesicoureteral reflux (VUR) causes the dilation** (urine recycles between upper tracts and bladder), not bladder-neck obstruction — neck surgery failed.
- Now: any prenatally dilated bladder — a finding, not a diagnosis.

Gestation	Normal	Abnormal
10–14 weeks	Sagittal length ≤6 mm; <10% of crown–rump length	**≥8 mm**; ≥13% of crown–rump length
Later (functional)	Empties within 45 min	No emptying in 45 min
15–40 weeks	Length ≈ gestational age (GA) − 5 mm	> GA + 12 mm = megacystis

- **Causes**: obstructive (lower urinary tract obstruction (LUTO)) — posterior urethral valves (PUV) ~50%, urethral atresia/stenosis, anterior valves; non-obstructive — PBS, megacystis–microcolon–intestinal hypoperistalsis syndrome (MMIHS), cloacal anomalies. ~1/3 undiagnosed. Male:female 8:1.
- **Aneuploidy ~1/5** (trisomy 13/18) → karyotype; outcome uniformly poor.

- Euploid 7–15 mm → most resolve; **>15 mm** → none (all obstructive).
- Early (<18 weeks) resolves more often; resolution before 23 weeks → no postnatal surgery; after 23 weeks → most still need intervention. **Persistent megacystis + oligohydramnios = LUTO.**

MMIHS

- Visceral smooth-muscle myopathy (**ACTG2**); female ~70%; prenatally often megacystis alone with normal/increased fluid.
- Postnatal bowel obstruction + big bladder → exclude Hirschsprung disease (ganglion cells on biopsy). Total parenteral nutrition (TPN) complications are the leading cause of death.
- Urology: clean intermittent catheterisation (CIC) or vesicostomy short-term → urodynamics once stable (acontractile) → most need **lifelong CIC**. VUR uncommon.

> **KEY POINT** Female fetus with megacystis cannot have PUV → think MMIHS, cloacal anomaly, urethral atresia.

4 · BLADDER DUPLICATION

- Complete/incomplete; sagittal/coronal; **complete sagittal commonest**. External genital duplication up to 90%, lower gastrointestinal (GI) duplication up to 42%; VUR, renal ectopia/dysplasia common. A bladder without urethra → obstruction → ipsilateral dysplasia.
- Incomplete: halves communicate via one urethra — may need nothing. Complete: two full-thickness bladders, each with ureter and urethra.
- Work-up: karyotype, US, videourodynamics, genitogram, voiding cystourethrography (VCUG), renal scan, endoscopy.
- **ALGORITHM**
  relieve obstruction, prevent infection → later: both sphincters competent → join distal urethras; one incompetent → close that bladder neck, excise its urethra; vaginas joined + vulvoplasty
- **Sphincter competence, not anatomy, decides reconstruction.** Asymptomatic → may leave.

5 · BLADDER DIVERTICULA

- Mucosa herniates through detrusor defect. VCUG is gold standard (shows VUR, residual); dynamic — repeat if suspicion persists.
- **Hutch**: paraureteral, primary (congenital) vs secondary (acquired). Enlarging paraureteral diverticulum disrupts Waldeyer sheath → ureter pulled out → VUR → later obstruction.
- Large diverticulum compressing bladder neck → retention.

	Primary paraureteral	Secondary (acquired)	Syndromic posterolateral/multiple
Cause	Focal congenital wall deficiency at ureteric hiatus	Outlet obstruction, neurogenic bladder, infection, prior surgery	**Ehlers–Danlos, Williams, Menkes**
Wall	Smooth; single	Trabeculated; multiple "pop-offs"	Attenuated lax muscle; giant (> bladder)
Treatment	Observe if small; excise if VUR; endoscopic dextranomer/hyaluronic acid injection still possible	**Treat outlet first** (may remodel); excise ± reimplant if symptomatic	Observe; resect for urinary tract infection (UTI)/stones; recurrence common → vesicostomy option; postoperative urodynamics

6 · URACHAL ANOMALIES

	Patent urachus	Sinus	Cyst	Vesicourachal diverticulum
Open	Whole canal	Umbilical end	Isolated segment	Bladder apex
Presents	Neonatal umbilical urine, delayed cord healing	Draining umbilicus	Infection (*S. aureus*), painful mass; commoner in adults	Incidental
Imaging	US; fistulogram/VCUG	Sinugram	US; CT/MRI if extensive	Incidental
Treat	Excise with cuff	Complete excision	Drain → excise	Only if narrow neck causes stones/UTI

- Differential: patent **omphalomesenteric duct**; both need excision.
- **Malignancy**: urachal adenocarcinoma unreported in children but lethal in adults. Asymptomatic incidental remnant: "excise all" vs "only if symptomatic" (increasingly adopted) — discuss with family.

Excision (open)

ALGORITHM
infected → drain + antibiotics first → balloon catheter fills bladder → small transverse infraumbilical incision → feeding tube in tract → circumscribe at umbilicus (preserve umbilicus) → extraperitoneal dissection to dome → excise with **bladder cuff** → two-layer closure → catheter overnight

- Laparoscopic/robotic: feasible <6 months; good view of whole tract but intraperitoneal — risk of spilling infected/malignant material and urine leak.

7 · ACQUIRED LESIONS

Lesion	Key features	Treatment
Haemangioma	Klippel–Trénaunay, Sturge–Weber; solitary <3 cm; haematuria	Fulguration/Nd:YAG laser; partial cystectomy if large
Fibroepithelial polyp	2nd commonest paediatric bladder tumour after rhabdomyosarcoma	Endoscopic resection (curative)
Nephrogenic adenoma	Female 5:1; after infection, stones, surgery, augmentation; distal tubular origin	Transurethral resection; **80% recur** → surveillance cystoscopy
Cystitis cystica	Almost all female; mimics tumour	Long-term antibiotic prophylaxis
Eosinophilic cystitis	Male; no typical appearance	**Biopsy mandatory**; resection + steroids, antihistamines, antibiotics
Haemorrhagic cystitis	Cyclophosphamide/ifosfamide; BK virus, cytomegalovirus, adenovirus; radiotherapy	Prevent with mesna; oral pentosan polysulfate; avoid traumatic irrigation

8 · MCQ ONE-LINERS

1. Urachus becomes the median umbilical ligament.
2. Persistent megacystis with oligohydramnios: LUTO.
3. Fetal bladder empties every 15–20 min — repeat scan before calling it absent.
4. Normal amniotic fluid + absent bladder filling → exstrophy.
5. Congenital megacystis: massive VUR is cause, not consequence, of dilation.
6. First-trimester megacystis ≥8 mm; >15 mm never resolves.
7. Commonest cause of fetal megacystis: PUV (~50%).
8. ~1/5 megacystis fetuses aneuploid (trisomy 13/18).
9. MMIHS gene: ACTG2; female predominance; normal amniotic fluid.
10. Leading cause of death in MMIHS: TPN complications.
11. Commonest bladder duplication: complete sagittal.
12. Hutch diverticulum: paraureteral.
13. Multiple giant diverticula: Ehlers–Danlos, Williams, Menkes.

4. Secondary diverticulum: treat the outlet first.
5. Commonest organism in infected urachal cyst: *S. aureus*.
6. Urachal excision must include a bladder cuff.
17. Urachal adenocarcinoma: not reported in children.
18. Nephrogenic adenoma recurs in 80% of children.
19. Eosinophilic cystitis: biopsy mandatory.
20. Cyclophosphamide cystitis: mesna prevents.

43 Exstrophy–Epispadias Complex

Reading time ≈ 63 min (slow pace)

1 · SPECTRUM AND EPIDEMIOLOGY

- One spectrum of the bladder exstrophy–epispadias complex (BEEC): **glanular epispadias** (mildest) → **classic bladder exstrophy (CBE)** → **cloacal exstrophy** (multisystem). Five entities: cloacal exstrophy; male and female CBE; male and female epispadias.
- CBE ≈ **1 in 46,000**; male:female ≈ 2.3:1. CBE is >50% of the complex.
- Recurrence in a family ~1 in 100; offspring of an affected parent **1 in 70** (~500-fold). Inconsistent twin concordance → multifactorial.
- Modifiers: maternal smoking → cloacal; preconception folate → milder (epispadias); high-dose first-trimester progesterone and in vitro fertilisation (IVF) raise risk.
- Genetics: **22q11.2 duplication** = commonest variant; **ISL1** = genome-wide susceptibility gene for CBE; p63 under-expressed in exstrophic urothelium (no mutation).

2 · EMBRYOLOGY

- **Core defect**: the **cloacal membrane** (bilaminar, infraumbilical) is not reinforced by mesodermal ingrowth. That mesoderm normally forms lower abdominal muscles and pelvic bones.
- Normally: mesenchyme ingrows → urorectal septum divides cloaca → membrane perforates → genital tubercles fuse **cephalad** to the membrane.
- **Marshall–Muecke**: overdeveloped membrane blocks medial mesenchymal migration; **timing of rupture** decides the phenotype — early → cloacal, later → bladder exstrophy, then epispadias. Other theories (body-wall fold failure, caudal genital hillocks, abnormal body-stalk insertion, pelvic-ring non-rotation) converge on absent infraumbilical midline mesoderm.
- Why one lesion gives three phenotypes: the earlier and more extensive the rupture, the more of the cloaca (bladder **and** hindgut) lies open; rupture after urorectal septation exposes only the bladder; rupture limited to the genital tubercle region gives epispadias.

3 · ANATOMY OF CLASSIC BLADDER EXSTROPHY

Bony pelvis

- Not simply a wide symphysis: three-dimensional computed tomography (3D CT) shows **external rotation of the posterior pelvis (~12°) and anterior segment (~18°)**, acetabular retroversion, **anterior pubic segment 30% short**; posterior segment normal; mean diastasis **4.8 cm**.
- Consequences: short pendular penis, wide hips, waddling gait, outward-rotated feet — little disability, improve with growth.
- Spine: occult variants common; significant dysraphism uncommon (cf. cloacal).

Pelvic floor and abdominal wall

- Levator ani displaced posteriorly, outward-rotated, flattened; puborectal sling supports twice normal area. After closure, continent patients have the shortest intersymphyseal distance and deepest bladder neck → rationale for **deep posterior placement, osteotomy and pelvic fixation**.
- Triangular fascial defect bounded inferiorly by the **intrasymphyseal band** (divergent urogenital diaphragm) tethering the posterior vesicourethral unit to the pubic rami; contains **no striated sphincter**. Radical incision to the levator hiatus is essential.
- Umbilicus low; short umbilicus–anus distance; omphalocele rare (common in cloacal).
- **Indirect inguinal hernia** common (especially males, often bilateral): patent processus, large rings, no canal obliquity. Excise sac and repair transversalis fascia at closure; explore the other side. Osteotomy lowers incidence and recurrence.

Anorectum

- Anus anteriorly displaced; imperfect early anal continence.
- **Rectal prolapse before closure** → transient, resolves with closure (an indication to close). **Prolapse after closure** → suspect outlet obstruction → urgent cystoscopy.

Male genitalia

- **Anterior corpora ~50% shorter**, ~30% wider; posterior corpora normal → penis short from diastasis **and** true corporal deficiency. Lengthening is therefore only apparent.
- Prostate never circumferential — urethra lies anterior; PSA (prostate-specific antigen) below age norms. Vasa, seminal vesicles, testes normal; testes mostly retractile.
- Cavernous nerves lateral, usually preserved (potency retained); retrograde ejaculation may follow closure or bladder neck reconstruction (BNR).

Female genitalia

- Vagina short (~6 cm), normal calibre; orifice stenotic and anterior; **cervix in anterior vaginal wall**; bifid clitoris; divergent labia; tubes/ovaries normal.
- Levator hiatus twice normal width → **uterine prolapse**, even nulliparous; predicted by adult diastasis width, **not prevented by osteotomy**.

Bladder and upper tracts

- Newborn bladder is **immature, not neuropathic**: normal muscarinic receptors, raised collagen (type III) that normalises after successful closure, no dysinnervation.
- Exposed plate develops polyps, squamous metaplasia, cystitis glandularis (commoner after delayed/secondary closure) → **adenocarcinoma risk** → cytology/cystoscopy into adulthood. Delayed closure reduces uroplakin and increases metaplasia → argues for early closure.
- Urodynamics: most bladders compliant and stable before BNR; compliance and stability fall after BNR → lifelong follow-up. Caveolae deficient in bladders that later need augmentation.
- Upper tracts usually normal (duplex commonest anomaly). Deep pouch of Douglas → ureter enters laterally without obliquity → **vesicoureteral reflux (VUR) in 100%** of closed bladders → reimplant at BNR.

4 · VARIANTS

Variant	Feature	Management
Skin-covered (split symphysis) — commonest	Musculoskeletal defect, urinary tract intact	Formal closure at birth
Duplicate exstrophy	Everted mucosal patch + closed pelvic bladder with ureters	Excise patch, close wall
Superior vesical fissure	Rupture only at top of membrane; sphincter intact	Toilet-trains normally; no BNR
Pseudoexstrophy	Musculoskeletal defect only; low umbilicus	—
Epispadias with bladder prolapse	Prolapse may show only on Valsalva	Repair ± osteotomy

> **KEY POINT** Superior vesical fissure inflated the belief that variants do better. All variants except superior vesical fissure need formal closure and follow-up like CBE.

5 · PRENATAL DIAGNOSIS

- Only ~25% diagnosed prenatally; often mislabelled omphalocele/gastroschisis.
- Signs: **absent bladder filling**, low-set umbilicus, widened pubic rami, diminutive genitalia, **lower abdominal mass enlarging** with gestation; cord insertion–genital tubercle distance <5th centile; pubic diastasis >10 mm.

- Fetal magnetic resonance imaging (MRI) if CBE vs cloacal unclear or spinal anomaly suspected. Purpose: counselling and **delivery at an exstrophy centre**.

6 · NEWBORN CARE AND SELECTION FOR CLOSURE

ALGORITHM
tie cord with 2-0 silk (no clamp — traumatises mucosa) → cover with non-adherent plastic wrap → irrigate with saline and replace at each nappy change → cardiopulmonary assessment, renal ultrasound → transfer to exstrophy centre; counselling with multidisciplinary team

- **Assess**: template size and quality (≥**3 × 3 cm** suitable), pubic diastasis and pelvic malleability under anaesthesia, urethral plate, penile length, associated anomalies.
- Apparent size ≠ potential capacity: a plate that **bulges on crying or indents under anaesthesia** is adequate.
- **Inadequate template**: <3 cm, polyp-covered, fibrotic/inelastic. Precluding primary closure: penoscrotal duplication, hypoplastic bladder, significant bilateral hydronephrosis; ectopic bowel in plate (relative).
- Small plate → **delayed primary closure at 6–9 months with osteotomy**: safer than closing a small plate badly (dehiscence ruins future continence). No growth by 6–12 months → excise bladder; non-refluxing colon conduit or ureterosigmoidostomy.
- Delayed closure: continence procedures equally successful but later, and a **catheterisable channel is needed far more often**; capacity losses steepest when closure slips beyond 9 months. Early closure also yields more differentiated urothelium (uroplakin II).
- Once surface trauma stops, a small bladder closed with modest outlet resistance enlarges — the rationale for early closure in a robust infant with a good template.

7 · SURGICAL PHILOSOPHIES

Repair	Stages	Distinguishing feature	Weakness
Modern staged repair of exstrophy (MSRE)	Newborn closure → epispadias repair 6–12 months → BNR + reimplant 4–7 years	Converts exstrophy to complete epispadias; balanced outlet grows bladder	Multiple operations
Complete primary repair of exstrophy (CPRE, Mitchell)	Newborn closure + penile disassembly + bladder-neck tailoring	Aims for continence without BNR	Hypospadias; glans/soft-tissue loss; early reimplantation
Kelly (radical soft-tissue mobilisation)	Closure at birth → mobilisation 3–6 months (penoscrotal urethrostomy) → hypospadias repair ~3 years	Avoids osteotomy: pelvic floor, pudendal vessels and nerves freed off ischium/pubis, wrapped round urethra	Fistula, stenosis
Erlangen (single stage)	Total repair at ~8 weeks	Closure, reimplants, neck plasty, epispadias, pubic closure in one sitting	—
Mainz	Closure → ureterosigmoidostomy (sigma pouch) at ~2 years	Bladder remnant = seminal receptacle	Colonic malignancy

- **Erlangen**: if the plate is small, bladder-only closure first; no osteotomy (obturator-foramen coaptation); epidural 5 days; mostly continent at ~20 years. **Mainz**: oral alkalinisation; in girls genital reconstruction with anterior uterine fixation.
- **Common principles**: radical mobilisation of the posterior vesicourethral unit; tension-free closure of wall and pelvis (osteotomy if needed); strict selection for newborn closure; combined closure + epispadias repair only in selected patients.

KEY POINT Erlangen, Kelly and CPRE claim to avoid BNR, yet almost all patients still need an outlet procedure. Compressing reconstruction adds ischaemic and outlet risk without removing that need.

8 · PRIMARY CLOSURE (MSRE STAGE 1)

- Anaesthesia: general + **epidural** (catheter to T10–T12; bupivacaine with adrenaline).

ALGORITHM
outline 2-cm mucosal strip from distal trigone past verumontanum to mid-shaft (male) / vaginal orifice (female) → incise round bladder–skin junction → separate rectus fascia from bladder; ligate umbilical vessels; strip peritoneum from dome → **divide urogenital diaphragm fibres sharply down to levator hiatus** → free corpora from inferior rami → close bladder, neck and posterior urethra onto the shaft; **taper over 12-Fr sound** → suprapubic Malecot (4 weeks) + ureteric stents; **urethra unstented** → pressure on greater trochanters → **No. 2 nylon horizontal mattress** pubic suture, knot away from neourethra → neoumbilicus (tubes exit)

- **Tapering**: enough resistance to aid bladder growth and prevent prolapse, not enough to harm the kidneys; site of later BNR.
- **Ureteric stents** 10–14 days: post-closure swelling can obstruct ureters (transient hypertension). Urethra not stented — secretions → necrosis.
- Short plate causing dorsal angulation → lengthen or use paraexstrophy skin flaps (stricture-prone; avoid if possible).
- **Commonest error**: inadequate urogenital diaphragm dissection — fibres found intact in most failed closures; the unit is pulled anteriorly when the pubes close → prolapse.

9 · OSTEOTOMY AND IMMOBILISATION

Indications

- Aims: close pelvic ring, relieve tension on bladder, urethra and wall, place unit deep with pelvic floor in midline, reduce hernias.
- **Newborn <72 h** with malleable pubes that meet on medial rotation of the trochanters → closure without osteotomy (sacroiliac ligament laxity). **If in doubt, osteotomise.**
- Osteotomy if: closure **>72 h**, diastasis **>4 cm**, stiff pelvis, large template, **failed closure**, **cloacal exstrophy**. Diastasis >6 cm → staged pelvic closure with fixator.
- Most referred dehiscences/prolapses had no osteotomy initially.

Type	Position	Features
Combined bilateral anterior innominate + vertical iliac (preferred)	Supine	Pins under vision; best malleability; least recurrence of diastasis
Bilateral posterior iliac (original)	Prone (turn)	Worse approximation, more recurrence
Anterior innominate alone	Supine	Less mobility

- Technique: Gigli-saw transverse innominate cut between anterior superior and inferior spines → vertical closing-wedge iliac cut lateral to sacroiliac joint, posterior cortex as hinge → pins → urological repair → pubic suture → **external fixator**.
- Aftercare: radiograph at 7–10 days, crank fixator if needed; **modified Buck traction ~4 weeks + fixator 4–6 weeks**. Without osteotomy: **modified Bryant traction** (hips 90°) 4 weeks. Tunnelled epidural 2–3 weeks.
- Pubic suture (No. 2 nylon mattress between rami fibrocartilage) gives the strongest load-to-failure, but every repair is weaker than an intact symphysis — hence the need for rigid immobilisation.

Immobilisation (CBE)	Failure overall	With osteotomy	Without osteotomy
Spica cast	47%	30%	**66%**
External fixation + traction	**7.5%**	8%	— (few)

- Osteotomy alone does not secure closure — the immobilisation method matters independently. Failure is reduced by Buck traction + fixator **>4 weeks** and increased when a non-paediatric orthopaedic surgeon osteotomises; osteotomy type and epidural do not matter. Options: mummy wrap, spica, Bryant (no osteotomy only), Buck (with fixator), fixator.
- Complications: pin-site inflammation (near-universal; oral antibiotics), transient femoral nerve palsy (bed rest, knee immobiliser; recovers by 12 weeks), delayed iliac union, peroneal palsy, abductor weakness. Partial diastasis recurrence from **early pin loosening** (soft infant bone) and **ischiopubic undergrowth**; still net beneficial, as an uncorrected diastasis widens with growth.
- Inadequate immobilisation → dehiscence, prolapse, loss of tubes and stents.

10 · AFTER PRIMARY CLOSURE

- Keys to success: adequate **bladder drainage, ureteric drainage, immobilisation, pain control**.
- At 4 weeks before tube removal: calibrate outlet, ultrasound, clamp for residual, culture. **Continuous antibiotic prophylaxis** (all have VUR).
- Ultrasound at 3 months then 6–12-monthly; **yearly cystoscopy + cystogram under anaesthesia** for capacity (awake capacity unreliable).
- Capacity **<30 mL at 1–2 years** → warn that a continence procedure may not be feasible.
- **Outlet too tight** (retention, infection, dilatation): dilation/catheterisation → incise stricture → early reimplant → revision. Recurrent urinary tract infection (UTI) + distended bladder → cystoscopy for **intrapubic stitch erosion**.
- Keep residuals <50 mL; staged repair does not harm renal function.

11 · EPISPADIAS REPAIR (STAGE 2)

- Moved before BNR because urethroplasty **increases bladder capacity** (outlet resistance); done at 6–12 months (better capacity if before 12 months).
- **Testosterone enanthate 2 mg/kg intramuscularly at 5 and 2 weeks** before every urethroplasty.
- Four goals: correct dorsal chordee, reconstruct urethra, reconstruct glans, close skin.
- **Modified Cantwell–Ransley** (preferred): urethral plate strip to tip → reverse MAGPI (meatal advancement and glanuloplasty) brings meatus to tip → glans wings → divide suspensory ligaments → plate freed from corpora but **distal 1 cm left attached to glans** → tubularise over 8-Fr stent → corporotomies at maximal curvature if needed → **corpora rotated over the neourethra** (urethra ventral) → glans closure → Z-plasty at base; stent 10–12 days.
- **Chordee**: dorsomedial corporal incision and anastomosis (cavernocavernostomy) or, in adolescents, dorsal dermal graft. Suspensory-ligament remnants and scar must always be excised and corpora freed further from the inferior rami — little is achieved at initial closure.
- **Skin**: ventral prepuce split into lateral flaps or buttonholed to the dorsum; base Z-plasty prevents upward tethering.
- Results: downward-deflected penis, always-catheterisable urethra; early fistula ~1 in 6, many close by 3 months. **Coronal fistula is rare** because the urethra lies ventral under apposed raw glans; fistulae occur at the base where the urethra rises between the corpora. Scarce tissue → tunica vaginalis flap.
- Abandoned: in-situ dorsal tubularisation, free grafts, transverse island flaps. Moving the tube beneath the corpora gives fewer fistulae, downward deflection and an easy catheter channel.
- For males, genital appearance often matters more than continence to adolescents.

- **Mitchell–Bägli complete disassembly**: plate depends on spongiosum → ischaemia; most left **hypospadiac**. Glans/corporal ischaemic loss linked to closure **without osteotomy** (pudendal compression) → if ischaemia seen, perform immediate osteotomy. Now modified to keep plate attached distally.
- Postoperatively: control bladder spasms (epidural + oxybutynin) — spasm causes extravasation and fistula.

Other closure techniques

- **CPRE**: penis disassembled into two hemicorpora-hemiglans and urethral wedge; urethra moved ventrally; perineal membrane dissected to set unit deep. ~Half need reimplantation in the first year; simultaneous bilateral reimplantation reduces later VUR. Girls closed after 72 h risk retention. Complications not more frequent than MSRE but more serious (glans loss, fistula, dehiscence).
- **Mitchell**: osteotomy (unlike Kelly) if >3 days old or wide diastasis; bladder neck, urethra and vagina mobilised as one unit in girls. If the urethra falls short (most), the meatus is left hypospadiac at the base. Voided continence without outlet surgery only ~1 in 5; 65–80% need an outlet procedure; most end on clean intermittent catheterisation (CIC).
- **Kelly stage 2**: plate moved beneath corpora → "flake" of pubis with corpora → levator incised → pudendal neurovascular bundles freed in ischiorectal fossa → muscles wrapped round urethra.
- **Combined closure + epispadias repair** (modified Cantwell–Ransley): for failed or delayed closure, boys >6 months; select by phallic size, groove depth, template and scarring; avoids hypospadias; about two-thirds dry after later YDL; risk of ischaemic penile loss; experts only.
- Kelly: complete unaided voided continence <1 in 5, about half partially continent; glans loss reported.

12 · FEMALE EXSTROPHY CLOSURE

- Diastasis narrower → osteotomy less often (if >4 cm or stiff).
- Steps: dissect bladder, urethra and vagina deep to levator hiatus as one unit, keeping urethrovaginal septum (blood supply) → single-layer bladder closure → suprapubic tube + stents 4 weeks, nothing through neourethra → pubic suture → **monsplasty, join clitoral halves**, Y–V labioplasty for introitus.

13 · BLADDER NECK RECONSTRUCTION (STAGE 3)

Prerequisites

- Capacity **≥100 mL** (under anaesthesia), stable urodynamics, motivated child (usually 5–7 years), voiding programme started ≥6 months earlier, psychologist assessment.
- VUR managed with prophylaxis until BNR; earlier reimplant for pyelonephritis, worsening hydronephrosis or falling function.

Modified Young–Dees–Leadbetter (YDL)

ALGORITHM

transverse bladder-neck incision with vertical extension → **cephalotrigonal / Cohen cross-trigonal reimplant** (moves orifices off the trigone) → posterior mucosal strip **15 × 30 mm** from mid-trigone to posterior urethra → denude lateral muscle triangles → small edge incisions, no cephalad transverse cut (avoids denervation, ischaemia) → tube over strip → **double-breast muscle flaps** → 8-Fr guide removed → suspend to rectus fascia

- Radical dissection behind the pubic bar is needed for narrowing and suspension.
- Post-op: suprapubic tube 3 weeks; clamp trials; failure to void → urethral catheter 5 days and retry; frequent ultrasound.

Outcomes and predictors

- MSRE: ~**70–75%** dry by day (>3 h) in both sexes; CPRE followed by YDL lowest (~56%).
- **Predictors**: template size at birth, **successful primary closure** (more than who did it), and **capacity at BNR** (most important determinant). Capacity >100 mL → higher continence, dry in half the time.

Category (after BNR)	Definition
Continent	Dry day and night, dry interval ≥3 h
Socially continent	Dry 3 h by day, occasional wet nights
Wet	Dry <3 h by day, wet at night

- High "continence" in failed-closure and large cohorts is achieved mostly by **bladder-neck closure + catheterisable stoma**, not urethral voiding.
- Prolonged post-BNR outlet obstruction needing catheter or suprapubic drainage is fairly common. Urodynamics: continent patients have higher capacity and compliance; high end-filling pressure → non-obstructive hydronephrosis.
- Closure <72 h or with osteotomy → continence likely; delayed closure without osteotomy ~10%.
- Daytime dry within ~12 months; **not dry by 2 years = failure**; night dryness 2–3 years (desmopressin/oxybutynin help).
- Each failed closure lowers potential: after two closures <1 in 5 void continently.

Age	Step
Birth	Closure ± osteotomy
6–12 months	Epispadias repair
2–4 years	Yearly cystogram + cystoscopy
5–7 years	BNR + reimplant

14 · FAILURES AND COMPLICATIONS

Failed closure

- Forms: dehiscence, **bladder prolapse** (counts as failure), outlet obstruction/stricture, soft-tissue loss, vesicocutaneous fistula.
- Causes: incomplete pelvic-diaphragm mobilisation, inadequate immobilisation, infection, distension, tube malfunction.
- Expert-series complications of primary closure: major dehiscence/prolapse ~3%; minor urological ~11% (febrile UTI, stones, outlet obstruction, stitch erosion); major orthopaedic ~6% of those osteotomised (non-union, leg-length inequality); femoral palsy ~2%.
- **Evaluate a failed closure**: template quality, perivesical scarring, diastasis, prior osteotomy, ≥4 months since failure.
- **Reclosure**: wait ≥4–6 months; **osteotomy** (repeat is safe), rigid immobilisation, radical division of urogenital diaphragm fibres. With prolapse/dehiscence → combine epispadias repair with reclosure after testosterone. After CPRE tissue loss → grafts/tissue expansion first.
- Stricture warning signs (most within 60 days): UTI, enlarging bladder, stones, long dry intervals, rectal prolapse. All have VUR → obstruction endangers kidneys → dilation/incision → urethroplasty → diversion; no outlet surgery until fixed.
- Diversion needed for obstruction >6 months → bladder usually ends in augmentation. Vesicocutaneous fistula usually signals failed closure (lower midline tract, widened diastasis).
- Secondary closure bladders are smaller and slower-growing; voided continence after reclosure only ~1 in 6. Bulking after reclosure disappointing → augmentation or continent stoma.
- Failure after reconstruction elsewhere → refer to an expert centre; fellowship-trained paediatric urologists achieve better reclosure success.

Failed BNR

- Causes: low outlet resistance, small capacity, poor compliance. Failure = no 3-hour dry interval within 2 years.
- Near-miss (dry >2 h) → bulking (dextranomer better in epispadias); no benefit if dry <1 h.
- **Redo YDL** only if patulous neck, capacity >100 mL, stable urodynamics. Otherwise **bladder-neck transection + augmentation + Mitrofanoff**; artificial sphincter rarely (needs capacity). Stones ~30%.
- Pubertal "prostatic" continence is unlikely (prostate absent anteriorly) — pelvic-floor maturation.

Failed genital reconstruction

- Urethrocutaneous fistula commonest. Glans/corporal loss after disassembly → tissue expansion, grafts, ultimately **radial forearm phalloplasty**. Over-aggressive lengthening only denervates.

15 · DIVERSION AND AUGMENTATION

- Indications: plate unclosable, failed closure without growth, failed BNR, upper-tract decompensation.
- **Ureterosigmoidostomy** (no stoma): needs anal continence. Problems: pyelonephritis, hyperchloraemic acidosis, ureteric obstruction, **colonic adenocarcinoma ~3 decades later → yearly colonoscopy** in adult life. Mainz sigma pouch lowers pressure. Severe renal impairment → colon conduit.
- Bladder too small → **non-refluxing colon conduit** (undiversion possible).
- Failed BNR → **augmentation (ileal/sigmoid) + appendiceal channel ± bladder-neck closure**; catheterised continence in great majority; stones commonest complication. Move early to diversion if BNR capacity unlikely.
- **Continent diversion indications**: failed closures not growing after reclosure; newborn closures not growing; inadequate capacity or wetness after BNR; upper-tract decompensation.
- Augmentation does not compromise long-term glomerular filtration rate (GFR).

16 · LONG-TERM ISSUES

- **Malignancy**: augmented exstrophy bladders develop multifocal adenocarcinoma disproportionately (inherent risk); smoking and immunosuppression add. Endoscopic surveillance from 5 years after reconstruction.
- **Male sexual function**: libido, erection and orgasm usually preserved; concerns are length, appearance, chordee. Adolescent: deglove, excise scar, release suspensory ligament, dorsal graft (acellular dermis) for chordee, tissue expansion. Routine PSA/digital rectal examination as other men.
- **Male fertility**: impaired — verumontanum injury at closure/BNR, retrograde ejaculation, low volume, oligo-/azoospermia; fewer BNR operations → better counts. ~1/4 achieve paternity, half with assisted reproduction (intracytoplasmic sperm injection (ICSI)).
- **Female**: stenotic vertical introitus → inverted-U **perineal flap vaginoplasty** (cutback increases prolapse); monsplasty with rhomboid flaps or lateral tissue expanders when recurrent diastasis has split mons and clitoris (topical oestrogen 2 weeks beforehand). **Uterine prolapse** → sacral suspension (presacral fixation both sides); consider prophylactic suspension at adolescent reconstruction.
- **Pregnancy**: fertility preserved; prolapse common; high-risk → **planned caesarean section with a urologist present**.
- Psychosocial: acting-out behaviour, sexual anxiety, anxiety and suicidal ideation in older males; continence achieved after age 5 → more acting-out → early counselling, lifelong psychological support and transition to adult urology.

17 · EPISPADIAS

	Male	Female
Incidence	1 in 117,000	1 in 484,000
Classification	Glanular, penile, **penopubic** (~70%, incontinent)	Davis: vesicular → subsymphyseal → retrosymphyseal (incontinent)
Genitalia	Dorsal chordee; short pendular penis as in CBE	Bifid clitoris, flat glabrous mons
Symphysis	Widened (less than CBE)	Closed or narrow band
VUR	30–40%	30–75%
Presentation	At birth	May present only with incontinence

- Continence depends on how proximal the meatus is (penopubic nearly always incontinent; glanular never).
- Treat as exstrophy (osteotomy + closure) if diastasis >3 cm, poor intrasymphyseal bar, little sphincter, small capacity.
- **Male**: small bladder → urethroplasty (modified Cantwell–Ransley) first to raise capacity → BNR (YDL + suspension + reimplant) at ~100 mL (4–5 years); good capacity → one stage. Usually no osteotomy. Disassembly in isolated epispadias causes no ischaemia but often hypospadias.
- **Female** (6–12 months, lithotomy):

ALGORITHM
vertical incision from mons base through urethral wall at 9 and 3 o'clock → excise urethral roof towards bladder neck (dorsal wedge taper) → inverting closure over 10-Fr Foley → denude medial hemiclitoris and labia minora → mons fat over suture line → unite clitoral halves and labia minora → clitoral hood → catheter 5–7 days → BNR at 4–5 years

- Old procedures (plication, Marshall–Marchetti suspension, muscle transplants) failed — raised resistance without correcting anatomy. Urethral closure increases capacity. Continence after BNR ~**87%**; epispadias girls do as well as CBE girls.
- After BNR, continence in complete epispadias: males 47–82%, females 70–87%. Capacity is the dominant predictor; continence unrelated to puberty.
- Epispadias outperforms exstrophy: bladder never exposed or scarred, so it grows better.

18 · CLOACAL EXSTROPHY

- Incidence 1 in 200,000–400,000. **OEIS**: Omphalocele, Exstrophy, Imperforate anus, Spinal defects.
- Anatomy: **two hemibladders flanking exstrophied ileocaecal bowel** (terminal ileum may prolapse), rudimentary hindgut, separated hemiphalluses/split scrotum, wide asymmetric diastasis (~twice CBE).

Feature	CBE	Cloacal
Spinal defects	Uncommon	**85–100%** (lipomeningocele commonest)
Omphalocele	Rare	Near-universal
Upper-tract anomalies	Uncommon	41–60%; solitary kidney commonest
Osteotomy	Selective	Always
Continence	Mostly voiding	Mostly augmentation + CIC

- **Neurospinal**: from disrupted dorsal mesenchyme, not neural-tube non-closure; only ~1 in 5 has a sacral skin sign → image all. Autonomic bladder nerves arise more medially → at risk in dissection (neuropathic bladder after repair). Watch for cord tethering lifelong.
- Other: müllerian duplication in most girls (preserve for reconstruction), clubfoot, short gut (~1/4) even with normal length.

- **Intravesical phallus** (rare, 46,XY): erectile tissue on the plate → biopsy before resecting apparent aphallia.
- **Prenatal** (only ~15%): non-visualised bladder (commonest), infraumbilical defect, omphalocele, myelomeningocele; **"elephant trunk" sign** = prolapsed ileum.
- **Newborn**: stabilise; neurosurgical evaluation (MRI spine for all); gender team.
- **Gender**: historic female assignment of 46,XY abandoned (male psychosexual shift) → assign congruent with **karyotype**; phalloplasty enables male rearing.
- **Bowel**: never discard hindgut — tubularise caecal plate with end colostomy (absorbs water, reduces short gut); keep unused hindgut for vaginoplasty/augmentation; pull-through if forms solid stool.

ALGORITHM
Stage 1 (newborn): excise/close omphalocele → separate caecal plate from hemibladders → join hemibladders (= bladder exstrophy) → colostomy → Stage 2: bladder closure + **combined osteotomy** + fixation 6–8 weeks → 5–7 years: YDL only if >100 mL (rare), usually augmentation + continent stoma

- Osteotomy (always): combined ~90% success > posterior > anterior; nearly sixfold odds of successful closure; does not improve continence. Diastasis >10 cm → fixator cranked over 1–2 weeks before closure; interpubic plate or bar; acellular dermal patch reduces stitch erosion.
- Staged closure ~4-fold odds of success vs single-stage; predictors of success: staging, osteotomy, immobilisation >4 weeks, paediatric urologist. Spica fails in most.
- Abdominal wall: osteotomy → relaxing fascial incisions → bioprosthetic patch.
- Primary closure fails in ~31–43% (mainly prolapse); reclosure needs osteotomy + Buck traction with fixator for 6 weeks.
- **Continence**: augmentation (hindgut, ileum or stomach) + CIC; median two procedures; most via continent stoma, chosen by outlet, dexterity and orthopaedic status. YDL success tracks neurological status.
- **Neurological level** is the main determinant of disability. Male fertility compromised; female fertility preserved but cervical prolapse worse than CBE → caesarean section.

19 · MCQ ONE-LINERS

1. Core embryological defect: failed mesodermal reinforcement of the cloacal membrane.
2. Offspring risk of an affected parent: 1 in 70.
3. Commonest genetic variant in BEEC: 22q11.2 duplication.
4. Anterior pubic segment 30% short; posterior segment normal.
5. Anterior corpora 50% shorter — penile lengthening is apparent.
6. Intrasymphyseal band contains no striated sphincter.
7. VUR in 100% of closed exstrophy bladders.
8. Rectal prolapse after closure → suspect outlet obstruction.
9. Cervix lies in anterior vaginal wall in female exstrophy.
10. Uterine prolapse is predicted by diastasis width, not prevented by osteotomy.
11. Newborn exstrophy bladder: immature, not neuropathic.
12. Commonest exstrophy variant: skin-covered (split symphysis).
13. Superior vesical fissure: sphincter intact, no BNR needed.
14. Template ≥3 × 3 cm suitable for newborn closure.
15. Cord tied with silk, not clamped.
16. Osteotomy if closure >72 hours or diastasis >4 cm.
17. Preferred osteotomy: combined anterior innominate + vertical iliac.
18. Spica without osteotomy fails in about two-thirds.
19. Commonest closure error: inadequate urogenital diaphragm division.
20. Urethra not stented after closure; ureters stented.
21. Recurrent UTI after closure: look for intrapubic stitch erosion.
22. Testosterone before urethroplasty: 2 mg/kg at 5 and 2 weeks.
23. Epispadias repair increases bladder capacity.
24. Complete disassembly leaves most boys hypospadiac.
25. BNR capacity threshold: 100 mL.

26. Most important determinant of continence: capacity at BNR.
27. YDL mucosal strip: 15 × 30 mm.
28. BNR failure: no 3-hour dry interval by 2 years.
29. Ureterosigmoidostomy cancer: adenocarcinoma ~30 years later; yearly colonoscopy.
30. Exstrophy pregnancy: planned caesarean with urologist present.
31. Male epispadias: ~70% penopubic.
32. Female epispadias classification: Davis.
33. OEIS = omphalocele, exstrophy, imperforate anus, spinal defects.
34. Cloacal exstrophy "elephant trunk" sign: prolapsed terminal ileum.
35. Cloacal exstrophy gender: assign by karyotype.

44 Prune-Belly Syndrome

Reading time ≈ 27 min (slow pace)

1 · DEFINITION AND TRIAD

- **Prune-belly syndrome (PBS)** (Eagle–Barrett / triad syndrome): constellation of anomalies of highly variable severity. **Triad**: (1) deficient abdominal wall musculature; (2) bilateral intra-abdominal testes; (3) anomalous urinary tract (hydronephrosis, renal dysplasia, dilated tortuous ureters, large bladder, dilated prostatic urethra).
- Incidence ~**1 in 29,000–44,000** live births (≈ bladder exstrophy); **95% male**; more in twins, African ancestry, young mothers. Name "prune-belly" given by **Osler** MCQ.
- **Survival follows the kidneys, not the belly**: degree of renal dysplasia (plus pulmonary hypoplasia) determines outcome; the abdominal wall defect is the least prognostic element.

2 · GENETICS AND AETIOLOGY

- For heredity: male predominance, affected siblings/cousins, twins (1 in 23 PBS children is a twin). Against: most sporadic, normal karyotype, most twins discordant. Chromosomal associations (Turner, trisomy 13/18, monosomy 16, Beckwith–Wiedemann) are the exception.
- No inheritance pattern established. Multiplex families: variants in **smooth muscle genes** — **MYOCD (myocardin)** and **FLNA (filamin A)** MCQ.

Embryogenesis theory	Mechanism
Early in utero urethral obstruction	Severe dilation, fetal ascites, oligohydramnios
Lateral plate mesoderm defect	Mesoderm forms ureters, bladder, prostate, urethra, gubernaculum
Intrinsic urinary tract defect	Ureteric dilation → fetal ascites
Yolk sac defect	Primary yolk sac anomaly

> **KEY POINT** Weakness of the obstruction theory: most patients have **no demonstrable obstruction postnatally**. Whatever the cause, the tract is **dilated, collagen-rich and largely non-obstructed** — that observation drives management.

3 · GENITOURINARY PATHOLOGY

Kidneys

- Dysplasia in ~**50%**, often asymmetric. Worst with outlet obstruction **without urachal decompression**. US: echogenic parenchyma with cortical cysts.
- Degree of dilation **does not correlate** with dysplasia; calyces often well preserved. Non-obstructive hydronephrosis is the rule; **infection, not obstruction, is the main threat** to function.

Ureters

- Dilated, tortuous, redundant; **distal worse than proximal**. Histology: smooth muscle loss, raised collagen:muscle ratio, fewer myofibrils. **Proximal ureter has more normal muscle** — preserved in reconstruction.
- Vesicoureteric reflux (VUR) ~**75%**; obstruction (pelviureteric junction (PUJ) / vesicoureteric junction (VUJ)) uncommon.
- A **failed pump, not a blocked pipe**: walls do not coapt, sparse muscle cannot propel the bolus → stasis → infection. Surgery targets stasis and reflux.

Bladder

- Massive, **thick-walled but smooth** (no trabeculation — unlike an obstructed bladder), collagen excess; **urachal pseudodiverticulum**; urachus patent at birth in **25–30%**.

- Trigone splayed, orifices lateral and high → VUR. Wide bladder neck into dilated posterior urethra.
- Urodynamics: **normal compliance, large capacity, delayed/absent sensation**; residual from weak detrusor plus relative outlet resistance = **"unbalanced voiding"**. ~Half void normally, but balance may deteriorate with growth → periodic reassessment.

Prostate, urethra, genital ducts

- **Prostatic hypoplasia** (mesenchymal–epithelial defect) → dilated posterior urethra, angulation on voiding, ejaculatory failure. Distal posterior urethral obstruction (atresia, valve, stenosis) in ~20%.
- Vasa/seminal vesicles often atretic; epididymis poorly attached; bladder neck incompetence → retrograde ejaculation.
- **Anterior urethra** usually normal. **Urethral atresia** — lethal unless urachus patent. **Megalourethra** is commoner in PBS than any other syndrome.

	Scaphoid megalourethra	**Fusiform megalourethra**
Deficient	Corpus spongiosum only	Corpora cavernosa + spongiosum
On voiding	Ventral urethra balloons	Whole phallus dilates
Severity	Milder	**More severe** (erectile bodies)

Testes

- Bilateral **intra-abdominal**, over the iliac vessels beside the dilated ureters. Mechanical theory doubted (pseudo-prune boys may have descended testes).
- Infant germ cells resemble age-matched controls → the **abdominal environment** destroys later spermatogenic potential, so time in the abdomen is the modifiable variable (basis of early orchidopexy).

4 · ASSOCIATED (EXTRAGENITOURINARY) ANOMALIES

~**75%** have non-urinary anomalies; nearly **half premature**.

System	**Features**
Abdominal wall	Medial and inferior segments most deficient; wrinkled skin, flank bulge; flared costal margin; weak cough; heals well
Pulmonary (>50% survivors)	Pulmonary hypoplasia from oligohydramnios (newborn death); pneumothorax/pneumomediastinum; recurrent pneumonia, atelectasis
Cardiac (~10%)	Patent ductus arteriosus (PDA), atrial/ventricular septal defects (ASD/VSD), tetralogy of Fallot
Gastrointestinal (≥30%)	Incomplete midgut rotation → malrotation, volvulus, atresia; splenic torsion; lifelong constipation
Orthopaedic (up to 65%)	Talipes equinovarus (commonest), hip dysplasia, scoliosis, **lateral knee dimples**; limb hypoplasia/amputation (bladder compressing iliac vessels)

> **KEY POINT** Two mechanisms explain most non-urinary morbidity: **oligohydramnios** (lung hypoplasia, limb deformity, knee dimples) and **absent abdominal wall pressure** (weak cough, atelectasis, constipation).

5 · PRESENTATION AND SPECTRUM

- **Prenatal**: mimics posterior urethral valves and megacystis–megaureter; massive bladder, hydroureteronephrosis, irregular abdominal circumference, urachal diverticulum. Prenatal dilation does **not** predict postnatal function. Dilated prostatic urethra with anhydramnios suggests urethral atresia.
- **Neonate**: wrinkled flaccid abdomen is diagnostic on sight; impalpable testes, knee dimples.

Woodard category	Kidneys / lungs	Course	Urological management
I	Dysplasia; **pulmonary hypoplasia**; oligohydramnios, Potter features, urethral atresia	Stillbirth or early death (~20% perinatal mortality overall)	Supportive; simple bladder drainage only
II	Full triad; minimal/unilateral dysplasia; no hypoplasia	Risk of progressive renal insufficiency	Individualised — **the controversial group**
III	Incomplete/mild; no dysplasia; stable function	Majority; good	Surveillance; surgery for urinary tract infection (UTI)/deterioration; orchidopexy in year 1

- The abdominal wall is **not** a sorting criterion; appearance does not predict prognosis.
- **RUBACE score** (Renal, Ureter, Bladder, Abdominal wall, Cryptorchidism + Extragenitourinary): severity score for survivors (categories II–III) to compare outcomes; renal is the heaviest domain MCQ.
- **Incomplete ("pseudo-prune")**: slight/unilateral flaccidity, unilateral undescended testis; diagnosis delayed (even to adulthood with renal failure, hypertension) → recurrent UTI → **higher late renal failure**. Subtle phenotype harms kidneys through missed surveillance.
- **Female PBS** (~5%): abdominal wall defect ± urinary anomalies, normal gonads; outlet obstruction and anorectal anomalies frequent; mortality similar to males.

6 · EVALUATION

ALGORITHM

Team (neonatology, nephrology, urology, cardiology, orthopaedics) → **cardiorespiratory first**: chest X-ray (pneumothorax, hypoplasia) → creatinine/blood urea nitrogen (BUN)/electrolytes (trend over days–weeks; day-1 value is maternal) → renal–bladder ultrasound (US) → antibiotic prophylaxis + circumcision → selective imaging

- **Voiding cystourethrogram (VCUG)** only if renal insufficiency or suspected outlet obstruction, under antibiotic cover, strict asepsis (static system easily infected). VUR in up to 70%.
- **Dimercaptosuccinic acid (DMSA)** for differential function at **4–6 weeks**; **mercaptoacetyltriglycine (MAG3)**/diethylenetriaminepentaacetic acid (DTPA) for drainage — unreliable with massive dilation and poor function. Magnetic resonance urography (MRU) for anatomy/dysplasia.
- Early intervention only for true outlet obstruction: catheter, preferably **percutaneous suprapubic tube**.

7 · MANAGEMENT PHILOSOPHY

	Conservative	Early reconstruction
Premise	Dilation non-obstructive, may improve with growth	Stasis, reflux, infection drive renal loss
Operate when	Proven obstruction, intractable infection	Progressive dilation, high-grade VUR, recurrent upper tract infection, declining function
Best for	Category III	Selected category II
Weakness	Progressive uropathy in some	Retrospective, heterogeneous evidence

- Delay major reconstruction to **≥3–6 months** (pulmonary maturity). Comprehensive single-stage surgery (urinary reconstruction + orchidopexy + abdominoplasty + circumcision) does not add morbidity.
- Correction of high-grade VUR is prudent — it is no less important in PBS.

8 · URINARY TRACT SURGERY

- **Temporary diversion** (sepsis, renal failure, urethral atresia): **Blocksom cutaneous vesicostomy**; excise urachal diverticulum; make the stoma **larger than usual** (stenosis common because of low intra-abdominal pressure). Rarely pyelostomy/distal ureterostomy — **avoid proximal ureterostomy** (wastes the good proximal ureter).
- **Ureteroplasty + reimplantation**: excise the distal ureter, keep the upper few centimetres, taper/infold, transtrigonal reimplant; tunnelling in the abnormal bladder is difficult.
- **PUJ**: secondary kink → ureterolysis; true obstruction → pyeloureteral anastomosis to the normal upper ureter.
- **Reduction cystoplasty**: capacity and residual recur → only for large urachal diverticulum or as part of wider reconstruction. **Clean intermittent catheterisation (CIC)** (urethral or appendicovesicostomy) empties better long-term.
- **Internal urethrotomy** (Otis to 24–30 Fr): short-term gains only; abandoned in infants.
- Urethral atresia/hypoplasia: progressive dilation (antegrade via vesicostomy) or hydrodistension. **Megalourethra repair**: subcoronal incision → degloving → excise/infold → reconstruct over catheter.
- **Circumcision** reduces UTI; routine with any reconstruction.

9 · ORCHIDOPEXY

- Early surgery: germ cells present in infancy, normal pubertal hormones, malignancy risk. Broad mesorchium → inguinal approach usually fails.
- **Bilateral transabdominal orchidopexy at ~6 months** (neonate–6 months vessels still mobilise) = approach of choice, often combined with vesicostomy, reconstruction or abdominoplasty.
- Later: **Fowler–Stephens** (spermatic vessel division; testis survives on vasal collaterals), **staged Fowler–Stephens** (collaterals enlarge between stages → **more successful than single-stage**), or microvascular autotransplantation.
- **Laparoscopic** (primary or Fowler–Stephens) when no other abdominal surgery planned; lax wall offers little resistance to port insertion — take care.

10 · ABDOMINOPLASTY

- Mild laxity may improve; corset helps. Cosmetic/psychosocial benefit certain; functional gains (cough, defecation, emptying) plausible but confounded. From ~6 months, timed with other surgery; infants may need postoperative ventilation.

Technique	Incision / umbilicus	Key step	Note
Randolph	Transverse, 12th rib to 12th rib via pubis	Full-thickness excision of lower (worst, per EMG) wall incl. peritoneum	Waistline, but lateral bulge persists
Ehrlich	Vertical midline; umbilicus on inferior epigastric pedicle	Vest-over-pants advancement of each side to opposite flank	Good long-term results
Monfort	Vertical ellipse xiphoid–pubis; umbilicus circumscribed in situ	Vertical fascial incisions lateral to superior epigastric arteries → central fascial plate; lateral flaps advanced over it	**Transperitoneal** — ideal with simultaneous reconstruction/orchidopexy; thickens wall
Dénes	Fusiform; umbilicus buttonholed	Eccentric fascial incision → double-breasted overlap	Modified Ehrlich
Furness	Midline, limited	Double-breasted plication; **peritoneum not opened**	When no intra-abdominal surgery

ALGORITHM

Monfort sequence: excise skin ellipse, umbilicus spared → incise central plate at lateral rectus borders → genitourinary surgery → score peritoneum over lateral muscle → suture plate edges to scored line → overlap lateral flaps over drains

11 · LONG-TERM OUTCOME AND FOLLOW-UP

- **Renal predictor**: nadir/baseline creatinine **<0.7 mg/dL at 1 year** → stable function through childhood unless pyelonephritis or obstruction supervenes.
- Up to **30%** develop chronic renal failure in childhood/adolescence (mainly those with poor initial function). Drivers: **recurrent pyelonephritis, dysplasia, polyuria**.
- **Transplantation**: outcomes equal age-matched controls; native tract can be retained; ensure good bladder emptying first.
- **Bladder**: train all — timed, double/triple voiding, Credé/Valsalva; failure → intermittent or nocturnal CIC. Lifelong surveillance and prompt UTI treatment.
- Constipation in >half; asthma in older patients; correcting musculoskeletal problems improves quality of life most.
- **Fertility**: normal puberty, libido and erections; **azoospermia usual, no spontaneous paternity** (testicular histology, ductal atresia, prostatic hypoplasia, retrograde ejaculation). Paternity via sperm retrieval + intracytoplasmic sperm injection (ICSI). Affected females can carry normal pregnancies.

> **KEY POINT** Long-term care is a contest against **stasis**: whatever lowers residual urine and infection protects the kidneys; dysplasia and polyuria set the ceiling surgery cannot raise.

12 · MCQ ONE-LINERS

1. PBS triad: abdominal wall deficiency, bilateral intra-abdominal testes, urinary tract anomaly.
2. Main determinant of survival: renal dysplasia (plus pulmonary hypoplasia), not abdominal wall.
3. Genes implicated in multiplex families: MYOCD and FLNA (smooth muscle).
4. Degree of hydronephrosis does not correlate with dysplasia.
5. Ureter: distal worse than proximal; preserve proximal in reconstruction.
6. VUR present in ~75%.
7. Bladder: large, smooth, normal compliance, urachal pseudodiverticulum.
8. Megalourethra more common in PBS than any other syndrome.
9. Fusiform megalourethra lacks corpora cavernosa and spongiosum — more severe than scaphoid.
0. Urethral atresia is lethal unless a patent urachus decompresses the bladder.
1. Commonest orthopaedic anomaly: talipes equinovarus.
2. Lateral knee dimples: sign of oligohydramnios.
3. Woodard category I: pulmonary hypoplasia + dysplasia — supportive care.
14. Controversy over management centres on Woodard category II.
15. Pseudo-prune: delayed diagnosis, higher late renal failure.
16. Neonatal VCUG only with renal insufficiency or suspected outlet obstruction, under prophylaxis.
17. DMSA for differential function after 4–6 weeks.
18. Temporary diversion of choice: Blocksom vesicostomy with a wide stoma.
19. Reduction cystoplasty: capacity recurs; CIC empties better.
20. Orchidopexy of choice: bilateral transabdominal at ~6 months.
21. Staged Fowler–Stephens more successful than single-stage.
22. Monfort abdominoplasty preserves umbilicus in situ and allows transperitoneal surgery.
23. Randolph: transverse incision excising lower full-thickness wall.
24. Creatinine <0.7 mg/dL at 1 year predicts stable childhood renal function.
25. No spontaneous paternity reported; ICSI has achieved paternity.

45 Posterior Urethral Valves

Reading time ≈ 45 min (slow pace)

1 · DEFINITION AND CLASSIFICATION

- **Posterior urethral valves (PUV)**: commonest cause of bladder outlet obstruction in infant boys. Ablation relieves the obstruction but **not the embryological injury** to bladder and kidney → lifelong joint urology–nephrology care.
- Lower urinary tract obstruction (LUTO) ~2.2/10,000 births; PUV commonest (~1.4/10,000), then urethral atresia, then prune-belly. Commoner in Black and minority ethnic groups.
- Term coined by **Hugh Hampton Young** (1919); first endoscopic resection by Randall MCQ.

Young type	Anatomy	Status
I (95%)	Leaflets from verumontanum run distally and anteriorly, fuse in midline just proximal to external sphincter	The lesion treated in practice; obstructing
II	Folds from verumontanum to bladder neck	Non-obstructing; probably not real — secondary bladder-neck change of distal obstruction
III (5–10%)	Annular membrane with central aperture (persistent urogenital membrane); resembles congenital stricture	Obstructing; contentious but accepted

- **COPUM** (congenital obstructing posterior urethral membrane): one oblique membrane, perforated by the first catheter into apparent leaflets — provocative, unproven.
- Genetics: mostly sporadic; chromosome 11 duplications in siblings; **copy number variant burden** predicts higher nadir creatinine and progression to chronic kidney disease (CKD) stage 5 — links genotype to renal prognosis, not to presence of the valve. Whole-genome loci: **TBX5, PTK7** MCQ.

2 · PATHOPHYSIOLOGY

Bladder and urethra

- Obstruction → detrusor hypertrophy, then **fibrosis replacing smooth muscle** → reduced contractility on strip studies.
- The bladder is the **driver** of the disease: renal dysfunction, VUR and voiding dysfunction are all mediated through it. High-pressure hypertrophy that protects emptying in infancy fibroses the detrusor and causes decompensation later — so relieving the outlet does not close the cycle.
- Voiding cystourethrogram (VCUG) triad: **dilated posterior urethra, hypertrophied bladder neck, flattened verumontanum** (reflux into ejaculatory ducts) — regresses after ablation.

Upper tract and kidney

- High storage pressure is transmitted to ureter, pelvis and glomerulus. Ureteric dilation has three causes: **pressure transmission, vesicoureteral reflux (VUR) (up to 70%), polyuria**. Chronic dilation → loss of peristalsis and coaptation → stasis, infection.
- Renal injury = **obstructive uropathy** (relievable, partly reversible) + **dysplasia** (fixed at birth; sets the ceiling).
- Tubular injury + reduced medullary blood flow → lost concentrating gradient → **nephrogenic polyuria** and post-obstructive diuresis after catheterisation.
- Renin–angiotensin: plasma renin rises before creatinine; angiotensin II induces transforming growth factor-β1 (TGF-β1) → fibrosis. Angiotensin-converting enzyme inhibitors (ACEIs) reduce fibrosis experimentally and lower TGF-β1/tumour necrosis factor-α (TNF-α) clinically — possible role in selected children. TGF-β1 and TNF-α fall after successful ablation (candidate biomarkers).

- Fetal sheep obstruction models: hydronephrosis appears rapidly, but irreversible dysplasia is present by term — the window for renal rescue closes early in fetal life.

Organ / lesion	Effect	Natural history
Lung — hypoplasia	Respiratory failure	Commonest cause of neonatal death
Kidney — obstructive uropathy	Reversible insufficiency	Improves, may recur with bladder dysfunction
Kidney — dysplasia	Irreversible insufficiency	Progressive renal failure, hypertension
Kidney — tubular injury	Salt and water wasting	Nephrogenic diabetes insipidus; worsens with age
Bladder	Poor sensation, hypercontractility, low compliance → myogenic failure	Lifelong, changes with age
Ureters	Poor coaptation, stasis	Most retain hydronephrosis — **not** a sign of residual obstruction

"Pop-off" mechanisms

- **VURD (vesicoureteral reflux and dysplasia)**: unilateral high-grade VUR into a non-functioning kidney, thought to act as a pressure reservoir protecting the other kidney.
- **Refuted**: most VURD boys with normal creatinine at 2 years lose it by 8–10 years; glomerular filtration rate (GFR) was already abnormal early. Pop-off bladders have worse urodynamics and more need for clean intermittent catheterisation (CIC); the contralateral kidney often has congenital scarring.
- **Urinoma** (forniceal rupture, subcapsular or perirenal) and **urinary ascites**: protective vs harmful debate — management is the same (relieve obstruction). Large **bladder diverticula** act similarly.

> **KEY POINT** Every pop-off is a false reassurance: a normal creatinine in a growing infant hides an abnormal GFR. Follow VURD as closely as any PUV child.

3 · DIAGNOSIS

Antenatal

- Ultrasound (US): **thick-walled distended bladder, bilateral hydroureteronephrosis, "keyhole sign"** (dilated posterior urethra), **oligohydramnios**; raised renal echogenicity/cortical cysts = dysplasia. Sensitivity ~95%/specificity ~80% for LUTO, but only ~50% accurate for the cause (atresia, prune-belly, VUR, megaureter mimic).
- Fetal magnetic resonance imaging (MRI) is an adjunct (lungs, cysts). ~Half diagnosed antenatally; antenatal diagnosis has **not** been shown to improve renal outcome.
- Of postnatal diagnoses, half present as neonates and half with UTI.
- Rule: male fetus with thick bladder and bilateral dilation → postnatal US and **VCUG before discharge**.

Postnatal

- **US** first: bladder wall thick if **>3 mm distended / >5 mm empty**; perirenal fluid = urinoma.
- **VCUG** (definitive) once stable: contrast crosses an elevated bladder neck → dilated, elongated posterior urethra → **abrupt funnelling** at the valve; trabeculated bladder, diverticula; high-grade VUR in ~half. Technique: 5–8 Fr feeding tube (coudé if it curls); **withdraw the catheter during voiding** or the posterior urethra is not assessed.
- Contrast-enhanced voiding urosonography (ceVUS): radiation-free alternative in experienced hands.
- **Mercaptoacetyltriglycine (MAG3)** renogram at **2–3 months** (tubular maturity) for differential function; slow washout from a dilated system is easily misread as ureterovesical junction (UVJ) obstruction.

- Labs: first 48 h creatinine is maternal; follow trend and **nadir creatinine** (1 month, 6 months, 1 year). Plateau may not occur until late in year 1.

4 · FETAL INTERVENTION

- Decisions hinge on **oligohydramnios, degree/laterality of hydronephrosis, gestational age**. Favourable for intervention: oligo- (not an-)hydramnios, severe hydronephrosis, normal karyotype, no cortical cysts, favourable urine biochemistry.

Fetal urine (after 20 weeks)	Favourable
Sodium / chloride	**<100 / <90 mEq/L**
Osmolality	**<200 mOsm/L**
β2-microglobulin	**<6 mg/L** (most sensitive)

- Low solute = tubules still reabsorbing. **Triple serial sampling** 48–72 h apart beats a single sample; rising values are ominous. Predictive accuracy is disputed (systematic review found poor prediction).
- Society for Fetal Urology (2016) proposed a standardised LUTO severity classification. Argument for caution: some fetuses do well despite severe obstruction, others survive to birth only to fare badly.
- **Vesicoamniotic shunt**: treats oligohydramnios → may prevent **pulmonary hypoplasia**; **no renal benefit**. Nitinol (Somatex) stent dislodges less than double-pigtail (Harrison).
- **Fetal cystoscopy** with laser ablation: no randomised evidence; no better than shunting.
- Both carry **40–50%** pregnancy loss — mandatory in counselling. Deliver at a centre with neonatal surgery.
- **Counselling** by the paediatric urologist: likely diagnosis, alternative diagnoses, possible outcomes, course of pregnancy — and the start of a lifelong relationship into adulthood.

> **KEY POINT** Fetal intervention can rescue the lung, not the kidney: amniotic fluid deficiency is still in progress, but dysplasia is already established.

5 · NEONATAL PRESENTATION AND INITIAL MANAGEMENT

- Stigmata of oligohydramnios (Potter facies, club feet), weak stream, palpable bladder. **Pulmonary hypoplasia** is the commonest cause of early death and usually what delays valve surgery.

ALGORITHM
Stabilise airway/ventilation → **5–7 Fr feeding tube** per urethra (coudé, stylet-curved tip, or rectal finger guiding it anteriorly if it curls) → confirm position on US → minimal balloon fill (spasm, ureteric orifice occlusion) → creatinine, electrolytes, fluid balance → US, then VCUG → antibiotic prophylaxis → ablation when stable

- **Post-obstructive diuresis**: concentrating defect → large salt and water losses after drainage. †Monitor urine output, weight and electrolytes closely and replace losses intravenously.
- **Hyponatraemia** (often triggered by urinary tract infection (UTI)): transient pseudohypoaldosteronism → sodium supplements.
- **Urinoma/ascites** (3–10%): forniceal rupture (subcapsular fluid distorting the kidney) or transudation/bladder rupture (ascites). Leak site often unidentifiable. Resolves with bladder drainage; percutaneous tap only for respiratory distress or tense distension.
- Pulmonary hypoplasia mechanism: oligohydramnios + huge kidneys and bladder → uterine compression of fetal chest → restricted diaphragm → lungs lack the intraluminal fluid volume, pressure and flow needed for alveolar growth.
- **Delayed presentation** is common (many have a normal prenatal US): recurrent UTI, lower urinary tract symptoms (LUTS), overflow incontinence, gross haematuria, renal failure, rarely ejaculatory

dysfunction. Over 40% of ablation series present after 6 months; late presenters are **not** necessarily milder. Screen with US (thick bladder, distal ureteric dilation), confirm with VCUG.

6 · SURGERY

Primary valve ablation (preferred)

- Goal: restore urethral flow and **normal bladder cycling** — superior to passive diversion experimentally.
- Instruments: **7.5–9 Fr infant cystoscope** with holmium:YAG laser or Bugbee electrode; 9.5 Fr resectoscope with cold (Collins) knife only if the urethra allows. Blind hooks (Whitaker insulated hook) and Fogarty catheters are alternatives without endoscopes.
- Incise at **5 and 7 o'clock ± 12 o'clock** (12 alone may suffice). Valves are thin and avascular → **incise, don't resect**.
- Small urethra: vesicostomy, or **PADUA** (progressive augmentation by dilation of the urethra anterior) — up-size the urethral tube 5 → 8–10 Fr over 3–5 days.
- Catheter ≥24 h. **Repeat VCUG within 1 month**: valve gone, posterior urethral calibre and bladder wall improving.

Complication	Cause / prevention	Management
Urethral stricture	Over-resection, hot loop, oversized instrument	†Dilation / urethrotomy; urethroplasty if dense
Meatal injury	Resectoscope through small meatus	Choose smaller scope / PADUA
Residual valve	Incomplete incision	†Persistent funnel on follow-up VCUG → re-ablate
†Sphincter injury	Ablating distal to valves	Incontinence — avoid by precise incision

- Trade-off: a 7.5 Fr scope gives poor vision once the working channel is filled; the resectoscope gives excellent vision but risks meatal injury.
- **Bladder neck incision** (rationale: elevated hypertrophied bladder neck, persistent residual after ablation): no benefit over controls, risks retrograde ejaculation; botulinum toxin to bladder neck fails — not recommended.

Vesicostomy

- For: very small/premature infant, failed decompression by catheter or ablation, persistent deterioration. Lowers storage pressure; bladder still fills and empties through stoma at low leak point pressure → **does not defunctionalise**. Same final outcome as primary ablation; bladder remains usable for transplant.
- **Blocksom technique**: 2 cm transverse incision midway pubis–umbilicus → separate recti → mobilise peritoneum off dome → identify urachus, excise with small cuff of dome → detrusor to fascia, mucosa to skin. Key: **bring the dome** (not anterior wall) to skin so the posterior wall stays taut and does not prolapse.

Upper tract diversion (ureterostomy, pyelostomy)

- Theory: decompress kidneys bypassing a thick-walled UVJ. Reality: **no renal protection**; biopsies at diversion all show dysplasia; complex undiversion needed.
- Loop ureterostomy still lets the bladder fill and cycle, but prolonged defunctionalisation remains a risk.
- Residual indication: worsening function or upper tract dilation ± sepsis **despite full lower tract drainage**, after nephrology consultation.

Circumcision and nephrectomy

- UTI risk 50–60% in PUV vs ~1% in normal boys; circumcision reduces it by >80%. Offer to all boys with PUV — always before considering reimplantation.

- **Nephroureterectomy** of a VURD kidney is rarely needed (polyuria is better than anuria). If unavoidable, **keep the ureter** for later ureteral augmentation.

7 · VUR IN PUV

- VUR in 50–80%; it is **secondary** (high bladder pressure). Resolves after ablation/vesicostomy in 25–40%, more with low grade and functioning kidney. Reflux itself does not predict renal outcome.
- Reimplant only for recurrent febrile UTI despite circumcision and optimal bladder management; bladder rehabilitation must precede it.
- Treat the **bladder**: residual obstruction, anticholinergics, emptying regimens, urodynamics. Reimplantation into an unrehabilitated thick bladder → stricture and persistent VUR; endoscopic injection safer but less successful.

> **KEY POINT** VUR in valves is a pressure gauge, not a lesion: fix the pressure, not the reading.

8 · BLADDER DYSFUNCTION AND VALVE BLADDER SYNDROME

- Patterns through life (overlapping): **detrusor hyperreflexia (infancy) → improving compliance (childhood) → large hypocontractile/atonic bladder (adolescence)**.
- Continence delayed (~5.5 vs 2.9 years); history underestimates dysfunction → **uroflow and post-void residual (PVR)** routinely in every toilet-trained PUV boy. New incontinence after dryness = red flag for renal deterioration.
- **Valve bladder syndrome** (Mitchell): worsening hydroureteronephrosis and function without residual obstruction.

ALGORITHM

Compensated high-pressure voiding → fibrosis, weaker contraction → rising PVR + growing urine volume (nephrogenic polyuria) → upper tracts store urine and refill the bladder → bladder never empty → pressure back-transmitted → renal decline → more polyuria

- Driven by **polyuria, insensitivity, residual urine**, on a poorly compliant collagen-rich wall. A continuum, not inevitable — effective drainage alters the course.

Treatment ladder

1. **Behavioural**: adequate fluids, timed and double voiding, biofeedback; don't force toilet training.
2. **Drugs** (follow the contractility phase):
   - **α-blockers**: relax bladder neck/sphincter → lower PVR.
   - **Oxybutynin** 0.1 mg/kg twice daily: lowers pressure, raises capacity. **Stop if PVR/capacity rise** — myogenic failure can follow.
   - **Mirabegron** (β3-agonist): lowers overactivity, improves compliance; early safety data.
   - Logic: α-blocker targets functional bladder-neck obstruction (early high-pressure outlet phase); antimuscarinic targets the thick overactive wall; both harm once hypocontractility sets in.
3. **CIC** if PVR rises despite medication.
4. **Overnight bladder drainage**: catheter for 7–10 h nightly when dilation, UTI or function fail behavioural measures — gives the bladder and kidneys a period of genuine emptiness; improves hydronephrosis, continence, UTI.
5. **Mitrofanoff appendicovesicostomy** if urethral CIC is hard (high bladder neck, sensate urethra).
6. **Augmentation** (rare): small, high-pressure, refractory bladder. **Ureterocystoplasty preferred** (megaureter/VURD unit available; avoids mucus, acidosis, stones of ileum).

- **Videourodynamics** serially (contractility changes with age), especially when the clinical picture changes (new UTI, worsening dilation); weigh radiation and infection risk.
- Even after Mitrofanoff, ~a third still show renal deterioration — dysplasia and polyuria continue regardless of emptying.

- Augmentation is now rare because of better understanding of bladder dysfunction, behavioural therapy and timely overnight drainage.

9 · PROGNOSIS FOR RENAL FUNCTION

- Lifetime end-stage renal disease (ESRD) **20–50%**. Risk factors: age at diagnosis, dysplasia ± VUR, **nadir creatinine in year 1**, recurrent UTI, bladder dysfunction.

Nadir creatinine	ESRD risk
1 year <0.8 mg/dL	Minimal (best single predictor)
1 year >1.2 mg/dL	High
<0.4 / 0.4–0.69 / 0.7–0.99 / >1.0 mg/dL	**0% / 2% / 27% / 100%**

- **0.7 mg/dL is the watershed** — risk leaps rather than grades. 1-year value beats 1-month value; creatinine velocity over first 30 days also predicts.
- US: echogenicity, cortical cysts, lost corticomedullary differentiation, smaller **renal parenchymal area** → worse.
- Age paradox: ESRD risk higher in the post-ultrasound era and in those presenting <1 year — **survival bias** (severe cases that once died before diagnosis now survive). Late presenters may have more azotaemia, so late ≠ mild.
- VCUG ratios (posterior urethral height:width) and deep-learning US models are emerging predictors; nadir creatinine remains the standard.

10 · CKD, ESRD AND TRANSPLANTATION *→ CH 61*

- Obstructive uropathy is a leading cause of paediatric transplantation. Urologist belongs on the transplant team.
- Pre-transplant: **videourodynamics in every candidate**; optimise reservoir with CIC/overnight drainage; native nephrectomy only for proteinuria or severe polyuria.
- Thick bladder → more **ureteric obstruction**, but graft and patient survival equal to other causes if bladder managed; late graft loss linked to bladder dysfunction.
- Outcome similar whether initial treatment was ablation, vesicostomy or supravesical diversion.
- Augmentation: old dogma "always before transplant" is gone — post-transplant augmentation has similar outcomes; augmentation raises UTI risk but does not harm the graft if CIC compliance is good. Transplant into a vesicostomy-drained bladder is a safe bridge until the child can manage CIC.

11 · LONG-TERM FOLLOW-UP, SEXUAL FUNCTION AND FERTILITY

- PUV is a lifelong illness: chronic bladder dysfunction, UTI risk and renal sequelae require adult urologists conversant with valve pathophysiology after transition.
- Neurodevelopmental disorders (attention-deficit/hyperactivity disorder) more frequent → early cognitive assessment.
- Adult LUTS 2–3× population. Erectile dysfunction and infertility overall similar to population, but worse in those with incontinence or renal insufficiency.
- †Lifelong: blood pressure, creatinine, urine protein, urine culture when febrile, US, uroflow/PVR; structured transition to adult urology.

12 · RELATED URETHRAL LESIONS

- **Anterior urethral valves**: commonest congenital anterior urethral obstruction but 25–30× rarer than PUV; often with a **urethral diverticulum** (bulbar, penoscrotal or penile); distal penile bulge on voiding, urine expressed by compressing the shaft. Premature infant → vesicostomy; most → endoscopic ablation (Bugbee/laser), usually definitive; large diverticulum → open repair over

catheter. Up to 80% bladder dysfunction; most (~78%) keep normal renal function. Pretreatment azotaemia + VUR + UTI raise poor renal outcome risk 25-fold.

- **Urethral atresia**: membrane at distal prostatic urethra; survival only with decompression (urachus, shunt); vesicostomy, then PADUA.
- **Urethral duplication** (Effmann I blind, II complete, III caudal duplication; "Y"-type to perineum): sagittal; the **ventral** channel holds sphincter and verumontanum → move it to the glans. Type I blind accessory → observe or Bugbee coagulation; two glanular openings → incise septum + dorsal urethroplasty; "Y"-type → staged preputial/buccal flaps; meatus in anal sphincter → anterior sagittal trans-ano-rectal approach (ASTRA).
- **Urethrorrhagia**: peripubertal boys, blood spots on underwear after clear urine (lesion distal to external sphincter). Mechanism: incomplete sphincter relaxation (or meatal stenosis) → turbulent flow → negative intraluminal pressure → mucosal sinus engorgement → bleeding. Up to 92% resolve → bowel/bladder history, US, uroflow, PVR; treat voiding dysfunction. Cystoscopy only if atypical (stricture symptoms, heavy bleeding) — stricture eventually found in 14–28%, and instrumenting inflamed urethra may itself provoke stricture.
- Anorectal malformation: most boys have a rectourinary fistula; US and VCUG for associated anomalies; at posterior sagittal anorectoplasty catheterise first, excise fistula flush with urethra, catheter ≥1 week.

13 · LANDMARK TRIALS

Trial	Design	Result	Take-home
PLUTO	Randomised controlled trial (RCT): vesicoamniotic shunt vs conservative in LUTO	Trend to better 28-day survival; very poor long-term survival and renal function in both	Shunt may help lungs, not kidneys
Circumcision RCT	Circumcision + prophylaxis vs prophylaxis alone	Febrile UTI 3% vs 49% over 2 years	Circumcise boys with PUV
Oxybutynin RCT	Oxybutynin to toilet training vs observation post-ablation	Better hydronephrosis/VUR; no change in creatinine, UTI, continence	Selective, monitored use

14 · MCQ ONE-LINERS

1. Commonest cause of infant male bladder outlet obstruction: PUV.
2. Young type I (95%): leaflets from verumontanum fusing anteriorly.
3. Type II is non-obstructing and probably not a true entity.
4. Type III: annular membrane, persistent urogenital membrane.
5. Antenatal US sign of dilated posterior urethra: keyhole sign.
6. Commonest cause of early death in PUV: pulmonary hypoplasia.
7. Bladder wall thick: >3 mm full, >5 mm empty.
8. Definitive diagnosis: VCUG — withdraw catheter during voiding.
9. Renogram timing: MAG3 at 2–3 months.
0. Neonatal creatinine reflects mother for first 48 hours.
1. Favourable fetal urine: Na <100, Cl <90, osmolality <200, β2-microglobulin <6 mg/L.
2. Most sensitive fetal urine marker: β2-microglobulin.
13. Vesicoamniotic shunting: possible survival/lung benefit, no renal benefit.
14. Pregnancy loss with fetal intervention: 40–50%.
15. Neonatal hyponatraemia: transient pseudohypoaldosteronism.
16. Ablation incisions: 5 and 7 o'clock ± 12 o'clock.
17. Hot-loop resection risks urethral stricture.
18. Repeat VCUG within 1 month of ablation.
19. PADUA: progressive urethral dilation 5 → 8–10 Fr over 3–5 days.
20. Vesicostomy: bring the dome to skin to prevent prolapse.
21. Vesicostomy does not defunctionalise the bladder.
22. High diversion: no renal benefit; biopsies show dysplasia.
23. Circumcision reduces UTI risk by over 80% in PUV.
24. VURD does not protect long-term renal function.
25. VUR in PUV: treat bladder, rarely reimplant.
26. Valve bladder syndrome coined by Mitchell.
27. Overnight catheter drainage: 7–10 hours nightly.
28. Stop oxybutynin if residual and capacity rise.

29. Preferred augmentation in PUV: ureterocystoplasty.
30. Nadir creatinine <0.8 mg/dL at 1 year: minimal ESRD risk.
31. Lifetime ESRD in PUV: 20–50%.
32. Every transplant candidate: videourodynamics first.
33. Duplication: ventral urethra is functional.
34. Anterior urethral valves: often with urethral diverticulum.
35. Urethrorrhagia: conservative; up to 92% resolve.
36. Upper tract dilation persisting after ablation is expected, not residual obstruction.

46 Neuromuscular Dysfunction of the Lower Urinary Tract in Children

Reading time ≈ 29 min (slow pace)

1 · DEFINITION AND CAUSES

- **Neuromuscular (neurogenic) lower urinary tract (LUT) dysfunction**: bladder and/or sphincter dysfunction from aberrant innervation — congenital or acquired, central or peripheral, transient or permanent.
- Commonest cause: **neural tube defect (NTD)**, used almost synonymously with spina bifida (SB). With survival now ≈90% to 10 years, **renal failure** becomes a leading cause of late morbidity/death; **10%** of SB deaths are urological.
- Pivotal advance: **clean intermittent catheterisation (CIC)**, Lapides 1971 [MCQ].

Congenital	Acquired
Open NTD (myelomeningocele); occult NTD (lipomeningocele, tethered cord); sacral agenesis; anorectal malformation (ARM)	Brain: cerebral palsy (CP), tumours, infarcts. Cord: trauma, transverse myelitis. Pelvis: extensive pelvic surgery

2 · MYELODYSPLASIA (NTD)

Classification and prevention

- **Myelomeningocele**: open; cord, roots and meninges protrude; most common and most severe; usually lumbosacral (lumbar + lumbosacral + sacral >90%). **Meningocele**: meninges only. **SB occulta**: closed, up to 15% of population, usually benign.
- Neural tube closes by **day 35** (from day 18). Prenatal detection: maternal α-fetoprotein + ultrasound (US).
- **Folic acid** cuts NTDs 50–70%: **400 µg/day** for all women of child-bearing age (tube closes before pregnancy is recognised), from ≥4 weeks before conception through the first month; **4–5 mg/day** for women with SB.
- Recurrence **5%** after one affected child; offspring risk 3.7% if one parent has SB (≈15% if both).
- Almost all open NTDs have **Chiari II malformation** and most hydrocephalus (ventriculoperitoneal shunt or endoscopic third ventriculostomy).
- **The bony level differs from the neurological level by 1–3 segments, and LUT function cannot be predicted from bony level or leg function** → urodynamics defines every bladder.

> KEY POINT **Management of Myelomeningocele Study (MOMS)**: prenatal repair (<26 weeks) reduced death or shunt need (relative risk 0.70) and improved motor/cognitive scores at 30 months, at the cost of prematurity (34 vs 37 weeks), uterine-scar thinning/dehiscence (⅓) and more re-tethering. Rationale: "two-hit" in-utero injury.

Pathophysiology

- Detrusor **overactive** (neurogenic detrusor overactivity, NDO — small, high-pressure bladder from pathological remodelling) or **underactive** (retention, overflow, urinary tract infection [UTI]). Outlet **competent, incompetent** (open bladder neck, denervated sphincter) or **dyssynergic**.
- **Detrusor–sphincter dyssynergia (DSD)**: sphincter fails to relax during a voiding/overactive contraction — the presumed driver of remodelling and renal risk. Diagnosed by electromyography (EMG), not fluoroscopy.

KEY POINT Every pattern = detrusor (over/under) × outlet (competent/incompetent/dyssynergic). The dangerous combination is **overactive or poorly compliant detrusor against a closed or dyssynergic outlet**: high storage pressure → vesicoureteric reflux (VUR) and hydroureteronephrosis. An incompetent outlet "pops off" — wet but kidney-safe.

Newborn evaluation

ALGORITHM
Birth → renal–bladder US before discharge + post-void residual, PVR (decides need for CIC) → serum creatinine after first week → urodynamics once stable, **safe from ≈3 months** (within 3 months: ICCS, SBA, EAU) → cystography (ideally videourodynamics; or voiding cystourethrogram, VCUG) → ± baseline dimercaptosuccinic acid (DMSA) scan (EAU) → US at 6, 12, 18, 24 months then yearly.

- Guidelines: International Children's Continence Society (ICCS 2012), British Association of Paediatric Urologists (BAPU 2016 — most selective, "no consensus" on early CIC), Spina Bifida Association (SBA 2018), European Association of Urology (EAU 2022).
- UMPIRE (Centers for Disease Control and Prevention [CDC] Urologic and Renal Protocol for the Newborn and Young Child with SB) baseline: US normal 56%, mild hydronephrosis 40%, moderate–severe 4%; dilating VUR (grade III–V) **7%**; DMSA defects ≈8%.
- Newborn urodynamics: most have detrusor contractions; >⅓ acontractile.
- **Cystatin C** beats creatinine in SB: low muscle mass keeps creatinine "normal" despite falling glomerular filtration rate (GFR). Gold standard = nuclear clearance study.
- High-grade VUR acts as a pop-off → urodynamic pressures underestimate risk.

Urodynamic risk factors

Category	Definition
Hostile / high	Detrusor leak point pressure (DLPP) or end-filling pressure **≥40 cmH_2O**; NDO with DSD (DSD removed in UMPIRE revision — poor agreement)
Intermediate	NDO, reduced compliance, pressure 25–39 cmH_2O
Abnormal but safe	Pressure <25 cmH_2O
Normal	Normal capacity, compliance >15 mL/cmH_2O, no NDO/DSD, minimal residual

- **McGuire (1981)**: DLPP >40 cmH_2O predicts VUR and upper-tract deterioration. Poor compliance independently predicts renal injury.

Tethered cord

- Conus fixed abnormally low (below L1–L2); axial growth stretches the cord → gait change, leg/perineal sensory loss, LUT deterioration. Secondary tethering after myelomeningocele repair is common (more after fetal repair).
- † Suspect re-tethering with new wetting, rising pressures or reduced compliance, new UTIs, worsening scoliosis, back/leg pain or orthopaedic deformity.
- Urodynamics **before untethering and 3–4 months after**; needle EMG clarifies re-tethering when post-repair scarring makes magnetic resonance imaging (MRI) equivocal.

3 · PROACTIVE (EARLY) MANAGEMENT

Priorities in order: **kidneys → continence → UTI avoidance → sexual function and fertility**.

- **Early intervention = CIC + antimuscarinic** (Bauer, 1980s) for hostile dynamics: DSD, poor compliance, high outlet pressure. All paediatric urology societies now endorse it for the hostile neonatal bladder; it gives **no benefit to the child who empties spontaneously**.
- CIC prevents overdistension and **bypasses DSD**; started in infancy, families accept it more easily. If CIC fails to clear hydronephrosis → overnight indwelling catheter drainage.

- Do **not** give antibiotic prophylaxis on CIC; treat bacteriuria only if symptomatic (fever, pain, new leakage).

Outcome	Expectant	Early CIC + antimuscarinic
Hydroureteronephrosis	18–80%	0–15%
Renal scarring	27–50%	4–12%
End-stage renal disease (ESRD)	≈30%	0–1.6%
Augmentation	—	18–24% fewer

> **KEY POINT** Why proactive won: it treats the cause (pressure, failure to empty) before the consequence (reflux, dilatation, scarring, renal failure). Waiting for hydronephrosis means waiting for damage.

4 · MANAGEMENT LADDER

ALGORITHM
CIC + antimuscarinic (± β3-agonist) → intradetrusor botulinum toxin → (vesicostomy as temporary bridge if CIC impossible) → augmentation cystoplasty ± continent catheterisable channel ± outlet procedure → *Ch 48*

Pharmacotherapy

- **Antimuscarinics** (first-line; detrusor M2 most numerous, M3 mediates contraction): ↑ capacity, ↑ volume at first contraction, fewer leaks. Side effects: dry mouth, constipation, flushing, blurred vision. No cognitive effect (randomised double-blind trial).
- Licensed in children (US Food and Drug Administration): **oxybutynin** (≥5 y; 0.1–0.6 mg/kg/day, max 5 mg/dose; oral, patch, **intravesical** — no first-pass, fewer side effects), **solifenacin** (≥2 y, weight-banded suspension), **fesoterodine** (≥6 y, >25 kg), **mirabegron** (β3-agonist, ≥3 y; may cause hypertension).
- Trospium (quaternary amine): poor brain penetration → few central effects.
- **α-Sympathomimetics** (ephedrine, pseudoephedrine) raise bladder-neck resistance for low-outlet incontinence; unlicensed; only a minority respond.

Botulinum toxin A

- Blocks acetylcholine (and ATP, substance P) release — acts on motor and sensory arms.
- Technique: general anaesthesia, rigid cystoscopy, **≈10 U/kg (max 360 U)**, ≈30 sites, **trigone spared**. Usual minimum age 2 y.
- After failed CIC + drugs: continence 65–87%, lower pressure, better compliance; lasts 6–10 months; repeat injections stay effective, no fibrosis or tachyphylaxis. Poor pre-existing compliance predicts poor response.

Surgery

- **Vesicostomy**: temporary; lowers pressure, reverses hydronephrosis and VUR, reduces UTI in infants whose families cannot manage CIC — a bridge, not definitive.
- **Augmentation cystoplasty** (ileum preferred) creates the definitive low-pressure reservoir for the bladder refractory to drugs and toxin; add a channel if urethral CIC is difficult and sling / bladder-neck reconstruction / artificial sphincter for sphincter weakness → *Ch 48*. Follow-up: B12 yearly from 2 y after ileal use; cystoscopy 10–15 y after.

VUR in the neurogenic bladder

- Secondary to hostile dynamics; treat the bladder first and 30–55% resolve. Grade I–III with complete emptying → prophylaxis; grade IV–V or poor emptying → CIC; poor compliance → add antimuscarinic.
- **Avoid Credé** with a reactive sphincter — reflex sphincter contraction raises voiding pressure and worsens VUR.

- Antireflux surgery indications as for normal bladders (breakthrough febrile UTI, persistent hydroureteronephrosis despite low pressure and emptying). With CIC, reimplantation succeeds ≈95% (Jeffs). At augmentation, **reimplant grade ≥III** (persists in 20–47% otherwise).

5 · BOWEL MANAGEMENT †

- † Neurogenic bowel (shared sacral innervation) causes constipation and faecal incontinence; social continence is as important as urinary continence.
- † Ladder: diet/fibre and timed toileting → oral laxatives ± suppositories → retrograde transanal irrigation/enemas → **antegrade continence enema** (Malone appendicostomy, often at augmentation).
- † Treat constipation before and after reconstruction — loaded bowel worsens bladder storage and UTI.

6 · LATEX ALLERGY †

- † SB children are at high risk of latex sensitisation from repeated early mucosal exposure (surgery, catheters) → anaphylaxis during surgery.
- † Latex-free environment **from birth** — gloves, catheters, theatre. Latex-free condoms are mandatory with latex allergy or avoidance.

7 · SEXUAL FUNCTION AND TRANSITION

- Function is **inversely proportional to neurological level**; most adults report sexual interest.
- **Central precocious puberty**: ≈50% of girls, 10–30% of boys (hydrocephalus, Chiari II).
- **Males** (sacral cord): erectile dysfunction (ED) ≈75% of adults; ejaculation 40–75% (some retrograde); lesion S1 or below → usually normal reproductive function. **Sildenafil** effective in ≈80% (randomised placebo-controlled trial).
- **Females** (hormonal control spared): 70–80% conceive; caesarean and late incontinence common. Avoid combined oestrogen pill (thrombosis with immobility) and intrauterine device (pelvic infection).
- Barriers: parental overprotection, social isolation. Offer SB-specific sexual education.
- † Plan structured **transition to adult urology** in adolescence: self-catheterisation independence, surveillance of reconstruction, fertility counselling.

8 · OTHER CAUSES

Sacral agenesis

- Absent sacral segments; † associated with maternal diabetes. Child may walk normally — look for **flat buttocks, short gluteal cleft**, absent palpable sacrum; diagnose on plain radiograph.
- Radiographic level does not predict bladder; ≈⅓ present late (incontinence, UTI). Urodynamics: areflexic bladder with weak sphincter, or NDO + DSD.

Spinal cord injury (SCI)

- Rare; six-fold commoner in adolescents.
- **Spinal shock**: underactive bladder, non-reactive sphincter → CIC. After ≈3–4 months: **suprasacral → NDO + DSD**; cauda equina → little recovery. Urodynamics after shock resolves.
- **Autonomic dysreflexia** (lesion **≥T6**): bladder filling → unopposed sympathetic surge → severe hypertension, stroke, seizures. Monitor blood pressure and have α-blockers ready during urodynamics/cystography.

Transverse myelitis

- Immune inflammatory myelopathy; acute retention >95%; treat with steroids, plasma exchange, intravenous immunoglobulin.

- Acute: areflexia ≈70% → after shock: NDO 59–90%, poor compliance ≈47%, DSD 17–80%. 57–73% need CIC.
- Examination, symptoms and ambulation do not predict upper-tract risk; **late CIC → poorer compliance** → early urodynamics and CIC.

Cerebral palsy

- Upper motor neuron, coordinated: continence achieved but late; incontinence and urgency; NDO 30–61%, DSD ≈12%; one-third normal urodynamics. Upper-tract risk **<5%**; urodynamics only if conservative therapy fails or DSD symptoms/recurrent UTI.

Anorectal malformation and pelvic surgery

- ARM: cord anomaly (tethering, fatty filum) in 18–50%, more with high lesions; skin markers rare → **spinal US before 3 months**. Neuropathic bladder comes from the **cord, not the posterior sagittal anorectoplasty (PSARP)**. Urodynamics before and after pull-through.
- Currarino syndrome (ARM + sacral defect + presacral mass): tethered cord >80%.
- Pelvic splanchnic (parasympathetic) injury → underactive bladder; hypogastric (sympathetic) → poor compliance, incompetent bladder neck. Sacrococcygeal teratoma (35–45% neuropathic) → urodynamics before and after resection; pelvic rhabdomyosarcoma survivors (esp. after radiotherapy) need close surveillance.

> **KEY POINT** Lesion level predicts pattern: **suprapontine** (CP) → coordinated overactivity, low renal risk; **suprasacral cord** (SCI, myelitis, tethering) → NDO + DSD, high risk; **sacral/peripheral** (cauda equina, pelvic surgery) → underactivity. Myelodysplasia is the exception — mixed and unpredictable.

9 · LANDMARK TRIALS

Trial	Design	Result	Take-home
MOMS	Randomised controlled trial (RCT), prenatal vs postnatal repair	Less shunting, better motor score; more prematurity, uterine dehiscence	Fetal repair option in selected centres
McGuire 1981	Myelodysplasia cohort	DLPP >40 cmH_2O → upper-tract damage	40 cmH_2O = safety line
Sildenafil in SB	Double-blind placebo RCT	≈80% improved erections	First-line for ED

10 · MCQ ONE-LINERS

1. CIC introduced by Lapides in 1971.
2. Commonest open NTD: myelomeningocele, usually lumbosacral.
3. Folic acid 400 µg/day; 4–5 mg/day if woman has SB.
4. Offspring risk with one affected parent: 3.7%.
5. Almost every open NTD has Chiari II malformation.
6. First urodynamics in myelomeningocele: within ≈3 months.
7. Dilating VUR in SB newborns: 7%.
8. DLPP ≥40 cmH_2O = hostile bladder.
9. Cystatin C preferred to creatinine because of low muscle mass.
10. DSD diagnosed by EMG, not fluoroscopy.
11. Early intervention = CIC + antimuscarinic.
12. Expectant management: ESRD ≈30%; early intervention 0–1.6%.
13. No antibiotic prophylaxis for children on CIC.
14. Detrusor contraction mediated by M3 receptors.
15. Mirabegron approved from age 3; may cause hypertension.
16. Botulinum toxin dose ≈10 U/kg, max 360 U, trigone spared.
17. Vesicostomy is a temporary bridge.
18. Credé is harmful with a reactive external sphincter.
19. Reimplant grade ≥III VUR at augmentation.
20. Central precocious puberty affects ≈50% of girls with SB.
21. Normal male reproductive function usually if lesion at S1 or below.
22. Avoid intrauterine device and combined pill in women with SB.
23. Autonomic dysreflexia: lesions at or above T6.
24. Transverse myelitis: acute retention in >95%.
25. CP upper-tract deterioration <5%.

16. ARM bladder dysfunction stems from cord anomaly, not PSARP.

47 Functional Disorders of the Bladder and Bowel in Children

Reading time ≈ 45 min (slow pace)

1 · NORMAL BLADDER AND BOWEL FUNCTION

- **Mature lower urinary tract (LUT) function = storage** (normal compliance, no detrusor overactivity, outlet closed) + **voluntary emptying** (coordinated detrusor contraction with relaxed smooth and striated sphincters).
- **Infancy**: voiding pressures often **>100 cm H_2O** (boys > girls) and interrupted voiding are normal, disappearing by **~2 yr**; infants arouse before voiding. Dyscoordination beyond toddlerhood is therefore **dysfunction, not immaturity**.
- **Rectum**, like the bladder, is a cloacal derivative and compliant store. Distension → **recto-anal inhibitory reflex (RAIR)**: internal sphincter (smooth, autonomic) relaxes to "sample" contents while the external sphincter (striated) contracts to keep continence.
- **Toilet training**: bowel → day bladder → night bladder. Structured and readiness-based methods reach the same end-point; **late initiation** is associated with incontinence, enuresis, urinary tract infection (UTI) and stool withholding.

> **KEY POINT** Bladder and bowel are one functional unit: shared pelvic space ("zero-sum" volume), and afferents converging in the same dorsal root ganglia (cross-organ sensitisation). A loaded rectum provokes overactivity and poor emptying; treating constipation improves lower urinary tract symptoms (LUTS); anticholinergics given for the bladder can recreate the constipation that drove the symptoms.

2 · DEFINITIONS — INTERNATIONAL CHILDREN'S CONTINENCE SOCIETY (ICCS)

Bladder and bowel dysfunction (BBD) = LUT dysfunction plus bowel dysfunction (constipation and/or faecal incontinence) in a neurologically normal child.

Group	Symptom	ICCS definition
Storage	Increased frequency	**≥8 voids/day**
	Decreased frequency	**≤3 voids/day**
	Urgency	Sudden immediate need to void
	Nocturia	Waking at night to void
Voiding	Hesitancy	Difficulty initiating stream when ready
	Straining	Valsalva to initiate and maintain stream
	Intermittency	Non-continuous stream
	Retention	Cannot void despite effort; distended bladder
Other	Holding manoeuvres; feeling of incomplete emptying; recurrent UTI	Incomplete-emptying sensation counts even if postvoid residual (PVR) is minimal; recurrent UTI is itself a symptom of BBD

- **Daytime urinary incontinence (DUI)**: leakage while awake, age **≥5 yr**. **Enuresis**: intermittent leakage while asleep, ≥5 yr — **monosymptomatic (MNE)** = no other LUTS; **non-monosymptomatic (NMNE)** = with any LUTS.
- **Faecal incontinence**: **retentive** (majority; overflow of loose stool around a rectal faecal mass) vs **functional non-retentive faecal incontinence (FNRFI)** (~1 in 5; no constipation; up to half also wet).
- † **Expected bladder capacity (EBC)** = (age in years + 1) × 30 mL (ICCS, to ~12 yr).

Daytime LUT conditions

Condition	Definition / mechanism	Key features
Overactive bladder (OAB)	Urgency, usually with frequency and nocturia, ± incontinence, without UTI or other obvious pathology	Urgency incontinence, holding manoeuvres; storage phase problem
Dysfunctional voiding	Habitual contraction of urethral sphincter/pelvic floor **during voiding** in a neurologically intact child; staccato flow	Learned behaviour; high PVR, UTI, secondary vesicoureteral reflux (VUR)
Underactive bladder	Must raise intra-abdominal pressure to initiate, maintain or complete voiding	Infrequent voiding, interrupted flow, PVR; often overlaps dysfunctional voiding
Voiding postponement	† Habitual deferral of micturition with holding manoeuvres until urgency incontinence	† Low voiding frequency; strong link with behavioural comorbidity (oppositional defiant disorder)
Giggle incontinence	Extensive emptying during/immediately after laughing; LUT normal otherwise	Distinguish from laughter-provoked leak in OAB, vaginal reflux, postponement
Vaginal (urethrovaginal) reflux	Vagina fills retrograde during voiding	Prepubertal girl; small leak just after voiding or on cough
Extraordinary daytime urinary frequency	† Toilet-trained child voiding very often (≥ hourly) with small volumes, no incontinence or nocturia	† Benign, self-limiting (months); exclude UTI and diabetes; reassure
Hinman syndrome (non-neurogenic neurogenic bladder)	End stage of dysfunctional voiding	Trabeculated bladder, secondary VUR, hydronephrosis, renal damage, hypertension

> **KEY POINT** **Trap:** dysfunctional voiding (neurologically intact) ≠ detrusor–sphincter dyssynergia (DSD; same discoordination with a neurological lesion). "Voiding dysfunction" is an old umbrella term — use "LUT dysfunction" (storage + voiding). Urgency incontinence does not by itself mean OAB: a voiding postponer also leaks with sudden urge, and treating him as OAB (anticholinergics) helps briefly then worsens emptying.

3 · CLINICAL PRESENTATIONS AND ASSOCIATIONS

- **Acute urinary retention** (peak 2–4 yr): usually genital inflammation, instrumentation or **faecal impaction**; exclude tumour and neurological causes. If comfortable → **analgesia, sitz bath and/or enema before catheterisation** — retention is thought to be a reflex to a pelvic noxious stimulus, and a catheter adds another.
- **Functional constipation**: ~14% worldwide; peaks 2–3 yr around toilet training. **Withholding cycle**: painful stool → withholding → more water absorbed → harder stool → more pain. Organic differential: anorectal malformation, Hirschsprung disease, coeliac disease, cystic fibrosis, hypothyroidism, hypercalcaemia, hypokalaemia, spinal causes, drugs.
- **Abdominal pain**: constipation causes ~half of acute presentations. **Alarm features** (weight loss, gastrointestinal bleeding, persistent fever, severe diarrhoea, vomiting) → gastroenterology.
- **Genital pain** is often referred from the bladder (boys grab the penile tip) — treat the dysfunctional organ, not the painful site.
- **Neurodevelopmental disorders** — attention-deficit/hyperactivity disorder (ADHD), autism spectrum disorder (ASD), intellectual disability (ID) — have the highest BBD rates and poorer behavioural response. Treating ADHD improves enuresis (not constipation); antipsychotics (risperidone) cause incontinence. Adapt urotherapy (short instructions, supervision).
- Anxiety, depression and stressful events associate with BBD. Quality of life is impaired and improves even with education alone — "wait to outgrow it" is not neutral. Childhood BBD predicts

adult OAB, incontinence and interstitial cystitis/bladder pain syndrome (IC/BPS) → plan transition.

4 · EVALUATION

Most BBD is functional; history and examination exist mainly to **exclude organic disease**. Children are often over-investigated.

Warning signs (licence further investigation)

Finding	Points to
Blood in stool without fissure, fever, vomiting, weight loss, severe distension	Organic gastrointestinal/systemic disease
Failure to thrive, raised blood pressure	Chronic kidney disease (CKD)
Lumbosacral dimple, hair tuft, vascular lesion, asymmetric gluteal cleft	Occult spinal dysraphism → spinal magnetic resonance imaging (MRI)
Weak/numb legs, abnormal gait; absent anal or cremasteric reflex	Neurogenic bladder/bowel
Abnormal anal position or small anus	Anorectal malformation
Urinalysis: low specific gravity, glucosuria, proteinuria	Diabetes insipidus, diabetes mellitus, CKD
Girls: constant dribble despite normal voiding	Ectopic ureter

History

- Ask **concretely** ("how many stools this week?", Bristol pictures) — children cannot answer "are you constipated?". Ask about bowels in **every** child with LUTS; families volunteer only the most bothersome symptom.
- **3-day bladder diary** (frequency, volumes, leaks, intake) and **bowel diary** (frequency, Bristol type, accidents, diet).
- Review prior treatment (laxatives without clean-out, alarm expected to work overnight), bother to the child, school toilet access, neurological history (tethered cord, pelvic surgery).

Bristol stool scale: 1–2 hard lumps/lumpy sausage (constipation) · 3–4 cracked/smooth sausage (normal) · 5–7 soft blobs → watery.

Examination

- Growth, behaviour, gait; back, lower limb neurology, abdomen (faecal mass, bladder); anogenital skin, anal position, fissure/skin tags. **Gush on separating labia** = vaginal reflux. Routine digital rectal examination is **not** needed to diagnose constipation.

Investigations (stepwise, non-invasive)

- **Symptom scores**: Dysfunctional Voiding Symptom Score (DVSS), Vancouver score (includes bowel). **ICCS response**: complete 100%, partial 50–99%, none <50% reduction.
- **Urinalysis** — false positives common. Specific gravity >1.005 makes diabetes insipidus unlikely. Persistent dipstick protein ≥1+ after urotherapy → first-morning protein:creatinine ratio (>**0.2 mg/mg** = abnormal). Pyuria/haematuria in girls may be vulvitis contamination.
- **PVR by bladder scan** within 5 min of voiding, bladder neither <50% nor >115% EBC. Renal ultrasound (US) not routine; DVSS ≥12 + UTI history predicts abnormal renal–bladder US. Thick bladder wall is common and nonspecific.
- **Uroflowmetry** (ideally 3 flows, volume >50% EBC): bell (normal), tower (OAB), staccato (dysfunctional voiding), interrupted (underactive), plateau (obstruction).
- **Pelvic floor electromyography (EMG)** with flow: active EMG during voiding = dysfunctional voiding; separates sphincter activity from abdominal straining. **EMG lag time <2 s** (pelvic floor quiet → flow) = detrusor already contracting → detrusor overactivity.
- **Invasive (video) urodynamics**: not routine in neurologically intact children; reserved for treatment failure or suspected obstruction/upper-tract risk.

Bowel — Rome IV functional constipation (child ≥4 yr)

ALGORITHM
≥2 of the following, ≥1/week for ≥1 month, IBS criteria not met: ≤2 defecations in toilet per week · ≥1 faecal incontinence episode per week · retentive posturing/volitional withholding · painful or hard stools · large rectal faecal mass · large-diameter stools that may block the toilet

- Abdominal pain is **not** a criterion; if pain persists after constipation resolves, think IBS.
- No routine laboratory tests (thyroid, coeliac, calcium) or **plain abdominal radiograph** (poor reliability) — X-ray only if impaction suspected and examination impossible, or in refractory cases. Rectal diameter **≥30 mm** on transabdominal US suggests constipation (European Association of Urology/European Society for Paediatric Urology, EAU/ESPU), but gastroenterology guidelines advise against routine US.
- Refractory cases: colonic transit study (radio-opaque markers), contrast enema, **anorectal manometry** (absent RAIR → rectal biopsy for Hirschsprung disease), colonic manometry (before surgery).

5 · MANAGEMENT OF DAYTIME INCONTINENCE — THE ALGORITHM

ALGORITHM
History + examination + urinalysis + PVR ± uroflow → exclude neurogenic (urodynamics, spinal MRI), diabetes, CKD, anatomical causes → screen and optimise neurodevelopmental disorder → **treat constipation first** → night-only symptoms? → treat as MNE → otherwise **standard urotherapy** (≥4 wk) → persisting → re-evaluate → **specific urotherapy** by diagnosis → refractory → third-line (botulinum toxin, neuromodulation)

Standard urotherapy (first line)

A structured, cognitive-behavioural programme — not simply "void more often". It teaches how, when and how often to void.

- **Elements**: information, instruction, lifestyle (fluids, diet, regular emptying), registration (diaries as feedback), support/follow-up.
- **Voiding posture**: hips and knees ~90°, feet supported (foot stool/seat insert), diaphragmatic breathing → pelvic floor relaxation.
- **Timed voiding** 6–7 times/day, anchored to events (meals, school recess) and a timer watch rather than nagging; school voiding plan.
- **Rewards for adherence, not dryness** — the child controls adherence but not the leaks.
- Resolves DUI in more than half — far above spontaneous cure.

Specific urotherapy and drugs by condition

Dysfunctional voiding — a learned outlet behaviour acting as a functional obstruction: hesitancy, abnormal stream, high PVR, thick wall/diverticula → proximal urethral dilatation, detrusor hypertrophy and overactivity → poor compliance → decompensation → secondary VUR, hydronephrosis, hypertension (Hinman). Diagnose by **staccato flow + active pelvic floor EMG**; add renal US, spinal MRI or cystoscopy as indicated.

Renal risk	Management
High (hydronephrosis, VUR, rising creatinine)	Video urodynamics; lower storage pressure; empty past the sphincter with clean intermittent catheterisation (CIC)
Low	Retrain the outlet: urotherapy (posture, breathing), double voiding, treat constipation, short-course antibiotic prophylaxis for recurrent UTI

- **Biofeedback = first-line specific therapy**: child watches own flow curve/perineal EMG and learns to relax the pelvic floor; vs urotherapy — fewer UTIs, better flow, normalised EMG.
- **α-blockers** (doxazosin): reduce PVR, including after biofeedback failure (lower-grade evidence).

- **Intrasphincteric onabotulinumtoxinA** (50–300 units) for resistant cases: lowers PVR, incontinence, UTI.

Overactive bladder — drugs only after urotherapy and bowel management.

Agent	Dose / point
Oxybutynin (non-selective; tertiary amine)	Immediate release (IR) **0.3–0.6 mg/kg/day** in 2–3 doses (max 15 mg); extended release (ER) 5–20 mg/day; patch 3.9 mg/day. Only United States-approved drug for paediatric non-neurogenic OAB (on observational data); liquid available. Dry mouth, flushing, constipation, cognitive effects. ER coating must not be crushed.
Tolterodine	1–2 mg twice daily or ER 2–4 mg; bladder-selective. ER randomised controlled trial (RCT) negative from under-dosing children >35 kg
Solifenacin (M3-selective)	2.5–10 mg once daily; long half-life — tablet can be split. Positive phase III RCTs (LION, LEOPARD)
Trospium (quaternary amine)	Does not cross blood–brain barrier; needs empty stomach
Mirabegron (β3-agonist)	Relaxes detrusor during storage, no anticholinergic effects; ER suspension; 25–50 mg if >35 kg. Approved in children ≥3 yr for **neurogenic** detrusor overactivity only; ≈ solifenacin in non-neurogenic OAB; safe add-on to antimuscarinic

- **Before and during antimuscarinics**: check PVR and stools — the drug can cause retention and constipation, which recreate LUTS.
- **Intradetrusor onabotulinumtoxinA** for refractory OAB: **5 units/kg, max 100 units** (higher doses no better, higher PVR); effect ~12 months; retention needing CIC is rare — counsel anyway.

Neuromodulation	Technique	Verdict
Parasacral transcutaneous electrical nerve stimulation (TENS)	Surface electrodes over S2–S3	More complete cure than oxybutynin; true-sham RCT negative → possible placebo effect
Transcutaneous / percutaneous tibial nerve stimulation (TTNS/PTNS)	PTNS: 34-gauge needle ~5 cm above medial malleolus; correct site → great-toe flexion	Conflicting/weaker than parasacral TENS in children
Sacral neuromodulation (SNS)	Implanted lead	Off-label in children; helps bladder and bowel; lead migration, malfunction, infection, revision

Other conditions

- **Underactive bladder**: timed and double voiding, biofeedback (better than urotherapy alone — many actually have dysfunctional voiding); interferential electrical stimulation improves flow; CIC if PVR persistently high (often poorly tolerated by sensate children).
- **Voiding postponement**: timed voiding, behavioural/psychological therapy of comorbidity; † avoid anticholinergics (they further reduce voiding frequency).
- **Giggle incontinence**: **methylphenidate** (central; effective but relapse on stopping, irritability, sleep disturbance) or **biofeedback** pelvic floor training (more durable).
- **Vaginal reflux**: positional — void with legs widely abducted, spine straight (sit backwards on the toilet), cough at end of voiding.
- **Extraordinary daytime frequency**: † reassurance; resolves spontaneously.

6 · NOCTURNAL ENURESIS

- Prevalence **5–10% at 7 yr**; ~15% resolve each year †; persists in ~2% of adults. Familial (loci on chromosomes 6 and 13).
- **Pathophysiology triad** — mismatch of: (1) **nocturnal polyuria** (relative antidiuretic hormone (ADH) deficiency), (2) **reduced nocturnal bladder capacity / nocturnal detrusor overactivity**, (3) **failure of arousal** from sleep. Each treatment targets one limb.

- **NMNE**: treat bothersome daytime LUTS (and constipation) first; minor daytime symptoms need not delay enuresis treatment. **MNE**: active treatment from **~6 yr**.
- Ineffective: night waking, fluid restriction alone, rewards, punishment, protective pants.

	Enuresis alarm	Desmopressin
Target	Arousal (conditioning)	Nocturnal urine output (ADH analogue)
Use	Every night; parent must wake child; motivated family; continue if improving after a few weeks until **14 consecutive dry nights**	**0.2–0.4 mg** tablet or 120–240 µg lyophilisate ~1 h before bed; stop if no effect after ~1 week
Response	50–70%; slow (weeks–months)	Immediate; rule of thirds (complete/partial/none); best with nocturnal polyuria and normal daytime volumes
Durability	**Durable**; overlearning (extra evening fluid after success until another 14 dry nights) lowers relapse	**Not durable** — effect ends with drug; use nightly with periodic drug holidays, or for special nights (camp)
Safety	Nil	Water intoxication/hyponatraemia if drinking excessively: † limit fluids from 1 h before dose until morning
Best for	Small nocturnal capacity, motivated family	Nocturnal polyuria; family wants rapid effect

ALGORITHM
MNE ≥6 yr → alarm or desmopressin → failure → check correct use → switch or **combine** (alarm + desmopressin) → re-evaluate: hidden daytime LUTS, constipation (impaction may be silent), sleep-disordered breathing, neurodevelopmental disorder; uroflow + PVR → **add anticholinergic** to desmopressin (second line) → refractory: imipramine, β3-agonist, specialist psychoactive drugs

- **Sleep-disordered breathing**: arousal failure + atrial natriuretic peptide → polyuria; ask about snoring; adenotonsillectomy helps about half.
- **Anticholinergic add-on**: check PVR and bowels first; initial response then relapse → suspect constipation.
- **Imipramine** 25–50 mg (+ desmopressin) works but is **cardiotoxic** (fatal in overdose, long QT) → electrocardiogram (ECG) if syncope, palpitations or family history of sudden death; keep away from siblings.

7 · FUNCTIONAL CONSTIPATION — TREATMENT

ALGORITHM
Faecal impaction? → **disimpact** (oral polyethylene glycol (PEG) or enemas) → education, diary, toilet sitting, **maintenance PEG** → review at 2 wk → failing → reassess adherence, dose, drug, impaction, mental health → **≥2 months** maintenance → wean when ≥1 month symptom-free, ≥3 stools/week, Rome IV no longer met → relapse → reinstate / paediatric gastroenterology

Laxative	Dose
PEG 3350 — disimpaction	**1–1.5 g/kg/day × 3–6 days** (>3 yr)
PEG 3350 — maintenance	**0.2–0.8 g/kg/day**, start 0.4 g/kg/day (>6 months)
Lactulose — maintenance	1–2 g/kg once or twice daily (>6 months; safe in infants)

- **PEG** = first choice (poorly absorbed osmotic agent). Electrolyte-free PEG is as effective with fewer adverse events. Daily enemas for 3–6 days are equivalent if PEG unavailable. Caution and electrolyte monitoring in renal impairment.
- **Titrate**: no stool in 24 h = stool accumulating → increase. If dose escalation causes watery stool and accidents, add a **stimulant** (senna, bisacodyl) for ejection with a smaller osmotic dose.

- **No dependence** with PEG or lactulose — "dependence" is relapse after abrupt stopping; wean gradually. Laxatives do not cure underlying dysmotility.
- **Non-drug**: scheduled toilet sitting after meals (gastrocolic reflex), foot support with knees above hips. **Normal** fluid, fibre (~10 g/day young child → ~20 g adolescent) and activity — extra water or fibre does not help, and probiotics are not recommended. These myths must not delay laxatives.

Refractory constipation

- Paediatric gastroenterology; psychological therapy; pelvic floor physiotherapy/biofeedback for dyssynergic defecation (bowel counterpart of dysfunctional voiding).
- **Transanal irrigation**: effective, but a quarter to a third stop (discomfort).
- Internal anal sphincter **onabotulinumtoxinA** (high sphincter pressures); TENS, PTNS; SNS effective but reoperation-prone.
- **Antegrade continence enema (ACE)** (Malone appendicostomy or caecostomy tube) when retrograde enemas work but an easier route is wanted; **colonic manometry first** — normal or segmental dysmotility empties well with ACE, diffuse dysmotility may need resection or diversion → *Ch 48*.

8 · RECURRENT UTI AND BBD

- In RIVUR/CUTIE, recurrence: **BBD + VUR 51% > BBD alone 35% > VUR alone 20%**. BBD outranks reflux as a risk factor. Wider caecal diameter (constipation) → more recurrence.
- Dysfunctional voiding, postponement and underactive bladder carry more UTI than OAB; secondary VUR often resolves with treatment of the dysfunction.
- **EAU/ESPU**: exclude BBD in every toilet-trained child with febrile/recurrent UTI; give prophylaxis during BBD treatment. **American Urological Association (AUA)**: prophylaxis for BBD + VUR (highest risk). ICCS: continue until LUT and bowel symptoms improve.

9 · LANDMARK TRIALS

Trial	Design	Result	Take-home
RIVUR + CUTIE (BBD analysis)	Prophylaxis vs placebo, children <6 yr after febrile UTI, with/without VUR	BBD + VUR highest recurrence; BBD alone > VUR alone	Treat BBD; prophylaxis for BBD + VUR
LION / LEOPARD	Phase III placebo-controlled RCTs, solifenacin	Improved all efficacy measures; mild–moderate side effects	Best paediatric RCT evidence for an antimuscarinic

10 · MCQ ONE-LINERS

1. ICCS increased frequency: ≥8 voids/day; decreased: ≤3 voids/day.
2. Daytime incontinence and enuresis are diagnosed from age 5 years.
3. † EBC = (age + 1) × 30 mL.
4. Recurrent UTI is itself an ICCS symptom of BBD.
5. EAU/ESPU: treat constipation before specific treatment of LUT dysfunction.
6. Dysfunctional voiding: staccato flow + active pelvic floor EMG during voiding.
7. Hinman syndrome = non-neurogenic neurogenic bladder, end stage of dysfunctional voiding.
8. EMG lag time <2 s suggests detrusor overactivity.
9. First-line specific therapy for dysfunctional voiding: biofeedback.
10. Standard urotherapy trial: at least 4 weeks; rewards adherence, not dryness.
11. Oxybutynin IR dose 0.3–0.6 mg/kg/day, maximum 15 mg/day.
12. Trospium is a quaternary amine — does not cross the blood–brain barrier.
13. Mirabegron approved in children ≥3 yr only for neurogenic detrusor overactivity.
14. Paediatric refractory OAB botulinum dose: 5 units/kg, maximum 100 units.
15. PTNS needle ~5 cm above medial malleolus; great-toe flexion confirms placement.
16. Giggle incontinence: methylphenidate or biofeedback; biofeedback more durable.
17. Vaginal reflux: void with legs abducted; leak just after voiding.

18. Rome IV: ≥2 criteria, weekly, for ≥1 month; pain is not a criterion.
19. Plain abdominal radiograph not recommended to diagnose functional constipation.
20. Absent RAIR on anorectal manometry → rectal biopsy for Hirschsprung disease.
21. PEG disimpaction 1–1.5 g/kg/day for 3–6 days; maintenance start 0.4 g/kg/day.
22. Enuresis triad: nocturnal polyuria, small nocturnal capacity/overactivity, arousal failure.
23. Alarm is durable; desmopressin is immediate but not durable.
24. Desmopressin tablet 0.2–0.4 mg 1 h before bed; risk hyponatraemia with excess fluid.
25. Imipramine displaced from first line by cardiotoxicity.
26. Recurrent UTI risk: BBD + VUR (51%) > BBD alone > VUR alone.

48 Lower Urinary Tract Reconstruction in Children

Reading time ≈ 49 min (slow pace)

1 · GOALS AND PRINCIPLES

- Order of priorities: **protect renal function → prevent urinary tract infection (UTI) → achieve continence**.
- Primary indication: bladder "hostility" (poor compliance, high storage pressure, poor emptying) unresponsive to medical management — most often spina bifida (neurogenic bladder); also exstrophy, posterior urethral valves, bilateral ectopic ureters.
- **Clean intermittent catheterisation (CIC)** (Lapides) is the single most important contribution to reconstruction: it allows aggressive correction of storage, because spontaneous voiding after reconstruction is a hope, not a requirement.
- Most important determinant of outcome: **commitment of patient and family**.

> **KEY POINT** Every reconstructive step is a storage–emptying trade. Raising capacity or outlet resistance converts a leaking but self-venting system into one that depends entirely on reliable emptying. A continent but uncatheterised reservoir causes hydronephrosis, sepsis or perforation — hence incontinence can be "protective", and CIC and motivation are prerequisites, not afterthoughts.

2 · PATIENT SELECTION AND PREOPERATIVE EVALUATION

Physiology to define

- **Storage pressure**: chronic filling pressure **>40 cm H_2O** impairs ureteral drainage → hydroureteronephrosis, falling glomerular filtration rate (GFR), acquired vesicoureteral reflux (VUR). First treat with anticholinergics + CIC; persistent poor compliance → **augmentation**.
- **Outlet**: continence needs outlet resistance > bladder pressure during all activity. Only one of bladder neck or external sphincter is needed. A deficient outlet → outlet procedure.
- **Detrusor behind a weak outlet**: raising outlet resistance may **unmask** high-pressure storage or uninhibited contractions → urodynamics with **bladder neck occlusion** (balloon catheter) before any outlet procedure.
- **Emptying**: no test predicts voiding after augmentation (neurogenic patients rarely void) → every patient must learn and practise CIC preoperatively.

Urodynamics — what each finding decides

Finding	Decision
Cystometric capacity + fluoroscopic bladder shape	Length of bowel segment
Detrusor leak point pressure, end-filling pressure, open bladder neck on fluoroscopy	Need for bladder neck procedure
VUR + history of UTI	Reimplantation at reconstruction
Sphincter electromyography (EMG; needle more accurate)	Dyssynergia

- Technique: fill at **≤10% of predicted capacity per minute** (fast filling falsely lowers compliance); body-temperature saline/contrast; catheter size affects leak pressure in infants. The Urologic Management to Preserve Initial Renal function (UMPIRE) protocol standardises paediatric neurogenic bladder care.

Upper tracts and renal function

- Ultrasound (US) in all. Hydronephrosis → nuclear renography **with a bladder catheter** (excludes primary obstruction, which is corrected at the same operation) and voiding cystourethrogram (VCUG)/video urodynamics for reflux.

- Serum electrolytes, blood urea nitrogen (BUN), creatinine; raised creatinine → nephrology and 24-h urine (creatinine clearance and **volume**). Impaired concentration → large urine volumes; the reservoir must hold **3–4 h** of output.
- **Renal insufficiency**: bowel absorbs chloride and acid → pre-existing acidosis **always worsens** with ileum/colon (stomach may be chosen). Oliguria → more mucus inspissation.
- **Liver failure**: ammonia reabsorption dangerous → liver function tests, blood gases.
- **Reflux**: often secondary; low-grade secondary reflux usually resolves after augmentation; correct high-grade reflux at reconstruction.

Bowel, dexterity and motivation

- **Bowel history**: short gut (cloacal exstrophy, resection, irradiation); chronic diarrhoea or faecal incontinence → avoid the ileocaecal valve. Identify children needing an antegrade continence enema (ACE) — do it at the same operation.
- **Catheterisation route**: examine the urethra (sensate urethra, wheelchair-bound child, prior bladder neck repair) — patients overwhelmingly prefer an abdominal stoma. † Assess hand function and body habitus (scoliosis, wheelchair) for independent CIC.
- **Commitment** proven by reliable preoperative CIC. Failure to catheterise → UTI, upper-tract deterioration or perforation despite a perfect operation.
- **Timing**: early when kidneys are threatened; otherwise elective, when incontinence becomes socially unacceptable. Address all needs (augment, outlet, channel, ACE, reimplant) in **one operation** where possible.

Preparation

- **Sterile urine** mandatory (culture days before; treat and document sterility). Mechanical/antibiotic bowel preparation now surgeon preference; enhanced recovery after surgery (ERAS) protocols (carbohydrate loading to 2 h, no prolonged clear diet) reduce 90-day complications. Hydrate well.
- **Ventriculoperitoneal (VP) shunt**: infection risk low with sterile urine; shunt failure increased after augmentation; remove pelvic drains early.

3 · MANAGING THE URETERS

- Reimplant into native bladder whenever possible (± **psoas hitch**: bladder fixed to psoas tendon lateral to iliac vessels with 3–5 non-absorbable sutures through muscle, not mucosa — avoid genitofemoral and femoral nerves; allows long 8–10 cm tunnel for dilated or scarred ureter). Transureteroureterostomy for a short ureter.
- Into bowel: **colon** — tunnel under a taenia; **stomach** — submucosal tunnel; **ileum** — no supporting wall, needs a **nipple valve** (most difficult).

4 · BLADDER AUGMENTATION (ENTEROCYSTOPLASTY)

Common principles

- Cystoscopy first; do other bladder steps before opening the peritoneum.
- **Native bladder**: preserve it but **bivalve widely** (sagittal, from just above the bladder neck anteriorly to just above the trigone; stellate for more circumference). A narrow anastomosis makes the segment a diverticulum.
- **Detubularise** on the antimesenteric border and **reconfigure into a sphere** — always. Intact bowel contracts at **60–100 cm H_2O**; detubularisation blunts contractions and a sphere gives maximal volume per surface area, so a shorter segment suffices.
- Length: ileum **20–40 cm**, colon **~20 cm**; err on too large rather than too small.
- **Never permanent sutures or staples** in the reservoir — nidus for stones. Absorbable sutures, mucosa inverted; suprapubic tube through native bladder; drain removed early.

Segments

- **Ileocystoplasty**: segment at least **15–20 cm proximal to ileocaecal valve** (terminal ileum absorbs vitamin B12 †and bile salts) → re-anastomose, close mesenteric window → open antimesenterically → U, S or W fold or cup patch → anastomose posteriorly first.
- **Ileocaecocystoplasty**: caecum + 15–30 cm ileum, opened through the valve and folded (Mainz: ileum twice caecal length). Gains: taenial reimplant, appendix for a channel. Loss of valve → diarrhoea in neurogenic bowel.
- **Sigmoid cystoplasty**: 15–20 cm; strong unit contractions → must be detubularised (U-fold better than end-closed Mitchell technique).
- **Gastrocystoplasty**: † wedge of gastric body on the right or left gastroepiploic pedicle. Reserved for **renal insufficiency with acidosis, short gut, heavily irradiated bowel**.
- **Jejunum**: avoided (severe salt-losing acidosis).

Segment	Advantages	Disadvantages
Ileum (default)	Least pressure activity, low basal tone; plentiful, mobile mesentery	Hyperchloraemic acidosis; B12 deficiency; mucus; hard to make antireflux
Ileocaecal	Taenial reimplant; appendix; imbricated valve as continence mechanism	Diarrhoea/faecal incontinence in neurogenic bowel; more mucus
Sigmoid	Taenial reimplant; no small-bowel loss; close to bladder	Strongest contractions; most mucus; more reaugmentation in some series
Stomach	No acidosis (loses acid); least mucus; fewer struvite stones; usable in renal failure	Hypokalaemic hypochloraemic alkalosis; haematuria–dysuria syndrome; rhythmic contractions; tumours
Ureter (ureterocystoplasty)	Urothelium + muscle: no mucus, no metabolic effect	Needs a massively dilated ureter of a non-functioning kidney (nephrectomy); limited volume
Autoaugmentation (detrusorectomy/myotomy)	Extraperitoneal; no bowel; bladder mucosa bulges as a diverticulum	Seldom adds significant capacity; compliance may not last

- **Ureterocystoplasty**: nephrectomy → ureter left attached, opened opposite its blood supply, reconfigured and sewn into the bivalved bladder.
- **Autoaugmentation**: detrusor excised off intact mucosa over the dome, edges hitched to psoas. Seromuscular enterocystoplasty covers the bulging mucosa with demucosalised sigmoid. Tissue-engineered bladder: no benefit in a multicentre trial — experimental only.
- Segment choice matters less than technique: use ileum unless there is a reason not to. Robotic intracorporeal augmentation mirrors open outcomes.

> **KEY POINT** Why is augmentation needed less now? Early anticholinergics, CIC, botulinum toxin and neuromodulation for the hostile neurogenic bladder lower augmentation rates (immediate vs expectant treatment: 17% vs 41%). Unproven by randomised trials, but the rationale for treating the bladder at diagnosis.

Incontinent alternatives

- **Cutaneous vesicostomy**: vents the bladder in infants with worsening hydronephrosis, febrile UTI with high-grade reflux or high pressure despite therapy — a bridge. Problems: stenosis, prolapse.
- **Ileovesicostomy**: 10–15 cm ileum from bladder dome to a rosebud stoma → low-pressure, catheter-free, low perforation risk; for those who cannot catheterise. Convertible later.

5 · CONTINENT CATHETERISABLE CHANNELS

Mitrofanoff principle (flap valve)

- Any supple, narrow tube tunnelled submucosally into a **low-pressure** reservoir is continent: rising reservoir pressure **compresses** the tunnelled segment. Appendix needs only a **~2 cm** tunnel.
- Nipple valves (Kock, Mainz, Hanna) are **effaced** by the same pressure — more failures and reoperations (continence ~85% vs >90–95%).
- Flap valves are the most reliable mechanism — but often give **no pop-off**: missed CIC → upper-tract damage or rupture.

Appendicovesicostomy — key steps

ALGORITHM

Mobilise caecum if high → detach appendix with a small caecal cuff (cuff to skin reduces stenosis; caecal flap extends a short appendix) → preserve mesoappendix → tunnel the **distal end** submucosally (posterolateral bladder; open or extravesical trough) → bring caecal end to skin without tension, twist or fascial compression → **skin flap** into stoma (V/rotational umbilical flap) → keep channel **short and straight**; fix bladder to anterior wall beneath fascia to prevent kinking → test catheterisation after every step and at different volumes → stent 6–12 French (Fr) for 10–14 days → surgeon does first catheterisation → CIC ≥4-hourly by day

- **Stoma site**: umbilicus (hidden, cosmetic) or right lower quadrant; † mark preoperatively in the sitting position, visible and reachable by the child (wheelchair users, obese).
- **Appendix unavailable** (appendicectomy, short, reserved for ACE) → alternatives. The appendix can be **split** for both ACE and channel.

Channel	Construction	Notes
Appendix (Mitrofanoff)	As above	First choice; minimal harvest morbidity; umbilical stenosis
Yang–Monti	1–2 cm ileal ring opened antimesenterically, retubularised **transversely** over 12 Fr (circumference becomes length); off-centre incision gives a long skin limb	Efficient; easy catheterisation; short (obese → double or spiral Monti); about twice the revision rate of appendix
Tapered ileum	Stapled along mesenteric border	Long enough; kinks if redundant
Ureter / fallopian tube	Native tube	More stenosis; ureteric discomfort; fertility (tube)
Imbricated ileocaecal valve (Indiana)	Tapered terminal ileum, valve plicated with permanent sutures	Simple, short learning curve; diarrhoea in neurogenic bowel
Bladder flap (Casale continent vesicostomy)	Anterior full-thickness bladder tube + intravesical mucosal flap valve	Extraperitoneal, spares appendix and bowel; consumes capacity; high stenosis

Channel complications and management

- **Stomal stenosis** — commonest (umbilical appendix > ileum), clustering in the first months to years: indwelling "L-stent" with topical steroid cream, or formal suprafascial revision; skin flaps and absence of tension reduce it.
- **Difficult catheterisation/false passage**: redundant extravesical length, kinking, pouch-like dilatation → subfascial revision.
- **Stomal incontinence** (rare): inadequate flap-valve length or persistently high reservoir pressure → urodynamics to tell which.
- Polyps/granulation tissue; necrosis of caecal extensions; reservoir stones (non-dependent drainage).

Continent diversion proper

- "Pure" continent diversion (Kock, Indiana, Mainz pouch) after cystectomy gives adult-equivalent results, but needs far more bowel (~40 cm ileum for Kock vs ~20 cm for augmentation) → keep any useful native bladder and repair the outlet instead. Orthotopic neobladder is rare (outlet problems prevent voiding).
- **Ureterosigmoidostomy and variants** (Mainz II sigma-rectum pouch: detubularised low-pressure rectosigmoid): urine passes with stool, but acidosis, pyelonephritis and lifelong adenocarcinoma risk persist → largely abandoned in children. Prerequisite: competent anal sphincter (manometry, EMG, upright retention of an oatmeal enema).
- **Bladder buttons** (low-profile balloon devices through a matured suprapubic tract): drainage only, for children unsuitable for augmentation/CIC.

6 · BLADDER OUTLET PROCEDURES

Every outlet procedure raises resistance at the cost of voiding and pop-off, and may create a hostile bladder → most neurogenic outlet repairs succeed only with **concurrent augmentation**, and need surveillance **≥5 yr** (many progress to augmentation or botulinum toxin within 2–5 yr).

Procedure	Mechanism	Voiding	Main problem / place
Fascial sling (rectus fascia 1 cm, ± circumferential wrap)	Coaptation + elevation of bladder neck	Rarely adequate → CIC	Better in girls; overcome by poor compliance — augmentation the only predictor of success; does not preclude a later artificial sphincter
Artificial urinary sphincter (AUS)	Inflatable cuff at bladder neck	**Preserved**	Infection, erosion, malfunction; best in patients who empty by voiding
Bladder neck injection (dextranomer/hyaluronic acid)	Submucosal bulking	Preserved	Poor durability; limited role
Young–Dees–Leadbetter	Tubularised trigone, ureters advanced	Possible	Exstrophy → *Ch 43*; poor in neurogenic muscle
Kropp	Anterior bladder tube tunnelled under trigone (flap valve)	Abolished	Highest continence; difficult urethral CIC (boys); new VUR
Pippi Salle	Anterior flap onlaid on trigonal strip	Abolished	Easier CIC; lower (night) continence; fistula at flap base
Bladder neck closure/division	Complete separation; continent stoma obligatory	Abolished	After failed repairs; no urethral access or pop-off

- **Sling technique**: incise endopelvic fascia ~2 cm; locate bladder neck by catheter-balloon traction; develop plane between bladder neck and vagina/rectum (open bladder in redo cases); fix fascia to rectus sheath. Best indication: marginal bladder neck, deficient sphincter, child already augmenting and on CIC.
- **AUS**: cuff at **bladder neck** in all girls and prepubertal boys (bulbar urethra only with mature spongiosum); children do not outgrow it. Sterile urine essential; omentum if cuff around bowel. Leave **deactivated ~6 weeks** so a pseudocapsule forms (less erosion). Most neurogenic children still need CIC (through the cuff). Device revisions are now uncommon and long-term continence high. Infection → remove, replace later. New hydronephrosis after AUS = unmasked detrusor hostility → anticholinergics or augmentation. Simultaneous vs staged with augmentation: no long-term difference.
- **Kropp steps**: 6 × 2 cm anterior flap based on bladder neck → tubularised over catheter → passed straight through submucosal trigonal tunnel → must open into true lumen to act as flap valve. Kropp/Salle both carry high delayed augmentation if not augmented concurrently.

- **Pippi Salle steps**: anterior full-thickness flap ~1 × 4–5 cm → two parallel trigonal mucosal incisions leave a midline strip → flap sewn onto strip in two layers (prevents fistula) → lateral mucosa closed over; widen flap base to avoid tip necrosis.
- **Bladder neck closure**: wide mobilisation, posterior lip rolled anteriorly, multilayer closure, **interpose omentum or rectus muscle** → low fistula rate. Only in reliable catheterisers.

> **KEY POINT** Choosing the outlet: a child who empties by voiding → AUS (the only option preserving voiding). A child on CIC who is being augmented → sling (girls) or flap-valve lengthening with a catheterisable channel. Failed repairs → bladder neck closure + Mitrofanoff. Always re-study the bladder afterwards.

7 · MALONE ANTEGRADE CONTINENCE ENEMA (MACE)

- † Principle: a catheterisable appendicocaecostomy allows antegrade colonic washout, emptying the whole colon on schedule → faecal continence in neurogenic bowel (spina bifida), anorectal malformations and refractory constipation.
- † Technique: appendix left on caecum (or reversed) and its base tunnelled/imbricated into the caecal wall as an antireflux valve; stoma at umbilicus or right iliac fossa; alternatives — Monti channel, percutaneous caecostomy button, left-sided (sigmoid) ACE.
- † Use: daily or alternate-day saline/tap-water flush (± stimulant), evacuation on the toilet within ~1 h.
- Complications: leakage, stomal stenosis, granulation, tube dislodgement, skin infection. In non-neurogenic constipation, anorectal and colonic manometry guide selection → *Ch 47*.
- Plan it with urinary reconstruction (split appendix or appendix for ACE + Monti for bladder).

8 · POSTOPERATIVE CARE

- Replace third-space losses.
- Suprapubic tube irrigated **≥3 times daily** (mucus); pelvic drain out early.
- **Cystogram at ~3 weeks** → clamp tube → CIC every 2–3 h by day + 1–2 times at night, lengthened gradually but **never beyond 4–5 h** by day.

9 · COMPLICATIONS OF AUGMENTATION AND THEIR MANAGEMENT

Further surgery is needed in up to one-third (10-year risk near half). Each complication follows from **bowel not being bladder**.

Metabolic

Segment	Derangement	Management
Ileum, colon	**Hyperchloraemic metabolic acidosis** — ammonium and chloride absorbed (near-universal rise in chloride; overt acidosis mainly with renal impairment)	Bicarbonate; **base deficit on blood gas** is the most sensitive marker; severe: chlorpromazine or nicotinic acid (block chloride transport)
Jejunum	Hyponatraemic, hypochloraemic, hyperkalaemic acidosis + hypovolaemia	Avoid jejunum
Stomach	**Hypokalaemic hypochloraemic metabolic alkalosis** during vomiting/dehydration; **haematuria–dysuria syndrome** (acid urine, sensate patients)	Fluids, H_2 blockers / proton pump inhibitors; rarely partial resection; warn families
Composite stomach + bowel	Near neutral	Most complex patients

- **Bone and growth**: chronic acid load buffered by bone → demineralisation and possible decline in height percentile even without frank acidosis → treat base deficit early.

- **Vitamin B12**: deficiency and megaloblastic anaemia after distal ileal loss; stores last years → measure B12 long term; oral replacement.
- **Gastrointestinal**: bowel obstruction uncommon (close mesenteric windows); chronic diarrhoea mainly after ileocaecal valve loss.
- † Drug toxicity: urinary-excreted drugs (e.g. methotrexate, phenytoin) can be reabsorbed.

Mucus, infection, stones

- **Mucus**: colon > ileum > stomach; blocks catheters, nidus for infection and stones → **daily irrigation**.
- **Bacteriuria** is expected with CIC. Treat only symptomatic infection (pain, new incontinence, fever) and **urea-splitting organisms**; may treat haematuria, foul urine, increased mucus.
- **Stones**: common; mostly **struvite**; risks — bacteriuria, mucus, incomplete emptying (highest with abdominal stoma), foreign bodies. Prevent: complete emptying, **daily irrigation ≥240 mL** saline (markedly reduces recurrence), treat urease producers. Stomach: fewer struvite, but uric acid stones.

Delayed spontaneous perforation

- Counsel **~9%** risk; mostly myelodysplasia; **outlet procedures double the risk**; risk with missed CIC, substance abuse.
- Cause: chronic overdistension → ischaemia of the bowel wall (high pressure + high outlet resistance), ± traumatic catheterisation, adhesions; recurrence common (field effect).
- **Presentation**: abdominal pain, distension, fever, sepsis, reduced urine output, shoulder-tip pain. Neurogenic patients have poor sensation → **present late**, with mortality; gastric augments present early (acid peritonitis).
- **Diagnosis**: high suspicion; contrast cystogram with full filling and post-drain film; false negatives → computed tomography (CT) cystogram/US.
- **Management**: **immediate laparotomy and repair** (standard). Conservative (catheter drainage, antibiotics, percutaneous drains) only if stable, sterile urine, contained leak — very low threshold for surgery. Repeated perforation → replace segment.

Malignancy

- Ureterosigmoidostomy: adenocarcinoma at the anastomosis, mean latency **~26 yr**, risk vastly increased (urine + faeces → N-nitroso compounds).
- Augmentation: adenocarcinoma reported from **4 yr**, highest after **gastrocystoplasty**; aggressive, often metastatic at diagnosis. Bladder cancer in spina bifida ~2–4%, mostly stage III–IV — part of the risk may be inherent to the underlying anomaly (exstrophy). Nephrogenic adenoma (benign) is the commonest tumour.
- Annual cystoscopy: no consensus on start or value; consider in high-risk groups (gastrocystoplasty, exstrophy, immunosuppressed transplant recipients); † investigate any haematuria.

Pregnancy

- Pedicle deflects laterally — not lost. Hydronephrosis common (nephrostomy preferred to stent), UTI, catheterisation difficulty. Vaginal delivery possible; caesarean after major bladder neck repair or with AUS; urologist present to protect pedicle/channel.

10 · FOLLOW-UP AND TRANSITION

When	Tests
3 months, 6 months, 1 year	Renal US; electrolytes, BUN, creatinine, urine culture (± urodynamics by 6 months)
Yearly, lifelong	Renal US, serum chemistry (± blood gas, B12), review CIC/irrigation compliance; urodynamics if new hydronephrosis, incontinence or VUR
Symptomatic	† Haematuria → cystoscopy; abdominal pain/sepsis → exclude perforation

- Compliance never assumed: early high-pressure contractions after augmentation → incontinence, hydronephrosis, perforation → repeat urodynamics; true reaugmentation is rare.
- Augmentation + bladder neck procedure → renal scarring and chronic kidney disease (CKD) stage ≥3 in a substantial proportion, driven by **catheterisation non-compliance**. Deaths from rupture. Adult (transitional) urology must police CIC, irrigation and upper tracts.

11 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Early vs expectant neurogenic bladder care	Cohort, hostile bladders treated at diagnosis vs expectantly	Augmentation 17% vs 41%	Treat hostile bladder early
Daily bladder irrigation	Augmented cohort, ≥240 mL saline daily	Marked fall in stone recurrence and incidence	Routine daily irrigation
Tissue-engineered bladder	Industry multicentre trial of seeded scaffold	No gain in compliance or capacity	Bowel remains standard; regeneration experimental

12 · MCQ ONE-LINERS

1. Pathogenic chronic filling pressure: >40 cm H_2O.
2. Fill paediatric urodynamics at ≤10% of predicted capacity per minute.
3. Reservoir must hold 3–4 h of urine output.
4. Bladder neck occlusion urodynamics unmask hostility before outlet surgery.
5. Ileum taken ≥15–20 cm proximal to the ileocaecal valve.
6. Typical augmentation: ileum 20–40 cm; sigmoid 15–20 cm.
7. Detubularisation always; intact bowel contracts at 60–100 cm H_2O.
8. Never permanent sutures/staples in the reservoir — stone nidus.
9. Ileum/colon cause hyperchloraemic acidosis; base deficit most sensitive marker.
0. Jejunum: hyponatraemic hyperkalaemic hypochloraemic acidosis.
1. Stomach: hypokalaemic hypochloraemic alkalosis and haematuria–dysuria syndrome.
2. Stomach preferred in renal insufficiency, short gut, irradiated bowel.
3. Mucus production: colon > ileum > stomach.
4. Augmentation stones are mostly struvite; daily irrigation ≥240 mL reduces them.
5. Delayed perforation ~9% (Indiana); outlet procedures double risk.
16. Perforation: cystogram with post-drain film; CT if negative; treat by surgery.
17. Ureterosigmoidostomy adenocarcinoma: mean latency ~26 years.
18. Earliest post-augmentation adenocarcinoma: 4 years.
19. Commonest tumour after cystoplasty: nephrogenic adenoma.
20. Mitrofanoff: appendix needs only a ~2 cm submucosal tunnel.
21. Flap valve tightens with pressure; nipple valve is effaced.
22. Commonest complication of continent channels: stomal stenosis.
23. Monti channels need about twice the revisions of appendicovesicostomy.
24. Monti: ileal ring retubularised transversely over 12 Fr.
25. Only outlet procedure preserving voiding: artificial urinary sphincter.
26. AUS cuff at bladder neck in girls and prepubertal boys; activate ~6 weeks.
27. Kropp: highest continence, difficult catheterisation; Pippi Salle: easier catheterisation.
28. Sling success predicted only by concomitant augmentation.
29. Bladder neck closure needs omentum/muscle interposition and a continent stoma.
30. Ureterocystoplasty needs a massively dilated ureter.

★ Putting it together: common paediatric pathways

Reading time ≈ 9 min (slow pace) · Integrates Ch 32–48 · use these as the skeleton of long-essay answers

A · ANTENATAL URINARY TRACT DILATION (UTD)

ALGORITHM

Prenatal: UTD A1 (mild) before 28 weeks → repeat ultrasound (US) after 32 weeks. UTD A2–3 → US every 4 weeks to delivery. Male fetus with a thick-walled bladder and bilateral dilation → suspect posterior urethral valves (PUV).

Postnatal: first US **after 48 hours** (earlier scans under-read dilation because the newborn is relatively dehydrated), except suspected PUV or bilateral severe dilation, which are scanned before discharge.

Then: normal (anteroposterior diameter <10 mm, all parameters normal) → transient, no follow-up · mild → repeat US · moderate–severe or dilated ureter → voiding cystourethrogram (VCUG), mercaptoacetyltriglycine (MAG3) renogram and consider continuous antibiotic prophylaxis (CAP) (→ *Ch 34*).

B · FIRST FEBRILE URINARY TRACT INFECTION (UTI)

ALGORITHM

Diagnose properly: catheter or suprapubic sample in the non-toilet-trained child; bag samples only to rule out.

Treat early: antibiotics within 48 hours (delay beyond 72 hours doubles the odds of new scarring); oral therapy is fine if the child is not toxic.

Image: US for all; VCUG if US is abnormal, infection is atypical or recurrent (American Academy of Pediatrics 2–24 months); dimercaptosuccinic acid (DMSA) scan at 4–6 months if scarring matters.

Prevent: treat bladder and bowel dysfunction (BBD) and constipation; CAP for the high-risk child; circumcision lowers infant UTI risk (→ *Ch 37, 41*).

C · NEWBORN WITH POSTERIOR URETHRAL VALVES

ALGORITHM

Stabilise breathing → 5–7 Fr feeding tube per urethra (minimal or no balloon) → creatinine, electrolytes, fluid balance (watch for post-obstructive diuresis) → US then VCUG (dilated posterior urethra, hypertrophied bladder neck) → antibiotic prophylaxis → **endoscopic valve ablation** when stable; vesicostomy if the urethra is too small.

Prognosis: nadir creatinine in the first year (<0.8 mg/dL good). Lifelong follow-up for valve bladder, CKD, blood pressure and continence; transplant → Ch 61 (book 1) (→ *Ch 45*).

D · NEUROGENIC BLADDER IN SPINA BIFIDA

Step	What	Why
Birth	US + post-void residual; creatinine after week 1; latex-free care	Detect retention and baseline function
≈3 months	Urodynamics (± video)	Identify a hostile bladder: detrusor leak point or end-filling pressure ≥40 cmH_2O, poor compliance, dyssynergia
Hostile bladder	Clean intermittent catheterisation (CIC) + antimuscarinic early	Protects the upper tracts

Step	What	Why
Failing	Intravesical botulinum toxin → vesicostomy (infant) → augmentation ± catheterisable channel ± outlet procedure	Low-pressure storage with reliable emptying (→ *Ch 46, 48*)
Always	Bowel programme; watch for re-tethering (new symptoms, changed urodynamics)	Continence and safety

E · DAYTIME WETTING AND BEDWETTING

ALGORITHM

History, examination, urinalysis, post-void residual ± uroflow → exclude neurogenic, anatomical and medical causes → **treat constipation first** → standard urotherapy for at least 4 weeks → specific therapy by diagnosis (antimuscarinic for overactive bladder, biofeedback for dysfunctional voiding) → refractory: botulinum toxin, neuromodulation.

Monosymptomatic bedwetting: enuresis alarm or desmopressin (no fluids from 1 hour before until morning); combine for partial responders (→ *Ch 47*).

F · CHOOSING A SEGMENT FOR BLADDER AUGMENTATION

Segment	Advantages	Disadvantages
Ileum	Default; compliant, reliable blood supply	Mucus, hyperchloraemic acidosis, B12 and bile-salt loss if terminal ileum used
Ileocaecum / colon	Appendix available for a channel; taenia for tunnelled reimplant	Loss of ileocaecal valve → diarrhoea in neurogenic bowel; acidosis
Stomach	Secretes acid — useful in renal failure; little mucus	Haematuria–dysuria syndrome, metabolic alkalosis
Ureter (ureterocystoplasty)	Urothelium — no mucus or metabolic problems	Needs a massively dilated ureter of a non-functioning kidney

KEY POINT Every reconstruction trades storage for emptying. A continent reservoir that is not reliably catheterised causes hydronephrosis, sepsis or perforation, so proven commitment to CIC comes before surgery. Spontaneous perforation of an augmented bladder presents with abdominal pain and sepsis and needs urgent diagnosis (→ *Ch 48*).